

Brāhmasphuṭasiddhānta

English Translation Reader

Brahmagupta (628 CE)

English-only reading text. The companion Indian-original reader carries source-page coverage.

Chapter I: Mean Longitudes of Planets

Verse 1: Salutation to that Brahman, by whose grace I have again come to know the computation of planets; now I proceed to reconcile it.

Verse 2: The planetary computation taught by Brahman had become inaccurate through the lapse of long time; that has been correctly established by Brahmagupta.

Verse 3: The years of a *yuga* are its *abda*; the days of a *yuga* are its *ahargaṇa*. A year consists of solar days; a month of lunar *tithis*.

Verse 4: A *sāvāna* year consists of *nakṣatra* days; a solar year is fourfold. The full moon at the fifteenth *tithi* is complete, at the fourteenth incomplete.

Verse 5: A thousand *caturyugas* are a day of Brahman; half is night. These are called a *kalpa*.

Verse 6: At the beginning of a *kalpa* arises Brahmā, in the middle Viṣṇu, at the end Rudra: these three lords of the three worlds.

Verse 7: In that *kalpa*, six *j*
=*ivas* — the *manus* — are established; there dwell the people called *p*
=*urva*.

Verse 8: Four world-ages in a *yuga*; a thousand *caturyugas* are a day of Brahman.

Verse 9: The *kr̥ta*, *trētā*, *dvāpara*, and *kali* are the ages; a thousand *caturyugas* are a day of Brahman.

Verse 10: In *kr̥ta*, dharma has four feet; in *trētā* three; in *dvāpara* two; in *kali* one.

Verse 11: A hundred human years is a *divya* year; by such divine years, a day of Brahman is known.

Verse 12: The thousand *caturyugas* of Brahmā, Viṣṇu, Indra, Rudra, and the *manus* are a day of Brahman.

Verse 13: Humans, gods, *gandharvas*, *rākṣasas*, serpents, and ancestors — all proclaimed at the start of the *kalpa*.

Verse 14: Planets, asterisms, months, seasons, years — these twenty-five *tattvas* to be known diligently.

Verse 15: By four measures the planets are computed; I declare their numbers in order.

Verse 16: By solar measure, the Sun's motion; by lunar, the Moon's; by sidereal, the planets'; by civil, the day's.

Verse 17: By these nine measures the world's transactions proceed, for planetary computation.

Verse 18: For humans, transactions by four measures: solar, lunar, sidereal, and civil, in order.

Verse 19: All planetary computation is accomplished by measures; without them, that knowledge does not proceed.

Verse 20: The solar, lunar, sidereal, and civil measures are for planetary computation; by these four, the world's transactions proceed.

Verse 21: Fifteen *tithis* in bright and dark fortnight make one month; a year of those.

Verse 22: Twelve months a year; with thirteen, deficient; with five extra, abundant.

Verse 23: Having stated past and future motions, the wise compute; by these mean quantities, planetary computation proceeds.

Verse 24: Knowing true from mean longitudes, then corrections must be applied for exactness.

Verse 25: The *tithi*, *antara*, *nakṣatra*, and planetary motions — all devised by measures in order.

Verse 26: Thus all measures declared in order by those who contemplate planetary computation.

Verse 27: The years of a *yuga*, its months and days. A year of solar days; a month of lunar *tithis*.

Verse 28: The solar year: fourteen and a half terrestrial years; mean motion reckoned beyond.

Verse 29: Fifteen *tithis* in each fortnight; a month of those, a year of months.

Verse 30: Twelve months a year; thirteen, deficient; five extra, abundant.

Verse 31: Past and future motions stated, the wise compute by mean quantities.

Verse 32: From mean to true, corrections applied, for exact planetary computation by the wise.

Chapter II: True Longitudes of Planets

Verse 1: He who desires the true from the mean longitudes of planets should understand the corrections; then the true is obtained.

Verse 2: The field of *manda* and

=s

=*ighra* apogees is the fifth of a circle (72°); there the *manda* motion, and

=s

=*ighra* beyond.

Verse 3: The *manda-phala* in that field, and

=s

=*ighra-phala* likewise; by their sum or difference, the true is known.

Verse 4: The arc from the *manda* apogee causes the *manda-phala*; from the

=s

=*ighra* apogee, the

=s

=*ighra-phala*.

Verse 5: The mean increased or decreased by *manda-phala*; that result increased or decreased by

=s

=*ighra* corrections.

Verse 6: Thence the true planet, by *manda* and

=s

=*ighra* process in order; thus planetary computation declared.

Verse 7: The Sun's motion sometimes without *manda*; together with

=s

=*ighra*, the true is known.

Verse 8: For the Moon, *manda* and

=s

=*ighra* corrections; by sum or difference, the true arises.

Verse 9: At apogee, onefold *manda* result; at perigee, double; in between, triple.

Verse 10: By four measures, sixteen corrections; knowing the corrected planet, compute the true.

Verse 11: In *tithis*, years, *yugas*, months, and days, planetary motion computed from means.

Verse 12: From mean to true measure of planets, the wise compute; all becomes more exact.

Verse 13: Thus true motion declared for planets in order; by them all computation proceeds.

Verse 14: For Sun, *manda-phala* subtracted,

=s

=*ighra-phala* added; for Moon, by both processes.

Verse 15: For all star-planets, *manda* and

=s

=*ighra* equations always; by sum or difference, the true arises.

Verse 16: Knowing eastward and westward motion by both processes, the true planet is always obtained by calculators.

Verse 17: Thus true motion declared for Sun-born planets and all star-planets in order.

Verse 18: He who desires true from mean of planets should understand corrections; it becomes more exact.

Chapter III: Three Problems: Time, Place, and Direction

Verse 1: I declare the three questions always in planetary computation: knowing time, place, and direction, one obtains the result.

Verse 2: Time, place, and direction — these three for a planet; joined with their knowledge, it becomes more exact.

Verse 3: From time, motion; from place, declination; from direction, amplitude — these three give results.

Verse 4: Time is called sky-wandering (*kha-cara*); place is earth-wandering (*bh = u-cara*); direction is direction-elephant (*dig-gaja*) — these three accomplish the planet.

Verse 5: In *tithis*, years, *yugas*, months, days, time is numerical; place is the difference of countries (*de = sāntara*).

Verse 6: The *dig-valaya* is direction — these three accomplish the planet; joined with knowledge, it becomes more exact.

Verse 7: Applying the difference of countries joined with time, then planetary computation: the true arises.

Verse 8: The sky's motion joined with time, with place, and with direction.

Verse 9: Thus joined with three questions, the planet becomes more exact; therefore their knowledge always by calculators.

Verse 10: Difference of countries, direction, and time — these three questions proclaimed; joined with knowledge, the planet becomes exact.

Verse 11: At fifteenth and fourteenth *tithi*, full moon remembered; the *tithi-antara* known with time.

Verse 12: Difference of countries joined with time; then planetary computation: the true produced.

Verse 13: The *dig-valaya* as direction — these three accomplish; joined with knowledge, exact.

Verse 14: Thus with three questions, the planet becomes exact; therefore their knowledge by calculators.

Verse 15: In *tithis*, years, *yugas*, months, days: time is numerical; place is difference of countries.

Verse 16: The *dig-valaya* as direction — these three accomplish; joined with knowledge, exact.

Chapter IV: Lunar Eclipses

Verse 1: I declare the lunar eclipse as it is; therefore all the characteristics of eclipse.

Verse 2: The node between Sun and Moon — there the eclipse conjunction, in bright and dark fortnight.

Verse 3: If Rāhu at full moon seizes the Moon, then eclipse conjunction; hence the full disk.

Verse 4: Knowing the Moon's eclipse, then compute the time; by *manda* and $=s$ *ighra* process, make exact.

Verse 5: When Rāhu seizes full Moon on full-moon day, total eclipse; of Sun, solar eclipse.

Verse 6: The Moon's eclipse declared by both processes; applying the true, then eclipse characteristics.

Verse 7: Thus lunar eclipse declared in order; therefore all characteristics of solar eclipse.

Chapter V: Solar Eclipses

Verse 1: I declare the solar eclipse as it is; therefore all characteristics of solar eclipse.

Verse 2: The node between Sun and Moon — there the solar eclipse; the conjunction auspicious or inauspicious.

Verse 3: If Rāhu seizes Sun there, Moon clear, then solar eclipse: total or partial.

Verse 4: Knowing Sun's eclipse, compute time; by both processes, make exact.

Verse 5: Solar eclipse characteristics by both processes; applying the true, then characteristics.

Verse 6: Thus solar eclipse declared in order; therefore all results of eclipses.

Chapter VI: Rise and Set of Planets

Verse 1: I declare rising and setting of planets in order; joined with knowledge, the planet becomes exact.

Verse 2: Rising first, setting always after; joined with knowledge, the planet exact.

Verse 3: Knowing ascendant (*lagna*) joined with direction, then planet's rising; more exact known by wise.

Verse 4: For planets, rising first, setting after; joined with knowledge, exact.

Verse 5: Beginning from Sun's rising, Moon etc. in order, their rising and setting I declare.

Verse 6: All rising and setting accomplished by measures; without them, knowledge does not proceed.

Verse 7: Thus eight risings of planets; the wise compute; thence more exact from computation.

Chapter VII: Constellations and the Moon's Cusps

Verse 1: The elevation of Moon's horns and asterisms — joined with knowledge, the planet exact.

Verse 2: When Moon's horn elevation on full-moon day, then full elevation of Moon's horns.

Verse 3: Twenty-four asterisms proclaimed by ancients — joined with knowledge, exact.

Verse 4: Thus Moon's horn elevation and asterisms — joined with knowledge, exact.

Chapter VIII: Mathematics (Gaṇitādhyāya) — The Crown of the BSS

Note: This chapter (corresponding to Chapter XII of the full BSS) contains Brahmagupta's most celebrated mathematical discoveries, including the rules for zero, the quadratic formula, the area of a cyclic quadrilateral, and the solution of Pell's equation.

8.1 Arithmetic Operations (Vyakta-gaṇita)

Verse 1: The product of two fortunes is a fortune; of two debts is a fortune; of a fortune and debt is a debt. The first as stated; the second by reversal of signs.

Verse 2: Zero, in multiplication with a quantity, is that quantity increased; zero joined with like signs, subtracted from unlike.

Verse 3: When the multiplier is debt, fortune, and zero, the multiplicand is zero; in multiplication of zero, the result is zero.

Verse 4: Zero multiplied by a quantity is zero; zero divided likewise. Zero joined with like, subtracted from unlike.

Verse 5: When zero is the divisor and the dividend a quantity: zero-divisor, zero quotient; the multiplier as before.

Verse 6: The gain of two fortunes is gain; of two like signs, like. Division: the result of like sign with like.

Verse 7: Debt by fortune gives debt; fortune by debt gives debt; division is by reversal. Signs determined in order.

Verse 8: Deficient and excess are pure; their square always positive. The square root done two ways by method.

Verse 9: The square: an equal quadrilateral field; twice that. From that, root extracted: the square root.

Verse 10: The cube: equal or unequal, with equal quadrilateral base. Its root: the cube root, made threefold.

Verse 11: From first, second cube; from second, third; from third, fourth — that remainder is the cube.

8.2 Fractions (Bhinna-gaṇita)

Verse 12: The fraction is two-fold: expressed and unexpressed; established separately, perform addition.

Verse 13: Divide the division and class of divisions; having reduced fractions, form the fraction.

Verse 14: Denominator multiplied by divisor, dividend divided by denominator; the quotient: mark of fraction-operation.

Verse 15: The denominator: divisor-form; the dividend: quantity; the quotient: result — thus fraction proclaimed.

Verse 16: Numerator by numerator, by denominator, denominator by numerators; having multiplied, divide by fraction.

8.3 Rules for Zero (=S =unya-gaṇita)

The Most Famous Verses in the History of Indian Mathematics

Brahmagupta (628 CE) was the first mathematician to establish systematic rules for arithmetic with zero as a number.

Verse 17: Zero multiplied by a greater quantity is that increased; zero multiplied is zero; zero divided likewise.

Verse 18: Zero joined with like signs, subtracted from unlike. When zero is divisor and dividend a quantity.

Verse 19: Zero-divisor, zero quotient; multiplier as before. This zero-computation declared, all by reason.

Verse 20: Zero — know as void (*kha*), field, vajra, quantities; addition, subtraction, multiplication — zero there.

Verse 21: In addition, zero multiplied by quantity; zero-multiplier (*kha-guṇa*) remembered; zero-divisor known.

Mathematical Commentary: Brahmagupta's rules for zero:

- $a + 0 = a$, $a - 0 = a$
- $a \times 0 = 0$, $0 \times 0 = 0$
- $0a = 0$
- $a/0$ undefined (*kha-hāra*, “zero-divisor”)

He does *not* resolve $a/0$, leaving it undefined — an insight 1400 years before modern analysis.

8.4 Algebra — Avyakta-gaṇita

Verse 22: The unknown (*avyakta*) is the multiplier-place of quantities; joined with the known (*vyakta*), form the result.

Verse 23: “So much as” (*yāvat-tāvat*), or *kālaka* (black), *n*
= *ilaka* (blue), *p*
= *itaka* (yellow), *lohita* (red), *harita* (green),
'sveta (white).

Verse 24: These colors (*varṇa*) are multipliers of unknowns; by them joined with knowns, operate with unknown.

Verse 25: Square of unknown multiplied by known, like by like; square of known multiplied by unknown.

Verse 26: Sum and difference of colors known; joined with knowledge, the result arises in reality.

Verse 27: Unknown joined with *r*
= *upas* (absolute terms), operate there; unknown disjoined from *r*
= *upas* — result remembered.

Verse 28: Unknown joined with known, known with unknown — joined with knowledge, result arises.

Verse 29: Color by color is first; second in multiplication; square of color; third is cube.

Verse 30: Equation (*sam*
=*ikaraṇa*) is division of known and unknown into *r*
=*upas*; one skilled in equation is devoted to math.

8.5 The Quadratic Equation (Madhyamāharaṇa)

The Quadratic Formula in Brahmagupta

Brahmagupta gives the complete solution to $ax^2 + bx = c$, equivalent to the modern quadratic formula, 1100 years before its European rediscovery.

Verse 31: Having made equal the square of unknown and *r*
=*upas* according to rule, when both are of one *r*
=*upa* — equation.

Verse 32: Multiplied by twice (the coefficient), from square of equal (coefficient), subtract square of unequal; root made twofold, joined or diminished by *r*
=*upas*.

Verse 33: That, beginning with two or three, divide by equal multiplied by two; obtained: “so much as” (*yāvat-tāvat*), the *r*
=*upa*.

Verse 34: This *madhyam*
=*aharaṇa* (elimination of the middle) declared by those skilled in reasoning; one skilled in equation devoted to math.

Verse 35: The “so much as” joined with *r*
=*upas*; multiplied by two, make the square there.

Verse 36: Square of *r*
=*upa* multiplied, both made equal; square root known — elimination of the middle.

Verse 37: From that, root of three etc., divide beginning with two or three; obtained: “so much as”, *r*
=*upa*.

Verse 38: Taking unequal from equal, equal from unequal; having made equal — equation.

Verse 39: “So much as” multiplied, with *r*
=*upas*; multiplied by two, make square there.

Verse 40: Square of *r*
=*upa* multiplied, both made equal; square root — elimination of the middle (*madhyam*
=*aharaṇa*).

Mathematical Commentary: In modern notation, Brahmagupta’s formula for $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

Equivalent to: $x = \frac{-b + \sqrt{b^2 + 4ac}}{2a}$

Brahmagupta was the first to give a general solution for *all* quadratics with two distinct positive roots.

8.6 Rule of Three (Trairā'sika) and Proportion

Verse 41: Three quantities of one kind — Trairā'sika (rule of three); their product known; result-quantity in order.

Verse 42: Argument (*pram* = *aṇa*) and result (*phala*); requisition (*icchā*) and result beyond; result of requisition by rule of three.

Verse 43: Requisition multiplied by argument, divided by result; obtained quotient: result by rule of three.

Verse 44: When reversed, inverse rule of three (*vyasta-trair* = *a* 'sika); operate by difference of argument and result.

Verse 45: Five-rule, six-rule, etc. — all from rule of three; inverse and direct, two-fold.

8.7 Arithmetic Progressions (Sreḍh=i)

Verse 46: First term joined with increase, doubled, divide by number of terms; multiply by terms, or half terms.

Verse 47: Mean multiplied by number of terms: sum in equal difference; even: mean doubled; odd: equal difference.

Verse 48: First joined with last, doubled, divide by terms; obtained: mean, from series operation.

Verse 49: Mean joined with number of terms: sum of series. Knowing difference of first and last, make multiplier.

Verse 50: Terms minus one, doubled, divide by terms; obtained: mean from series operation.

Verse 51: Thus sum of series (Sreḍh = *i-phala*) declared by skilled reasoners; therefore geometric progression.

Verse 52: In geometric progression, first term and increase known; born of multiplier; form result.

Verse 53: First term multiplied by common ratio (*guṇottara*) repeatedly; divide by ratio minus one; that quotient.

8.8 Geometry — Plane Figures (Kṣetra-vyavahāra)

Brahmagupta's Formula for the Cyclic Quadrilateral

Area with sides a, b, c, d : $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ where
 $s = \frac{a+b+c+d}{2}$.

This generalizes Heron's formula to quadrilaterals.

Verse 54: Root of triangular field area: product of half-base (*sam-dala*) and perpendicular (*koṭi*); square root of half and sides.

Verse 55: Square root of quadrilateral field area remembered; from square of sum of opposite sides, root produced.

Verse 56: Quadrilateral area from half-base and perpendicular; for irregular quadrilateral, make twofold (two triangles).

Verse 57: Square of half sum of opposite sides — called area; by sum of roots of two, form area.

Verse 58: Circular field square from half-diameter (*vy*
 $=s$
 $=ardha$); from fourth of diameter-square, area produced.

Verse 59: Knowing circular field area, compute diameter; area fourfold, its root: diameter.

Verse 60: Circumference multiplied by five, diameter by seven remembered. Thus circle multipliers form result.

(Note: $\pi \approx 22/7 \approx 3.143$ or $\sqrt{10} \approx 3.162$)

Verse 61: Diameter multiplied by half-chord (*jy*
 $=ardha$), squared, divided by four; remainder's root: chord (*j*
 $=iv$
 $=a$).

Verse 62: Square of chord subtracted from square of diameter; root of remainder: arrow (*'sara, sagitta*), half-diameter's chord-maker.

8.9 Brahmagupta's Formula for the Cyclic Quadrilateral

Verse 63: Quadrilateral area from half-base and perpendicular; from square of sum of opposite sides, root produced.

Verse 64: Square root of equilateral quadrilateral (*sama-caturbhuja*) area; square of half sum of opposite sides: area.

Verse 65: Square root of irregular quadrilateral area; beginning with two or three, form area.

Verse 66: Quadrilateral area from half-base and perpendicular; from half-diameter, form area.

Verse 67: Quadrilateral area from two triangles; by sum of their roots, form area. (*Brahmagupta's formula*: $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$)

Verse 68: Isosceles trapezium (*sama-t*
 $=t$
 $=iya-bhuja$) from half-base and perpendicular; field twofold, form area.

Verse 69: Quadrilateral area from sum of opposite sides; from pair of triangles, form area.

Verse 70: Base-height product, doubled, joined with root: equilateral and isosceles (*dvi-sama*) quadrilateral area.

Verse 71: Irregular quadrilateral, doubled, joined with root; with half unequal root: root of area.

Verse 72: This operation of fields declared by skilled reasoners; therefore operation of excavations in order.

8.10 Excavations (Khāta-vyavahāra)

Verse 73: Knowing excavated field area, compute depth; oblong quadrilateral — form area.

Verse 74: Volume of excavation accomplished, compute capacity; increase known from excavation operation.

Verse 75: Depth and capacity known from excavation operation; therefore operation of shadows in order.

8.11 Shadows and Gnomons (Chāyā-vyavahāra)

Verse 76: Knowing gnomon and shadow motion, compute time; midday time (*madhy* = *ahnak* = *ala*) from shadow operation.

Verse 77: Gnomon (*'saḥu*) of twelve aṅgulas; its shadow proclaimed; joined with knowledge, time exact.

Verse 78: Product of gnomon and shadow made, thence direction computation; beginning triangulation (*trijñ* = *ana*), declare in order.

Algebraic Operations

Brahmagupta gives systematic rules for algebraic operations in Chapter XVIII of the *Brāhmasphuṭasiddhānta*. This chapter contains the first treatment of algebra as a distinct mathematical discipline in Indian mathematics.

Addition and Subtraction of Unknowns

Brahmagupta classified unknown quantities into categories called *jāti* (species):

Verse 18.24. *Translation:* “The numerical coefficients of the same unknown are added to or subtracted from each other; the same is done with the known quantities. When the unknowns are different, they are not united.”

This means that the numerical coefficients of x cannot be added to or subtracted from the numerical coefficients of y or x^2 or x^3 or xy and so on because these terms belong to different *jāti* or they do not belong to the category of the “like.”

Multiplication of Unknowns

Again, regarding multiplication, Brahmagupta says:

Verse 18.25. *Translation:* “The product of the two like unknowns is a square; the product of three or more like unknowns is a power of that designation. The multiplication of unknowns of unlike species is the same as the mutual product of symbols; it is called **bhāvita** (product or factum).”

Rules for Zero, Positive and Negative Numbers

Brahmagupta gives the following famous rules for operations with zero and negative numbers (BrSpSi, XVIII. 30–36):

Verse 18.30: Addition

Verse 18.30. *Translation:* “[The sum] of two positives is positive, of two negatives negative; of a positive and a negative [the sum] is their difference; if they are equal it is zero. The sum of a negative and zero is negative, [that] of a positive and zero [is] positive, [and that] of two zeros [is] zero.”

Verse 18.31: Subtraction

Verse 18.31. *Translation:* “If a smaller [positive] is to be subtracted from a larger positive, [the result] is positive; [if] a smaller negative from a larger negative, [the result] is negative; [if] a larger [negative or positive is to be subtracted] from a smaller [positive or negative, the algebraic sign of] their difference is reversed—i.e. negative [becomes] positive and positive [becomes] negative.”

Verse 18.32: Subtraction with zero

Verse 18.32. *Translation:* “A negative minus zero is negative, a positive [minus zero] positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added.”

Verse 18.33: Multiplication

Verse 18.33. *Translation:* “The product of a negative and a positive is negative, of two negatives positive, and of positives positive; the product of zero and a negative, of zero and a positive, or of two zeros [are all] zero.”

Brahmagupta’s Laws of Signs

$$\begin{aligned} (+a) \times (+b) &= +ab & (-a) \times (-b) &= +ab \\ (-a) \times (+b) &= -ab & (+a) \times (-b) &= -ab \\ 0 \times a &= a \times 0 = 0 \text{ for any quantity } a. \end{aligned}$$

Verse 18.34: Division

Verse 18.34. *Translation:* “A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.”

Verse 18.35: Division by zero

Verse 18.35. *Translation (Colebrooke):* “A negative or a positive divided by zero has that [zero] as its divisor, or zero divided by a negative or a positive [has that negative or positive as its divisor]. The square of a negative or of a positive is positive; [the square] of zero is zero. That of which [the square] is the square is [its] square-root.”

Note: Brahmagupta’s statement that $a/0$ gives “kha-hara” (a fraction with zero denominator) was his way of expressing the concept of infinity. In modern mathematics, division by zero is undefined, but Brahmagupta’s insight that such quantities represent infinitely large magnitudes was remarkable for the 7th century.

Quadratic Equations

The geometrical solution of a quadratic equation in this country would take us to the Vedic *Sulba* period. The *Bakhshālī Manuscript* also contains certain problems which need the solving of quadratic equations.

Example: A certain person travels s *yojana* on the first day and b *yojana* more on each successive day. Another who travels at the uniform rate of S *yojana* per day, has a start of t days. When will the first man overtake the second?

This leads to the quadratic:

$$s + (s + b) + (s + 2b) + \dots + [s + (n - 1)b] = S(n + t)$$

$$ns + \frac{n(n-1)}{2}b = Sn + St$$

$$n^2 \cdot \frac{b}{2} + n \left(s - \frac{b}{2} - S \right) - St = 0$$

Sanskrit Verses: Brahmagupta's Rule for Quadratic Equations

Brahmagupta gives the following rule (BrSpSi, XVIII. 44-45):

Verse 18.44: The quadratic formula (first form)

Verse 18.44. *Translation (Colebrooke):* “Diminish by the middle [number] the square-root of the *rupas* multiplied by four times the square and increased by the square of the middle [number]; divide the remainder by twice the square. [The result is] the middle [number].”

Verse 18.45: The quadratic formula (second form)

Verse 18.45. *Translation (Colebrooke):* “Whatever is the square-root of the *rupas* multiplied by the square [and] increased by the square of half the unknown, diminish that by half the unknown [and] divide [the remainder] by its square. [The result is] the unknown.”

Modern Formulation: For the equation $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is equivalent to the modern quadratic formula for the positive root.

Bhāskara I and the Kuṭṭaka Operation

An indeterminate equation of the first degree of the type:

$$\frac{ax - c}{b} = y$$

(with x and y unknown) is known in Hindu mathematics by the name of “pulveriser” — *kuṭṭākāra*. In this equation, a is called the “dividend” (*bhājya*), b the “divisor” (*bhāgaḥāra*), c the interpolator (*kṣepa*), x the “multiplier” (*guṇakāra*), and y the “quotient” (*labdha*).

In the pulveriser contemplated in the above stanza:

a = revolution number of a planet

b = civil days in a *yuga*

c = residue of the revolutions of the planet (*śeṣa*)

Sanskrit Verse: Brahmagupta’s Rule for the Pulveriser

Verse 18.50. *Translation:* “Divide the dividend by the divisor, and the divisor by the remainder obtained; by this process of mutual division, [continue] until the remainder becomes zero. From this, by working backwards, one can find a particular solution.”

Verse 18.51: Condition for solution

Verse 18.51. *Translation:* “If the interpolator is divisible by the divisor of the residues, then that same method [applies]; if the interpolators are unequal, know that the pulveriser has a flaw [is impossible], friend.”

Verse 18.52: Working backwards

Verse 18.52. *Translation:* “The successive quotients obtained [in the mutual division] are called ‘kotis’ [vertical lines]; placing the *gunakaras* [multipliers] at the kotis, the multiplier is [found] combined with the interpolator.”

The Pulveriser Method

The method of the pulveriser (*kuṭṭaka*) involves the following steps:

1. Divide a by b and obtain the quotient and remainder.
2. Divide b by this remainder and obtain a new quotient and remainder.
3. Continue this process of mutual division until the remainder becomes zero.
4. Write down the quotients obtained in a column.
5. The number of quotients is made odd by adding the residue if necessary.
6. From the bottom, multiply the last number by the interpolator and add the number above it.

7. Continue this process until the top is reached.
8. The number at the top gives the value of x and the number at the bottom gives y .

General Form

The general form of the pulveriser is to find integer solutions to:

$$ax + c = by$$

where a , b , and c are given integers.

Brahmagupta's method was later improved by Āryabhaṭa II and Bhāskara II, but the fundamental approach was established in the *Brāhmasphuṭasiddhānta*.

Example: Pulveriser for planetary residues

Verse 18.53. *Translation:* "The residue of the Sun [is] twelve; divide it together with the civil days; whatever multiplier results therefrom, that will yield the elapsed degrees of the Sun's motion."

Example Problem: Find x such that $\frac{137x-10}{60} = y$ is an integer.

Here dividend $a = 137$, divisor $b = 60$, interpolator $c = 10$.

By mutual division: $137 = 2 \times 60 + 17$; $60 = 3 \times 17 + 9$; $17 = 1 \times 9 + 8$; $9 = 1 \times 8 + 1$.

Working backwards: $x = 50$, $y = 114$ is a solution.

Planetary Aphelions and Epicycle Corrections

Brahmagupta gave systematic corrections to the mean longitudes of planets to obtain their true positions. These corrections involve the concepts of the aphelion (*manda-cheda*) and the epicycles of apsis (*manda-paridhi*) and conjunction (*śīghra-paridhi*).

Verse: Mean to True Position

Verse U.Kh. 5. *Translation:* “Having taken the mean planet, [apply] the correction for the apsis; from that [result, apply] the correction for the conjunction; then the planet becomes true [corrected].”

Verse: Epicycle Dimensions

Verse U.Kh. 6. *Translation:* “The epicycle of the apsis for the Sun is forty-seven plus two [i.e., 49 for the *manda*]; that of the Moon is thirty-three. The epicycle of the conjunction [for the Moon] is seventy.”

Verse: Aphelion (Mandocca) Positions

Verse U.Kh. 7. *Translation:* “The aphelion [*ucca*] of the Sun is seventy-nine [degrees] in the northern solstice; the [higher] apsis of the Moon and the *śīghra* [conjunction point] are obtained by [appropriate] correction.”

Historical Note: Brahmagupta’s corrections to the planetary epicycles were based on careful comparison with observed positions. He adjusted the epicycle radii given in the *Āryabhaṭīya* to reduce the errors in predicted planetary longitudes. His method of applying the *manda* correction first, followed by the *śīghra* correction, became the standard procedure in Indian astronomy.

Uttara Khaṇḍakhādyaka: Advanced Planetary Theory

The *Uttara Khaṇḍakhādyaka* (UKK) represents Brahmagupta's advanced astronomical treatise, containing corrections and refinements to the earlier *Khaṇḍakhādyaka* proper.

Brahmagupta First to Use Second Differences

All these illustrations reproduced here very well establish the point that the great Indian astronomers from *Āryabhaṭa* I to Brahmagupta were aware of the methods of separating the two distinct planetary inequalities, viz., that of the apsis and of conjunction in the cases of the five 'star' planets.

In the *Khaṇḍakhādyaka*, Brahmagupta having given the "sines" and the equations of the Sun and the Moon at the interval of 15° of arc of the mean anomaly, in the *Uttara Khaṇḍakhādyaka* teaches, for the **first time in the history of mathematics**, the improved rules for interpolation by using the **second difference**.

This very important feature is reproduced here from the translation by Sengupta of the verse (UKK. 8):

Verse U.Kh. 8. *Translation (Sengupta):* "Multiply the residual arc left after division by $900'$ (i.e. by 15°), by half the difference of the tabular difference passed over and that to be passed over and divide by $900'$ (i.e. 15°): by the result increase or decrease, as the case may be, half the sum of the same two tabular differences; the result which, whether less or greater than the tabular difference to be passed, is the true tabular difference to be passed over."

The rule given here applies to the case of all functions hitherto considered in the *Khaṇḍakhādyaka*, which are tabulated at the difference of 15° of arc of the argument. They are:

- (i) the tabular differences of the Sun's equation,
- (ii) the tabular differences of the Moon's equation,
- (iii) the tabular differences of the 'sines'.

Illustration — To find the 'sine' of 57° .

Brahmagupta's table of sines in the *Khaṇḍakhādyaka* is as follows:

"Thirty increased severally by nine, six and one; twenty-four, fifteen and five, are the tabular differences of sines at intervals of half-a-sign. For any arc, the 'sine' is the sum of the parts passed over, increased by the proportional part of the tabular difference to be passed over." (KK.I.30; also III.6)

Table 1: Brahmagupta's Table of Sines

Verse on Lunar Correction (Evection)

Verse U.Kh. 10. *Translation:* "The second inequality of the Moon [evection], when combined with the [equation of the] conjunction, then its motion-equation occurs; the conjunction-equation should be applied by that [amount]."

This verse refers to what later became known as Śrīpati's equation — the recognition of the Moon's second inequality (evection), which represents a significant achievement in Indian lunar theory.

Now $57^\circ = 3420 \text{ minutes} = 900' \times 3 + 720'$. Thus three of the tabular differences are considered as passed over; the last one being 31 and the one to be passed over is 24.

The true tabular difference by the rule, for arc 57° :

$$\frac{31 + 24}{2} + \frac{720}{900} \times \frac{31 - 24}{2}$$

Hence the 'sine' of 57° :

$$= 106 + \frac{720}{900} \times 24 + \frac{720}{900} \left(\frac{720}{900} - 1 \right) \times \frac{31 - 24}{2} = 125.76$$

As worked out from the logarithm tables the same comes out to be 125.80.

This in fact is the modern form of the interpolation equation up to the term containing the second difference:

$$f(a + nh) = f(a) + n\Delta f(a) + \frac{n(n-1)}{2} \Delta^2 f(a)$$

Sanskrit Verse: Construction of the Sine Table

Verse Kh. I.29. *Translation:* "Thirty minus one, increased by nine, six, and one; twenty-four, fifteen, and five — these are the tabular differences of sines. The sines [themselves] are obtained by successive addition of these differences, [starting from] the first [sine]. These twenty-four sines [span] the quadrant."

Verse on the Epicycle Correction

Verse U.Kh. 12. *Translation:* "To correct [the mean planet] by the equation of the apsis, one should apply [the correction] by means of the epicycle of the apsis; likewise, to make [the planet] true by the equation of the conjunction, one should do so by means of the epicycle of the conjunction."

Brahmagupta thus takes a decidedly improved step here and is undoubtedly the **first man in the history of mathematics** who has done this.

Operations with Fractions

Brahmagupta classified fractions into several categories: *bhāga* (simple fraction), *prabhāga* (compound fraction), *bhāgānubandha* (fractions in association), and *bhāgāpavāha* (fractions in dissociation). He gave systematic rules for all operations with fractions.

Verse: Addition and Subtraction of Fractions

Verse Pā. 2.3. *Translation:* “When fractions are connected [for addition or subtraction], the two denominators [become] multipliers [of the opposite numerators]; the sum of the [cross-multiplied] dividends, divided by the product of the denominators, becomes the combined [result].”

In modern notation:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Verse: Multiplication and Division of Fractions

Verse Pā. 2.4. *Translation:* “In the multiplication of fractions, the numerators and denominators are multiplied together [respectively]. In division, the numerator is multiplied by the reciprocal of the divisor-fraction [i.e., the numerator and denominator are interchanged].”

Brahmagupta’s Fraction Operations

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d} \quad \text{and} \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

The Rule of Three (Trairāśika)

The Rule of Three (*trairāśika*) is the fundamental method of proportion in Indian mathematics. Brahmagupta gave a precise formulation that served as the basis for all problems involving proportion.

Verse: The Rule of Three

Verse Pā. 5.1. *Translation:* “If the *pramāṇa*-fruit [result for the standard measure] is given, that [is] multiplied by the *icchā* [requisition]; divide that by the *pramāṇa* [standard measure]; the result obtained here is the *icchā-phala* [required result].”

In the *trairāśika*, three quantities are given:

- **Pramāṇa** (p): the standard measure or argument
- **Phala** (f): the fruit or result corresponding to the standard
- **Ichchā** (i): the requisition or desired measure

The required *icchā-phala* (x) is:

$$x = \frac{f \times i}{p}$$

Verse: Inverse Rule of Three

Verse Pā. 5.2. *Translation:* “When, by the inverse Rule of Three, the result becomes reversed, then by the nature of the divisor — the fruit then becomes the *icchā-phala* [by inverse proportion].”

Direct Rule of Three: $x = \frac{f \times i}{p}$	Inverse Rule of Three: $x = \frac{f \times p}{i}$
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Verse: Compound Proportion (Pañcarāśika etc.)

Verse Pā. 5.3. *Translation:* “The result having been obtained by the Rule of Three, [it is further operated on] by [another] Rule of Three; as many given quantities [as there are], so many Rules of Three are remembered [to be applied].”

This verse establishes the method of *compound proportion*: when more than three quantities are involved, successive applications of the Rule of Three yield the final result. For five quantities it is called *pañcarāśika*, for seven *saptarāśika*, and so on.

The Sine Rule in Indian Trigonometry

Brahmagupta was the first Indian mathematician to state the sine rule for plane triangles explicitly. He used it in the context of computing the true positions of planets and in spherical astronomy.

Verse: The Sine Rule

Verse Br.Sp. 14.21. *Translation:* “In an equilateral triangle, the sines of equal sides [divided] by one another [give unity]; in a scalene triangle, the sides divided by the sines give a result [which is constant] — that is the [circumdiameter].”

In modern form, for any triangle ABC with circumdiameter D :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = D$$

Verse: Shadow and Sine Relations (Spherical Astronomy)

Verse Br.Sp. 16.8. *Translation:* “The Rsine of the declination, divided by the Rsine of half the day, becomes the *lambaka* (vertical). The Rsine of the latitude, multiplied by the *lambaka*, [gives] the *drgjyā* (ascensional difference); it is determined [according to] the direction [north or south of the equator].”

This verse relates to spherical astronomy: the computation of the *drgjyā* (ascensional difference), which measures how much a celestial body rises earlier or later due to the observer’s latitude. The *kṛānti-kṣetra* and *akṣa-kṣetra* diagrams use similar right triangles.

Further Algebraic Operations

Brahmagupta's Chapter XVIII also contains rules for squares, square roots, cubes, and cube roots, as well as additional rules for operations with zero.

Verse: Square and Square Root

Verse 18.46. *Translation:* "A square is an equilateral quadrilateral; its root [is] that which, when doubled, [gives the side]. That of which the square is the square is [its] square-root [i.e., the square root of a^2 is a]."

Verse: Cube and Cube Root

Verse 18.47. *Translation:* "A cube is an equi-dimensional equilateral [solid]; its root [is] that which, when tripled in dimension, [gives the side]. That of which the cube is the cube [is its] cube-root [i.e., the cube root of a^3 is a]."

Verse: Zero as a Number

Verse 18.36. *Translation:* "Zero, void, 'kha', 'path' — the learned know this as a number. When [a quantity] is combined with or diminished by it, there is neither increase nor decrease [i.e., $a + 0 = a - 0 = a$]."

Note: This verse (18.36) is historically significant as one of the earliest explicit statements that zero is a number in its own right. Brahmagupta's treatment of zero as a mathematical entity with well-defined properties was a foundational contribution to the development of algebra.

Summary of Brahmagupta's Mathematical Contributions

1. **Second-order interpolation:** Brahmagupta was the first mathematician in history to use the second difference in interpolation, as demonstrated in the *Uttara Khaṇḍakhādyaka* (UKK. 8).
2. **Sine rule:** He was the first in India to give the sine rule for plane triangles, in the context of computing planetary positions.
3. **Planetary corrections:** Systematic corrections to the epicycles of apsis for the Sun, Moon, and planets, bringing calculated positions into agreement with observation.
4. **Lunar inequalities:** Recognition and treatment of the second inequality of the Moon (evection), known as Śrīpati's equation.
5. **Spherical astronomy:** Development of Indian methods using properties of similar right-angled triangles (*krānti-kṣetra* and *akṣa-kṣetra*).
6. **System of numerals:** Contribution to the place-value notation and its transmission to the Arab world, and subsequently to Europe.
7. **Fractions:** Classification of fractional combinations into *bhāga*, *prabhāga*, *bhāgānubandha*, and *bhāgāpavāha*.
8. **Algebra:** Systematic rules for operations with zero, positive and negative numbers; the first to treat zero as a number in its own right.
9. **Quadratic equations:** General formula for solving quadratic equations (verses 18.44–45).
10. **Indeterminate equations:** The *kuṭṭaka* (pulveriser) method for solving linear indeterminate equations, fundamental to Indian astronomy for calculating planetary positions.

Editor's Note

- **Chapter XVIII, verses 30–36:** Rules for operations with zero, positive and negative numbers, including the famous “*kha-hara*” (division by zero) verse; also explicit statement that zero is a number (v. 36)
- **Chapter XVIII, verses 44–47:** The quadratic formula (two equivalent forms), square and square-root, cube and cube-root
- **Chapter XVIII, verses 50–53:** The pulveriser (*kuṭṭaka*) method for indeterminate equations, including conditions for solution and worked examples
- **Pāṭiṅaṇita, verses 2.3–2.4:** Rules for addition, subtraction, multiplication, and division of fractions
- **Pāṭiṅaṇita, verses 5.1–5.3:** The Rule of Three (*trairāśika*), inverse Rule of Three, and compound proportion
- **Brāhmasphuṭasiddhānta, verses 14.21, 16.8:** The sine rule for plane triangles and spherical astronomy relations

- **Uttara Khaṇḍakhādyaka, verses 5-12:** Planetary aphelions, epicycle corrections, sine table construction, second-difference interpolation, and lunar inequality

Chapter VIII: Brahmagupta as an Algebraist

Algebraic Rules

“Of the pulveriser, the divisor and dividend being mutually divided, one should operate the pulveriser with the reduced divisor and dividend.” — Kuṭṭaka (Pulverizer) method for solving $ax \pm c = by$.

“One should divide the dividend by the divisor and the divisor by the dividend alternately; the quotients should be written down separately, one below, one above.” — This describes the Euclidean algorithm steps for the kuṭṭaka.

“Having multiplied the chosen number (mati) by the one above it, add the quotient below; place the result above and remove the lower; proceed again as before.” — The iterative backward-substitution step to find the multiplier and quotient.

“The upper (number), called the ‘multiplier,’ divided by the divisor; the lower, called the ‘quotient,’ divided by the dividend: the remainders are respectively the āhargaṇa and the revolutions (or what is sought).” — Final step: $x = \text{ahargaṇa}$, $y = \text{revolutions}$.

Modern Mathematical Formulation

The kuṭṭaka method solves the linear Diophantine equation:

$$ax \pm c = by$$

where a = civil days in *yuga*, b = revolution-number of the planet, c = residue, and x, y are positive integers to be found.

Step 1 — Reduction: Compute $g = \gcd(a, b)$. Divide through by g :

$$a' = \frac{a}{g}, \quad b' = \frac{b}{g}, \quad c' = \frac{c}{g}$$

Step 2 — Euclidean Algorithm: Apply the mutual division to obtain the continued-fraction quotients $q_1, q_2, \dots, q_n$.

Step 3 — Back-substitution: Starting with a chosen *mati* m such that $r_n \cdot m - c' \equiv 0 \pmod{b_n}$, iteratively compute:

$$V_i = q_i \cdot V_{i+1} + V_{i+2}$$

Step 4 — Solution: The two final numbers give $x \equiv V_{n-1} \pmod{a'}$ and $y \equiv V_n \pmod{b'}$.

Varga-Prakṛti (Square-Nature)

“In the Square-Nature, when two solutions are found with the same multiplier (N) and interpolators k_1, k_2 : their interpolators being multiplied, the multiplier being doubled and combined with the two roots — the sum of their squares is (the new equation with) unity as interpolator.”

Brahmagupta’s Lemma (Composition): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then:

$$x = x_1y_2 \pm x_2y_1, \quad y = y_1y_2 \pm Nx_1x_2$$

solves $Nx^2 + k_1k_2 = y^2$.

“In the Square-Nature, if the interpolator is a square, being capable of being a square — dividing the roots by the square root of that interpolator, new roots are obtained.”

Second Lemma: If $Nx^2 \pm p^2d = y^2$, set $u = x/p$, $v = y/p$, giving:

$$Nu^2 \pm d = v^2$$

The original roots are p times those of the derived equation.

Cakravālā (Cyclic Method)

“The teacher (ācārya), first taking the square root of the multiplier nearest to it, divided by the interpolator (and) increased or diminished by unity — that root divided by the multiplier and the interpolator multiplied by it, combined with the previous pair of roots — that is the new interpolator; the product of the two divided by the multiplier, the sum of the squares of the previous roots — (these are the new roots).”

The Cyclic Method: Starting from an auxiliary $Na^2 + k = b^2$:

1. Choose P so $P^2 \approx N$ and $k \mid (P^2 - N)$
2. New interpolator: $k' = \frac{P^2 - N}{k}$
3. New roots: $a' = \frac{Pa+b}{|k|}$, $b' = \frac{Pb+Na}{|k|}$
4. Iterate until interpolator becomes $\pm 1, \pm 2, \pm 4$

Chapter X: Arabic and Indian Divisions of the Zodiac

Nakṣatras Enumeration

Chapter XI: Brahmagupta's Astronomy

Time Units

“Beginning with Caitra, all the months of a yuga proceed for the creation-epoch (sṛṣṭi-cchāda) at Lañkā simultaneously with the beginning of the day of Brahmā.”

All months begin simultaneously at the creation epoch at Lañkā (on the prime meridian).

“Two vipalas (atoms of time) make a nālikā; two nālikās make a prāṇa; thirty-six prāṇas make a vināḍikā; sixty vināḍikās make a nāḍī (ghaṭikā); sixty ghaṭikās make a day; thirty days make a month.”

Time hierarchy: 2 vipalas = 1 nālikā; 2 nālikās = 1 prāṇa; 36 prāṇas = 1 vināḍikā; 60 vināḍikās = 1 ghaṭikā; 60 ghaṭikās = 1 civil day; 30 days = 1 month.

Kalpa Constants

Epoch Calculation (Sṛṣṭi-saṃvatsara)

“Six manus have elapsed in half a kalpa; and of the seventh manu, twenty-seven caturyugis have gone by; of the twenty-eighth caturyugi, the three yugas kṛta, dvāpara and tretā have elapsed, and also 3179 years of the present kaliyuga.”

The total elapsed period at Brahmagupta's epoch (year 628 CE):

$$\begin{aligned}
 T &= 6 \times 71 \times 4,320,000 \\
 &\quad + 7 \times 4 \times 432,000 \\
 &\quad + 27 \times 4,320,000 \\
 &\quad + (1,728,000 + 1,296,000 + 864,000) + 3,179 \\
 &= 1,972,947,179 \text{ years.}
 \end{aligned}$$

Mean Longitude Calculation

General method:

1. Compute the *ahargaṇa* A = civil days elapsed since epoch
2. Multiply by planet's revolution-number R in a kalpa
3. Divide by civil days in kalpa D :

$$\text{Mean longitude} = \frac{A \times R}{D} \text{ (in revolutions, signs, degrees, etc.)}$$

Chapter XII: Brahmagupta as a Critic

“Know the lambana (difference of parallaxes in longitude) by means of five Rsines: the madhya-jyā, the udaya-jyā, the dṛk-kṣepa-jyā, the dṛg-jyā, and the dṛg-gati-jyā; and the avanati (difference in latitude) likewise.”

The *lambana* is computed from five Rsines:

- *madhya-jyā* = $R \sin z_m$ (zenith distance of meridian ecliptic point)
- *udaya-jyā* = $\frac{R \sin L \cdot R \sin \varepsilon}{R \cos \phi}$
- *dṛk-kṣepa-jyā* = $R \sin z_c$ (zenith distance of central ecliptic point)
- *dṛg-jyā* — related to observer's zenith distance
- *dṛg-gati-jyā* — related to the motion of the observer

where L = longitude of horizon ecliptic point, ε = obliquity of ecliptic, ϕ = latitude of place.

Chapter XIII: Astronomical Instruments

The Gnomon (Śaṅku)

“(The gnomon) two añgulas wide at the base, pointed as a needle (*śūla-agra*), twelve añgulas in length, and full of holes from the basic circular part to the pointed extremity — thus is the *śaṅku* (gnomon) described, O beloved.” — BrSpSi. XXII. 39.

The gnomon dimensions: base width = 2 añgulas (≈ 3.75 cm), height = 12 añgulas (≈ 22.5 cm), tapering to a point.

“When there is no wind, placing a jar full of water upon a tripod on the ground which has been made plane by means of eye or thread, one should bore a fine hole (at the bottom) so that the water may fall drop by drop.”

Procedure for testing level ground: Place a water-filled jar on a tripod on leveled ground. A fine hole allows water to drip steadily, confirming the surface is horizontal.

Madhyamādhikāra

Brāhmasphuṭasiddhānta: First Chapter — Madhyamādhikāra

*jayati praṇata-sura-asura-auragīndra-ratna-prabhā-cchurita-pāda-
pañkajaḥ |
kartā jagat-patir-aditya-tulya-yānām mahādevaḥ || 1 |*

“Victorious is Mahādeva, the Lord of the Universe, the Creator, the lotus-feet of whom are rubbed with the lustre of jewels from the heads of gods, demons, and serpent-kings who bow before Him — He whose chariot is equal to that of the Sun.”

*brahmagupto graha-gaṇitaṃ mahatā kālena yat khalvīmatam |
kṣīṇe 'dhunī tad-viparīta-vad-brāhmaṇair upetaṃ || 2 |*

“Brahmagupta (says): The science of the computation of planets, which was well-established in the past, has now become corrupted; being contrary (to the truth), it has been mixed up by the brāhmaṇas.”

*dhruva-bala-arka-suta-budhaj na jyotir-vidāṃ prīti-śad-gama-do |
poṣaṇārthaṃ vivṛtāntara-vyaktaṃ saha prabandhena sūcitam || 3 |*

“(I write this) for the pleasure of astronomers, who are learned in the (motions of) Dhruva (pole star), Bala (Mars), Arka (Sun), Suta (Saturn), Budha (Mercury), and J na (Jupiter); for the preservation (of knowledge), the obscure meaning is made clear along with the connected exposition.”

caitry-ādi-tad-eva-dvija-nodvan-māsa-sarveṣa yugasya |
sṛṣṭi-cchādo lañkāyāṃ samam pravṛtā dine-kalasya || 4 |

“Beginning with Caitra, all the months of a yuga — (which are) the creation-covering — proceed simultaneously at Lañkā with the beginning of the day of Brahmā (dina-kala).”

prāg-ry-udda-nāṭikā-dvau dvi-dvi-yaṣṭikā tri-nāṭikā ṣaḍ-ghnā |
ghaṭikā ṣaḍ-ghnā diva-lo divasānām triṃśatā māsaḥ || 5 |

“Two atoms (paramāṇu) make an aṇu; two aṇus make a lava; two lavas make a nimeṣa; three nimeṣas make a kṣaṇa; five kṣaṇas make a kṣathi; four kṣathis make a muhūrta; eight muhūrtas make a prahara; four praharas make a day (aho-rātra); thirty days make a month.”

Time scale: 2 paramāṇu = 1 aṇu → 2 aṇu = 1 lava → 2 lava = 1 nimeṣa → 3 nimeṣa = 1 kṣaṇa → 5 kṣaṇa = 1 kṣathi → 4 kṣathi = 1 muhūrta → 8 muhūrta = 1 prahara → 4 prahara = 1 day → 30 days = 1 month.

Verses 22-24: Solar and Lunar Month Relations

parivartah sapta-catuṣṣaṣṭir sūryasya dinasya mitih |
ravi bhagañānām mānaḥ sāvana-dinasya kuṭṭakair eṣa |

“The revolution-number of the Sun is $64 + 7 = 71$ (per caturyuga); this is the measure of solar days. By the pulverizer (kuṭṭaka), this (gives) the civil days.”

Rule: A solar year ≈ 365 days 15 min 30 sec. The revolution-number of the Sun = 4,320,000 per kalpa.

ravi bhagañārdha-badhā dvādaśāṃśena bhavati ravi-māsaḥ |
bhagañantaram ravi-dinaḥ śaśi-māsaḥ sūrya-māsena |

“The product of half the revolutions of the Sun divided by twelve becomes a solar month. The difference of revolutions (between Sun and Moon) is the lunar month exceeding the solar month.”

Formula:

$$\text{Solar month} = \frac{R_{\odot}/2}{12} \quad ; \quad \text{Lunar month} = R_{\text{Moon}} - R_{\odot}$$

adhi-māsāḥ śaśi-mśāv-indv-adinasya bhavati śaśi-divasāḥ |
śaśi-sāvana-divasa-antara-nivamāni tithiḥ śaśāñka-dinaṃ |

“The intercalary months (adhi-māsa) are the lunar months exceeding the solar months. The lunar days are (obtained from) the difference between lunar and civil days; the tithi (lunar day) is the measure of the difference between lunar and civil days.”

Key relations:

$$\begin{aligned} \text{adhi-māsa} &= \text{lunar months} - \text{solar months} \\ \text{śaśi-divasa} &= \text{lunar days} - \text{civil days} \\ \text{tithi} &= 1/30 \text{ of a lunar month} \end{aligned}$$

Verses 26-27: Epoch and Ahargaṇa

kalpa-parārdha manavaḥ ṣaṭ-kāśya gatāś catur-dvi-yugāni yugānām |
śruta-purāṇādi-nikṣepānām kṣayaḥ sūrya-dinaṃ cakṣuḥ |

“In half a kalpa, six manus have elapsed, and twenty-seven caturyugis of the (seventh) manu. (This gives) the destruction (elapsed period) of the sūrya-dina (solar days) — the eye (of calculation).”

The *ahargaṇa* (day-count) from the kalpa epoch:

$$A = D_{\text{manus}} + D_{\text{caturyugis}} + D_{\text{current yuga}}$$

nava-gaṇa-rāśi-munin nava samanvantara-devaḥ śakunta-pāte |
śarva-mateḥ sambatsarāt samkhyāṃ śaravac-chataṃ rūpaiḥ |

“O Muni of nine-digit signs! (This gives) the number of years from the śarva-mata (doctrine year), 106 (śara) increased by unity — 3179 years of kaliyuga (at the epoch śaka 550 = 628 CE).”

Brahmagupta’s epoch: year 3179 of kaliyuga = śaka 550 = 628 CE.

Appendix: Rational Geometrical Figures

iṣṭa-yaṣṭeḥ sama-cchedaṃ rūpeṇ a hīnam ubhayor mithaḥ |
ṣaḍ-bhāgena sama-j nā-try-akṣetrasya bhuje |

“The square of the optional side divided and then diminished by an optional number; half the result is the upright (koti), and that increased by the optional number gives the hypotenuse of a rectangle.”

Formula: For arbitrary rational m, n :

$$\text{Base} = m, \quad \text{Upright} = \frac{1}{2} \left(\frac{m^2}{n} - n \right), \quad \text{Hypotenuse} = \frac{1}{2} \left(\frac{m^2}{n} + n \right)$$

yasmin dve samānī bāhū anyad dvayam-asamānī ca |
tayor varga-yogo vibhājyo bhedenā sama-dvi-bāhu-kṣetre |

“Where there are two equal sides and two other unequal sides — the sum of their squares divided by their difference (gives the base of an isosceles triangle).”

For two unequal integers p, q : equal sides = $\frac{1}{2} \left(\frac{p^2+q^2}{p-q} + (p-q) \right)$, base = $\frac{p^2+q^2}{p-q}$.

sarva-dor-alpa-catustāya-kṣetra-phala-mūlaṃ dvi-bhuja-kṣetre |
dvayor dvayor ddhayor ghātaṃ mūlaṃ dvi-bhuja-kṣetrasya |

“The square root of the product of four sets of half the sum of the sides, each diminished by one side, is the exact area of a triangle (and) quadrilateral.”

Brahmagupta’s Formula:

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2}$$

This generalizes Heron’s formula for triangles (when $d = 0$).

Chapter III: Triprasnādhikāra (Conclusion)

— The chapter on the three problems relating to diurnal motion

Verses 25-27: Ghaṭikā computations and eclipse timing

Translation: (25) When the Sun, diminished by the day-radius (*dinadala*), is east or west, the ghaṭikās till the end of the tithi are obtained by subtracting that from thirty and the ghaṭikā-remainder is obtained. (26) The reverse applies at midnight for a lunar eclipse; slowly for a solar eclipse. From the rising of the Sun, by the same ghaṭikās twice. (27) The semi-diameter (*vyāsa*) of the Earth's shadow at the Moon's orbit is obtained: the day-diameter, circumference, accurate tithi-nata-kendra-jyā and the degrees of Jupiter or Kandu (the Moon), Indu — by 191 and the new nine-veda 466, divided by the square of the radius.

Verses 38-39: Sphuṭikaraṇa of Mars

Translation: (38) Multiplied by seven degrees, the day-multiplier, the daily motion in signs, divided by the obtained degrees, the sphuṭa of Mars is increased or decreased from the mark (starting point) at Mrga (start of Capricorn) and Karka (start of Cancer). (39) Its sphuṭa-circumference is 70 yojanas; the śīghra-sphuṭa circumference diminished by obtained degrees; 24330 diminished by veda-jina-akhi; this is the correction (*sphuṭikaraṇa*) of Mars.

Verses 40-42: Celestial latitude and shadow calculations

Translation: (4) The shadow-circle-radius multiplied by the hypotenuse, divided by the semi-diameter obtained by the Sun, the viśuvacchāyā, the yāmyā, the sum or difference of that at the tip of the bhuja. (5) The eastern or western shadow of the gnomon — taking the square root of the difference of the bhuja-squares, that eastern or western shadow is the koṭi between the tips of the bhuja. (6) Having made the shadow-tip in the middle of the direction, the gnomon according to direction is to be measured; the observer's shadow-tip situated in the middle of the direction moves in reverse.

Verses 64-66: End of Triprasnādhikāra

Translation: (64) At the end of Kṛṣṇa-caturdaśī, in the bird (śakuni) season, in the four-footed (catuspada) measure, the half-tithi, nāga, kistughna, from pratipadā etc. (66) The manifest two kalās divided, the moving karas and the hara-sa-gu-ra-ṇa-ṇaḥ 360, with one obtained sum, the remainder divided by hr̥ta is like a tithi and other.

[At this point the Triprasnādhikāra concludes. Several verses in the manuscripts are mutilated; Dvivedī notes that at some places only the commentary (ṭīkā) is available. The chapter contained 66 verses.]

Chapter IV: Candragrahaṇādhikāra

Chapter on the Computation of Lunar Eclipses

This chapter contains rules for computing lunar eclipses, including the dimensions of the Earth's shadow at the Moon's orbit, the times of first and last contacts, the phases of the eclipse, and the colours assumed by the eclipsed Moon.

Verses 1-4: Shadow dimensions and eclipse elements

Translation: (1) The Sun's jyā multiplied by the day-part-jyā, divided by the guṇākṣa-jyā or alternatively, the viṣuvacchāyā divided by twelve. (3) Twelve times the viṣuvacchāyā gives the shadow; or separately by the akṣa-lamba-jyā. Multiplied by the declination (krānti), the sama-maṇḍala-karṇa; as before, the separate shadow.

Verses 7-8: Eclipse magnitudes and durations

Translation: (7) The two akṣa-lamba-chāyā-jyā, their sum or difference gives the root; the viṣuvat-karṇa-chāyā, otherwise the hypotenuse is the śaṅku (gnomon). (8) The uccatana-jīvā-koṭi, the chāyā, the ha-jyā, the bhuja or the nata-jyā; the hypotenuse-chāyā-circle from the semi-diameter; similarly elsewhere.

Verses 14-18: Sphuṭa-tithi and lambana

Translation: (14) If the half-duration is greater, by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda). (16) From the sphuṭa-tithi-antā the parallax (lambana), with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without, from the self-vimarda-udaya, thus the corrected half-obscuration. (18) The Moon's latitude (sphuṭa-vikṣepa) multiplied by the half-duration, the corrected (sphuṭaḥ) bhūja multiplied by the half-duration is the root of the remainder as before.

Verses 16-20: Eclipse colours and phases

Translation: (16) At the end of obscuration, smoke-coloured, black; in partial eclipse, half-increased eating (grāsa); black-brown (sa-kṛṣṇa), coppery (tāmra), white-yellowish (śakli-kapīla-varṇa). (20) The measure of pure, half-duration-motion-desired, eating in the samakala-etc.; the breadth of the nadagūhara-āyāma desired one-fourth.

Thus ends the Fourth Chapter of the Brāhmasphuṭasiddhānta.

Chapter V: Sūryagrahaṇādhikāya

Chapter on the Computation of Solar Eclipses

This chapter treats the computation of solar eclipses, including the parallax corrections (*lambana*) that distinguish solar eclipse computation from lunar eclipse computation.

Verses 2-4: Lambana and eclipse elements

Translation: (2-3) Same as lunar eclipse — the shadow dimensions are computed from the Sun's declination. (4) The bhuja from the Sun's tip (arkāgra), the square deducted, 346450 and 144 multiplied by the square; the first and other, twelve times the viṣuvacchāyā, the product of the two.

Verses 5-8: Lambana-ghaṭikā and eclipse timing

Translation: (5) From the lambana-ghaṭikā-lagna, from the tātkālīka, diminished by three signs; if negative with addition, five times the asaṁṛt, divided by hṛdaya. (6) From the karaṇa-guṇa, from the semi-diameter, dviveda 48, divided; the lambana-nāḍis or the bāskara-vitribha-lagna-antaram. (7) If the lambana is negative, the positive result in its own lalipta-phala. (8) From the aṣajyā, from the vitribha-lagna, its own declination (krānti) uttarākṣa or of the Moon; if greater, otherwise the yāmyā.

Verses 14-17: Sphuṭa-tithi and corrected obscuration

Translation: (14) If the half-duration is greater, then by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda-ardha). (16) From the sphuṭa-tithi-anta, the lambana, with or without the asaṁṛt-sthityardha; that sphuṭa is by separation, at the end of the sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without; from the self-vimarda-udaya, thus the corrected (spaṣṭa) half-obscuration.

Verses 18-20: Deflection and sphuṭa computation

Translation: (18) Thus in the conjunction of planet and Moon, if greater than the measure of unity plus half, in the middle-intermission; diminished by the half-duration-obscuration, the cross (tārā). (19) The lambana like akāla-graha, asaṁṛt-svāvanati-lāptikā-sphuṭi, the tātkālīka-vikṣepī, the samānya-diśoḥ sum or difference. (20) The deflection (vikṣepa), the middle-difference, the above-shadow-planet situated below; if greater than half the unity-measure, not too distinct, hence the sphuṭa-statement.

Thus ends the Fifth Chapter of the Brāhmasphuṭasiddhānta.

Chapters VI-IX: Udayāsta, Candraśṛṅgonnati, Candrac-cāyā, Grahayuti

These chapters cover miscellaneous astronomical computations: rising and setting of planets, the elevation of the Moon's horns, the Moon's shadow, and conjunctions of planets.

Chapter VI: Udayastādhikāra (Rising and Setting of Planets)

Translation: (6) By twelve, the śītāṃśu-sita-jīvā, the earth-born, by nine; the day-increment-difference-ghaṭikā multiplied by six in kalāṃśa. (7) The graheka like hr̥syā-hr̥syā-yuti, by the difference of suttacāntara-obtained-days; subtracted or added in liptāmyaḥ, as before by the tātkālika-asakṛt. (8) Daily rising and setting thus, by the tātkālika-graha-vilagnais; the solar rising and setting at the end of the Pausa month, the Moon (śītāṃśu).

Chapter VII: Candraśṛṅgonnatyādhikāya (Elevation of the Moon's Horns)

Translation: (38) The latitude (unnati) of the Moon is found to disagree from the hypotenuse-śravaṇa-jñānāt; the rising and setting are caused by day, by prati-ghaṭika measure and from vā-jñānāt. (36) The akaicara-ghaṭikā, vyasta-jyā, multiplied by candra-māna, divided by vyāsa is pure; since the dṛk-sama is not true. (40) Previously taught as different, afterwards the other reverse; that time which is shadow, that asat-sphuṭa, suktimān before. (41) Who knows the uḍita-uḍita-astamita-avaśeṣa-kāla; how can he who knows the shadow-unnati of the Moon be an ārya-maṭa-jña.

Chapter VIII: Candraccāyādhikāya (The Moon's Shadow)

Translation: (14) The declination-jyā, the deflection (vikṣepa), the declination-cāpa: the sum or difference of the jyā of like or unlike directions of the declination. (16) Thus the bhūmi-niśa-vayo (latitude etc.) of the planet, by its declination, the candra-aṣṭaka-karma etc.; having done this in the eclipse, as in bhūmi, from that the svakrantijīvā.

Chapter IX: Grahayutyādhikāya (Conjunction of Planets)

Translation: (44) By the prati-ghaṭika and adhika-śaṅku, in the middle is shown the earth; the tripraśna-oktacā like the Sun's śaṅku, the remainder of the bhramaṇādi-krama. (46) The śaṅku, the eastern-western antaram, the viṣuvacchāyā-sum or difference, nṛtale; the yāmyottaraṃ guṇāhataṃ by svakrānti-leśaiḥ and karṇa. (47) The cāpāṃśāḥ of sada-yeḥ, saṅghuvakāpakramāṃśa-rūnāḥ; the vikṣepāṃśe vyastā, the vyasta-viśuddhā vya-sahaśāṃśaḥ.

[The chapters on Udayāsta, Candraśṛṅgonnati, Candraccāyā, and Grahayuti conclude here, with 12-70 verses each.]

Chapter XI: Tantraparīkṣādhikāya (Dūṣaṇādhikāya)

Chapter on the Critical Examination of Astronomical Systems

This celebrated chapter contains Brahmagupta's critique of various astronomical systems current in his time. He refutes the doctrines of Āryabhaṭa, Śrīṣeṇa, Viṣṇucandra, Lāṭadeva, Vijayanandi, and Pradyumna, as well as the foreign systems derived from Greece and Babylon (the Romaka and Puliśa Siddhāntas).

Verses 1-2: Introduction to the critique

Translation: (1) Those who know the brahma-vid (science) in a chapter-barred dual manner, they consider it a siddhānta; I shall declare their faults arising from the yuga-ādi differences. (2) They say the yuga is five years; making the Saṃhitā of the Sun and Moon; the adhimāsa, vamara-atra, sphuṭa-tithi-jñāna — that is false.

Verses 6-9: Critique of planetary positions

Translation: (6) The yuga-year-navadvyekṣa (nine), the bright (Moon) etc., the same pravṛttānyat; that is false, since the corrected yuga-position of the mandanodes is fixed. (7) The planet's release is in the ruṇāyā, the mandocca increases in śīghra; in the high position, the mandocca does not exist without (its own) motion. (8) In Ārya's eight hundred, the nodes move, in the Daśagītaka are fixed; the released two-nodes move in the circle, not fixed. (9) Āryabhaṭa knows the planets; what he has said is false; the Rāhu does not cause the eclipse, its pāta is not the eighth Rāhu.

Verse 63: Conclusion

Translation: (63) Thus the systems taught, even by reading, cause ignorance by faults; the examination of systems — this is the eleventh chapter.

Thus ends the Eleventh Chapter of the Brāhmasphuṭasiddhānta.

Chapter XII: Gaṇitādhyāya

The Twelfth Chapter — On Mathematics

The Gaṇitādhyāya is the most celebrated mathematical chapter of the *Brāhmasphuṭasiddhānta*. It contains 66 verses divided into ten sections (vyavahāras), covering:

1. — Preliminary operations (fractions, proportions)
2. — Mixture problems (verses 1-16)
3. — Arithmetic and geometric series (verses 17-20)
4. — Geometry of plane figures (verses 21-39)
5. — Circle geometry (verses 40-43)
6. — Excavation/volume problems (verses 44-46)
7. — Pile problems (verses 47+)
8. — Shadow of a gnomon etc.
9. — Rules for treating quantities (verses 50-51)
10. — Shadow problems (verses 52-66)

Section 1: Parikarmma (Preliminary Operations)

Translation: (1) He who knows separately the parikamma (operation), the saṃkalita (summation) etc., and the twenty vyavahāras with chāyāta — he becomes a calculator (gaṇaka).

Translation: (2) The inverted (viparīta) and the divisor-multipliers of the quantities, the divisor-parts (chedāṃśakāḥ) with common divisor; the saṃkalita etc. are to be added, the difference is vyavakalita. (3) The rūpāraṇī (unity) multiplied by the divisor, with parts added, of two or many; the pratyutpanna (product) becomes the division by the divisor-product of the fraction-product. (4) The pratyutpanna after conversion, the divisor-part of the divisor, the divisor-times-divisor; the two-part-sum of the dividend, the divisor of all is the divisor.

Translation: (5) The square of all-parts divided by the square of the divisor becomes the varga (square root); the root of all-śita-parts, the root divided by the divisor-term. (6) Having set up the varga-mūla (square root), the utpadyanyakṛti, the trigunottara-saguṇa, and the prathamā; the nottara-kṛti-antitya-guṇā, triguṇā and utara is the ghana (cube). [Cube completed.] (7) From the second chedāghana, the cube-root-square-times-trisaguṇa-sakṛti; subtracted, times the tripūrva, from the prathama-ghana, the ghana-mūla. [Cube root completed.]

Section 2: Miśrakavyavahāra (Mixture Problems)

Translation: (16) Divided by the prakṣepa-yoga (sum of contributions), obtained, the prakṣepaka (contributor) multiplied by the gains; the ūnādhika-uttara-ardha is true, the dvitonayā with its own phala-sum added.

End of Mixture Problems.

Section 3: Śreḍhīvyavahāra (Series)

Translation: (17) The pada (term) minus one, multiplied by the utara (common difference), added to the sum of the first and last terms; the first plus the last term from the first, the cube of the middle term is pada-gaṇa (number of terms) multiplied. (18) From the utara-hīna (decreased by common difference) dviguṇādi-śeda-varga, the ghanottara-aṣṭra-vadhe; added, the pada by the śeṣeṇam, the dviguṇottara hṛtaṁ gacchaḥ (progression). (19) If the eka-uttara (common difference one) eka (first term one) and the desired gaccha (number of terms) is the saṃkalita (sum); that viyut (without) gaccha multiplied, divided by three, is the saṃkalita.

Section 4: Kṣetravyavahāra (Geometry of Plane Figures)

The *kṣetra-vyavahāra* contains rules for computing the areas and diagonals of triangles, quadrilaterals, and other plane figures. This is one of the most mathematically significant sections of the work.

Verses 21-23: General area rules

Translation: (21) In triangles and quadrilaterals with equal and two sides etc., the vyāsa (diameter) is like āyata-catura (rectangle); the vyāsa-karṇa of the field is the sama-koṇi (right-angled diagonal) of the pair of fields. (22) Half the sum of the lambaka (altitude) and karṇa in sama-tribhuje (isosceles triangle) is the area; in sama-catura (square) it is the square; in sama-koṇi (right-angled) half the sum of the vyasta (crossed diagonals). (23) In asama-catura (unequal quadrilateral), the product of the rāśis (sides), their squares and the sum; their difference-varga (square), with equal-square-half added mutually.

Verses 32-35: Specific geometric formulas

Translation: (32) The karṇa and lamba (hypotenuse and altitude) — the lower karṇa-lamba; in proportion, the uddha-śūnyāṃsa-pāṭyāma. (33) The product of kṛti-yura-sahaśa-rādyo-bāhu-dhāto; the lambaka (altitude) twice; the product of kṛty-antara of unequal (sides) and dvitribhuja. (34) Divided by two desired quantities, from the sum of the dvi-ghaṇa-varga-phala; the side of a viṣama-tribhuja (scalene triangle) from the viṣama-phala is added to yā mūḥ. (35) The square of the desired bhuja divided by the undesired, that portion is the koṭi; of the āyata-caturastra (rectangle), the field has the desired hypotenuse.

Section 10: Chāyāvyavahāra (Shadow Problems)

The *chāyā-vyavahāra* deals with computations involving the shadow of a gnomon (*śaṅku*), problems of distance, height determination using shadows, and related topics.

Verses 1-4: Lamp and shadow calculations

Translation: (2) The *chāyāntara* (difference of shadows) with one added, divided by two, the *prāga-parayor-yuga-doṣam*; the *dinagata* (elapsed day), the *śeṣāṃśa-hṛtaṃ dvidalaṃ*, the *chāyāntara-vyekam*. (13) The distance between the lamp-surface and *śaṅku*-surface, the desired measure-*śaṅku-guṇa*; the lamp-wick-*ucchyā*, the *śaṅku* deducted, the *śeṣa-uddhṛtaṃ chāyā*. (4) The *chāyā* multiplied by the *chāyāgra-antara*, divided by *chāyāntara* is the *bhū* (earth/distance); the *bhū* is *śaṅku-guṇa chāyā*, divided, the lamp-wick-height.

Verses 44-45: Gnomon problems and *kuṭṭaka*

Translation: [End of shadow problems.] (45) The *guṇākāra* (multiplier) and *khāṇḍa-tulya* (equal to the section) are the *guṇyau* (multiplicands); the *gomūtrikā* (cow's urine) *kṛtī* multiplied; the *sahitāḥ pratyutpannaḥ* (combined product), the *guṇa* (quality) or equal to the *kāraṇa-meda*.

[The Gaṇitādhyāya concludes here with 66 verses total, containing the foundation of Indian algebra including the kuṭṭaka (pulverizer) method for solving linear Diophantine equations.]

Chapter XIII: Praśnādhyāya

The Sixth Problem-Chapter: On Mean and True Motion

This chapter presents mathematical problems (*praśna*) whose solutions demonstrate the computational methods developed in the preceding chapters. The title “” (sixth) refers to the chapter’s position within the *uttara* (secondary) section of the text.

Verses 1-2: Introduction

Translation: (1) I shall declare the *praśnādhyāyas* (problem-chapters) with solutions (*uttara*), increasing the intelligence of calculators; by which the teacher (*ācārya*) of the *tānta*-vids becomes one of intelligent mind. (2) The *adhimāsakāḥ* (intercalary months) with *vikalā* (seconds), the *avamarātra* (omitted days) with the *yugayāta*, by the *dyugaṇa* (day-count); he who knows the motion of planets in the wind’s path knows the *kāla-tantra* (science of time).

Verses 3-5: Time and planetary motion problems

Translation: (3) He who comprehends the *avamāti* with *vikalā*, the *adhimāsa-radhika-māsaka-anavamaiḥ*, the *grahamiṣṭhaṃ* in that, knows the *kāla-tantra*. (4) The *dyugaṇa* without *adhimāsa*, *avinādina garona candra-ākkah*; in that without *sphuṭa-tithi*, he who knows is the *kāla-tantra-jña*. (5) From the mean (*madhya*) to other *tithi* from *iṣṭa-madhyamā-cchadi-prahaṇas*, from the *iṣṭārka*vid-upāte *yom*, without knowing, he is not a *tantra-jña*.

[The Praśnādhyāya continues with detailed problem-solution pairs covering mean and true longitudes of planets, conjunctions, eclipses, and other astronomical computations. The full chapter contains approximately 48 verses.]

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Chapter XIV: Sphuṭagatyuttarādhikārah

Chapter XIV: On True Planetary Motions

This chapter deals with *sphuṭa-gati*—the computation of a planet’s true longitude from its mean longitude by applying the *manda* (eccentric) and *śīghra* (epicyclic) corrections.

V.1. The divisor of the *yuga* makes the cosine (*koṭijyā*) into the cosine, and the sine (*bāhuḥjyā*) into the sine. The hypotenuse (*kṣetra*) together with these—that is the *sphuṭagatikah*, the true-motion triangle.

V.2. The radius of the outer circle (*paramaṇḍala-karṇa*), made to produce the cosine-shadow (*koṭi-chāyā*) for computing the true — the hypotenuse-end, cosine-corner, and sine: that is the true-motion triangle.

V.3. The following cosine-chord (*anujakoṭijīvā*) at the radius, which the outer-circle-knower makes to produce — that which makes the *svocca* (own apogee) from its own equation (*sva-phala*) into the true — that is the true-motion triangle.

V.4. Of the field of the true-hypotenuse (*sphuṭa-pada*), sine and cosine are the results without the chord (*jyā*); without the chord, or with the equation (*phala*) added — that makes the true-motion computation.

V.5. He who makes the mean planets true by desired-day indicators (*iṣṭa-diva-yadaiḥ*) for the directions — that *sphuṭagatikah* which makes (the computation) with its own apogee-true (corrections).

V.6. Threefold: Saturn (*śani*), Moon (*indu*), Nait̥yama (Mercury), Gārā (Mars), Lakṣmī (Venus), the planet-group — gathered: Mars (*bhauma*), Aṅgiras (Jupiter), Guru — Durvarca, Vāt̥ya, Gārā — with Kai (Sun?). [Planetary name correspondences in numerical notation.]

[The chapter continues with more verses on true motion computations, including corrections for the manda and śīghra epicycles, and rules for finding planetary positions. The chapter concludes at printed page 181 with verse 46.]

V.46. This rule of mean-upper-line (*madhyotarekhā*) is the fifty-fifth chapter-method. It is to be known by the teachers of the knowers of the *tantra*; one becomes skilled in mean motion.

Here ends the Fourteenth Chapter, on Mean Motion, in the Brāhmasphuṭasiddhānta.

Chapter XV: Tripraśnottarādhikārah

Chapter XV: The Three Questions

This chapter treats the “three questions” of astronomy: relating to time (*kāla*), place

(*sthāna*), and direction (*dik*). It covers the determination of latitude from gnomon observations, shadow measurements, and the computation of time.

V.44. The sine of rising (*udaya-jyā*) and the sine of declination (*krānti-jyā*), multiplied by the diameter and divided: by twelve and by fire (3), by Śiti (cold, the zero-point) and by the equinoctial shadow — divided by that is the declination (*krāntiḥ*).

V.45. From the desired diameter (*iṣṭa-vyāsa*), from the shadow-ray (*chāyā-raśmi*), from Bhāskara's (the sun's) crossing-sign, from the double rising: the string (*sūtra*) — that sine is the measure of the particular declination (*krānti-viśeṣa-padam*).

V.46. In the six parts (of the day), the gnomon (*śaṅku*) is not known (in this way); if the gnomon is placed toward the east-west, the northern (deviation) is due to the southern excess, and that difference indicates greater latitude.

V.47. In the northern hemisphere, it is not thus; the leftward (deviation) is in the globe-string. The gnomon-base divided by the gnomon — the equinoctial shadow, twelvefold, is the multiplier.

[The chapter contains 60 verses on the three problems, concluding at printed page 193.]

Chapter XVI: Grahāṇottarādhikārah

Chapter XVI: On Eclipses (Advanced)

This chapter treats refined computations for eclipses (*grahaṇa*): the computation of lunar and solar eclipses, the graphical *parilekha* method, and the determination of contact (*sparsā*) and release (*mokṣa*) times.

V.15. The *parilekha* (graphical representation) of the planet, by the planet-rule (*graha-kalpa*) first, either curved or straight — the instantaneous position, the closing and opening (of the eclipse) — thus.

V.16. The sine of the particular (longitude) difference in minutes (*viśeṣa-puṇy-aṁśa-jyā*), the shadow by the measure, and the degrees of the second arc — at the edges according to direction, direct or reversed (*viparyayaḥ*).

V.20. The difference of those deflection-minutes (*valana-aṁśaka*), from the sine-east-measure and perpendicular — the shadow obtained by the radius (*trijyā*) — the *pragraha-kṣepa*, before the retrograde motion.

V.21. Divided by the semi-diameter (*vyāsāṅka*), the minutes of the lunar-node-difference are the multiplier. By the middle-strength-minutes (*madhyabala-lip*) the north-south direction goes to the left.

[This chapter contains 46–47 verses on eclipse computations.]

Chapter XVII: Śṛṅgonnatyuttarādhikārah

Chapter XVII: On Lunar Cusp Elevation

This chapter treats the elevation of the lunar horns (*śṛṅgonnati*)—the elevation of the moon’s cusps, and the determination of time from this knowledge. This is a short chapter of ten verses.

[The complete verses of this chapter are not fully preserved in the scanned pages of this section. The chapter is transitional between the astronomical and mathematical sections.]

[This short chapter of 10 verses completes the astronomical portion before the extensive mathematical chapters.]

Chapter XVIII: Kuṭṭakādhyāyah — The Algebra Chapter

Chapter XVIII: The Kuṭṭaka (Algebra) Chapter

This is the most important mathematical chapter of the BSS, containing 102 verses. It covers the *kuṭṭaka* (pulverizer) method, operations with positive and negative numbers, *avyakta* (algebra) computations, *saṅkramaṇa* (transposition), linear and quadratic equations in one unknown, multi-variable equations, the *bhāvitaka* (product-nature) equation, and the *varga-prakṛti* (Pell’s equation).

V.1. Since problems generally cannot be solved without the pulverizer (*kuṭṭaka*), I shall state the *kuṭṭaka* together with problems.

V.2. The teachers (*ācāryas*) of the experts of the *tantra*, who know the pulverizer, zero (*kha*), debt (negative, *ṛṇa*) and wealth (positive, *dhana*), *avyakta* (unknown), elimination (*haraṇa*), one-*varṇa* and *bhāvitaka* equations, and the *varga-prakṛti* (quadratic nature, i.e. Pell’s equation).

V.3. The remainder from the divisor (*hāra*) corresponding to the greater residue (*adhikāgra*), divided by the divisor (*cheda*) corresponding to the lesser residue (*ūnāgra*) — whatever is the quotient from the mutual division is to be placed separately, downwards (*adhaḥ*), one below the other.

[This is the initial step of the Euclidean algorithm applied to the divisors A and B. The quotients are written in a column.]

V.4. The remainder, multiplied by an assumed number (*iṣṭa*) such that, when combined with the difference of the residues (*agra-antara*), it is exactly divisible (*śudhyati*) — the multiplier (*guṇakaḥ*) is to be set down; and the quotient is the product of the last (*antya*) and the penultimate (*upāntya*).

[If r is the final remainder, find m such that $rm + (a - b)$ is divisible. Then the *guṇaka* and *labdha* are computed from the column of quotients by back-substitution.]

V.5. The upper one (*ūrdhva*), combined with the last (*antya-yutaḥ*), (is divided by) the divisor of the lesser residue (*hīnāgra-cheda*); the remainder, multiplied by the divisor of the greater residue (*adhikāgra-cheda*) and combined with the greater residue, becomes the residue (*agram*).

[The solution x to $ax + c = by + d$ (modulo the divisors) is constructed by this back-substitution. Let x_0 be the *guṇaka*; then $N = x_0 \cdot B_{\text{large}} + r_{\text{large}}$.]

V.6. Twice the *yuga* (epoch period) is the product of the divisors; the product of the divisors is the *yuga-gata* for both residues. Thus by the *kuṭṭaka*, the derivation of three etc.

V.7. The residue of the revolutions (*bhagaṇa*) etc. is the *agra*; divided by the divisor; and zero, divided by the residue of days — of these two, the residue of revolutions etc., divided by the day-born residue — that is the day-count (*dyugaṇa*).

V.8. By whatever multiplier the day-born and revolution-etc. residues of the circle etc., divided by their common divisor, are equal — the product of that is the day-count etc.

V.9. The mutual division of the dividend and divisor, whatever is the remainder — by that, divided, they become reduced (*nis-chedau*); those same, mutually divided.

V.10. The quotient is to be placed down, down; the remainder multiplied by the chosen multiplier (*iṣṭa-guṇakāra-saṅguṇam*) — it is exactly divisible so that one is subtracted (*eka-hīnam*). The multiplier is to be set down; and the result is from the last.

V.11. The end-residue (*agrāntam*), by the penultimate multiplied, combined with its upper (value), divided by the divisor without remainder (*niśśeṣa-bhāgahāreṇa*) — thus is the *sthira-kuṭṭakaḥ* (fixed pulverizer), the remainder.

V.12. From the desired revolution-etc. residue, by its own *kuṭṭaka*-multiplier, by its own divisor-multiplied — the remainder is the day-count, combined with the reduced multiplier and divisor.

V.13. Thus in equal (signs) and in unequal: debt (*rṇa*) and wealth (*dhana*), wealth and debt — whatever was stated, the reversal (*vyastatvam*) of debt and wealth is to be done to the dividend (*guṇya*) and additive (*prakṣepa*).

V.14. The multiplier is the divisor; the divisor is the multiplier; wealth is debt, debt is wealth — to be done. Square (*varga*) is root (*pada*); root is square; from the end, reversed (*viparītam*); that is the first.

V.15. He who knows the *kuṭṭaka* by the epoch-periods (*yu-gādi*) of planets, separately stated by two, three, four etc. — **he** truly knows the *kuṭṭaka*.

[Verses 16–29 contain illustrative problems (*praśnas*) on the pulverizer applied to astronomical computations.]

V.16. “The residue of revolutions etc. at the desired — when on a moon-day, on a sun-day, or a Jupiter-day, on a weekday — tell the signs.” He who tells this knows the *kuṭṭaka*.

V.17. “On a weekday, whatever is the residue of degrees, the residue of minutes — when that on a moon-day, of the sun, or of the moon — he who tells this is a knower of the *kuṭṭaka*.’

V.18. “The desired (planetary positions) in the tithi-measure-days, which are sun etc., again when in those desired planet-weekdays” — he who tells this is a knower of the *kuṭṭaka*.

[Verses 19–29 continue with astronomical praśnas: given planetary residues at a particular weekday, finding the elapsed days and other planetary positions. These were standard problems for Indian calendar-makers (pañcāṅga-kāras).]

Operations on Positive and Negative Numbers

This section contains Brahmagupta's famous rules for arithmetic operations involving positive (*dhana*), negative (*ṛṇa*), and zero (*śūnya*, *kha*, *ākāśa*). These are the **earliest explicit rules** for operations with signed numbers and zero in the history of mathematics.

V.30. The sum of two positives is positive; of two negatives is negative; of a positive and a negative is their difference. Zero is the sum of a positive and an equal negative; and of two zeros.

[In modern notation:]

$$\begin{array}{ll} (+) + (+) = (+) & (-) + (-) = (-) \\ (+) + (-) = \text{their difference} & a + (-a) = 0 \\ 0 + 0 = 0 & \end{array}$$

V.31. The lesser is to be subtracted from the greater; the positive from the positive, the greater negative from the lesser negative — their difference is reversed (*vyastam*). The negative becomes positive, the positive becomes negative.

[Subtraction rule: $a - b = -(b - a)$ when $b > a$. For signs: $(+a) - (+b) = -(b - a)$ and $(-a) - (-b) = +(b - a)$ when $b > a$.]

V.32. Zero subtracted from debt (*ṛṇa*) is debt; from wealth (*dhana*) is wealth. Zero (*ākāśa*) — when positive is subtracted from negative, or negative from positive, then it is additive (*kṣepya*).

[Sign rules for subtraction with zero:]

$$a - 0 = a \qquad 0 - a = (\text{sign reversed})$$

V.33. The product of two negatives is positive; the product of two positives is positive. The product of zero and a negative, of zero and a positive, or of two zeros — the product is zero.

[Multiplication of signed numbers:]

$$\begin{array}{lll} (-) \times (-) = (+) & (+) \times (+) = (+) & (+) \times (-) = (-) \\ 0 \times (-) = 0 & 0 \times (+) = 0 & 0 \times 0 = 0 \end{array}$$

V.34. Positive divided by positive is positive; negative divided by negative is positive. Zero divided by zero is zero. Positive divided by negative is negative; negative divided by positive is negative.

[Division of signed numbers:]

$$\begin{array}{lll} \frac{(+)}{(+)} = (+) & \frac{(-)}{(-)} = (+) & \frac{0}{0} = 0 \\ \frac{(+)}{(-)} = (-) & \frac{(-)}{(+)} = (-) & \end{array}$$

[Verses 30–34 are historically momentous. Brahmagupta is the first author in world history to state general rules for arithmetic operations with signed numbers and zero. Note that $\frac{0}{0} = 0$ is the only questionable rule from a modern perspective; Brahmagupta also stated (elsewhere) that division by zero leaves the fraction unchanged, i.e. $\frac{a}{0} = \frac{a}{0}$ as an unreduced entity.]

V.36. The sum, decreased by the difference and divided by two, is the *saṅkramaṇa* (transposition); or divided by the difference. The difference of squares, decreased by the sum of the difference and divided by two — the *viśama-karma* (unequal operation).

[If $x + y = S$ and $x - y = D$, then:]

$$x = \frac{S + D}{2}, \quad y = \frac{S - D}{2}$$

The “difference of squares” formula: $x^2 - y^2 = S \cdot D$.

[Verses 37–42 treat *karaṇī* (surds), *bhāvitaka* (product-form), and algebraic identities for powers of unknowns. These are not fully preserved in the scanned pages of this section.]

V.43. Divided by the difference of unknowns (*avyakta*), reversed, the difference of *rūpas* (absolute terms) — in the equation (*sama*), the unknowns (*avyakta*) are to be subtracted; the *rūpas* are placed below that.

[For equations $ax + b = cx + d$, divide by $(a - c)$ and rearrange:]

$$x = \frac{d - b}{a - c}$$

V.44. Of the *rūpas* multiplied by four and the square of the middle (*madhya*) — the square-root of that, decreased by the middle and divided by twice the multiplier (*varga-dvi-guṇa-uddhṛtam*) — is the *madhya* (unknown).

[This is Brahmagupta's verbal statement of the quadratic formula. If $ax^2 + bx = c$:]

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is algebraically equivalent to the modern formula $x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ for the equation $ax^2 + bx - c = 0$. Brahmagupta always writes the constant term on the right side, so his formula uses $+4ac$.

V.45. Of the *rūpas* multiplied by the square and combined with the square of half the unknown — whatever is the root, decreased by half the unknown, divided by the square — that is the unknown.

[Alternative form:]

$$x = \frac{\sqrt{a^2c + \left(\frac{b}{2}\right)^2} - \frac{b}{2}}{a}$$

[Verses 46–50 contain astronomical problems solved by the *kuṭṭaka* and quadratic methods.]

V.46. “From the degree-residue (*aṃśaka-śeṣa*) with one added, divided by twelve, multiplied by four and increased by eight, equals the degree-residue with one — when this happens, tell the day-count (*ahargaṇa*).”

[An astronomical problem leading to a linear equation in the *ahargaṇa* (elapsed days).]

V.47. “The *adhimāsa-śeṣa* (excess month residue), decreased by two, divided by three, increased by seven, multiplied by two — equals the *adhimāsa-śeṣa*. When this happens, tell the *yuga-gata* (elapsed in the *yuga*).”

V.48. “The *avama-avaśeṣa* (omitted day residue), decreased by one, divided by six, increased by three — equals the *avama-śeṣa* divided by five. When this happens, tell the *yuga-gata*.”

V.51. From the first unknown (*varṇa*), having subtracted the other unknowns, divided by the first measure (*ādyā-māna*), combined — two similar divisors, repeatedly, reversed — the *kuṭṭaka* (is applied) for many (unknowns).

[For multiple equations in many unknowns: eliminate all but one unknown, then apply the kuṭṭaka.]

V.52. “From the day-count increased by the elapsed revolutions, from that with its residue added, combined with their sum, or from their sum — the day-count” — he who tells this is a *kuṭṭaka*-knower.

V.53. “From the day-count decreased by the elapsed revolutions, from that with its residue subtracted, diminished by their sum, or from their difference — the day-count” — he who tells this is a *kuṭṭaka*-knower.

V.54. “With the quantities etc., with their residues, thus with the elapsed intercalary months and days subtracted, and with their residues — the *yuga-gata*” — he who tells this is a *kuṭṭaka*-knower.

[Verses 55–59 contain more astronomical applications and computational rules for calendar calculations.]

V.60. The *bhāvitaka* (product-nature) multiplied by the *rūpa*, together with the product of the unknown (*avyakta-vadhā*), divided by the desired (*iṣṭa*) — at the desired quotient: if less, add more; if more, add less; *kṣepya* (additive) to the *bhāvita* divisor; reversed.

[The *bhāvitaka* is the coefficient of the product of two unknowns, i.e. the term Bxy in the general quadratic form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.]

V.61. From the product of the sun's sign and degrees, having subtracted the signs and degrees multiplied by three and four, having seen ninety — making the sun (*sūrya*), the calculator (does this) up to a year.

V.62. In the *bhāvitaka*, the product by which unknown — having made that amount desired, the sum of the unknown (*varṇa*) multiplied by that becomes the *rūpas* (known quantities).

V.63. The product of the unknown-quantity and the *bhāvitaka* becomes the desired unknown-number. Thus it is accomplished even without the *bhāvitaka* equation; what was done from that?

V.64. The root is twofold: from the square of the desired, from the multiplier multiplied, increased or decreased by the desired. The first product is the multiplier-product; together with the last product — made last.

[For equations of the form $Nx^2 + k = y^2$ where N is the *guṇaka* (multiplier) and k is the *kṣepa* (additive).]

Varga-prakṛtiḥ (Pell's Equation)

This section treats the “square-nature” or Pell’s equation: finding integers x, y such that

$$Nx^2 + k = y^2$$

where N (the *guṇaka*) is a non-square positive integer and k (the *kṣepa*) is an integer. Brahmagupta gave the composition formula (*bhāvanā*) and the cyclic method (*cakravāla*), which were rediscovered in Europe only by Euler and Lagrange in the 18th century.

V.65. The product of the cross-terms (*vajra-vadha*) plus one is the first *prakṣepa*; equal to the product of the *prakṣepas*; divided by the *prakṣepa*-subtractor — the roots (*mūle*) are in the *prakṣepas*, *rūpas*.

[The “cross multiplication” identity (*bhāvanā*): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then $(x_1y_2 \pm x_2y_1, Nx_1x_2 \pm y_1y_2)$ solves $Nx^2 + k_1k_2 = y^2$.]

V.66. The *rūpa*-*prakṣepa* roots, separately, by the desired *kṣepya* and *sódhya*-roots — having done, the end and first roots which are in the *prakṣepa* — in subtraction or desired.

V.67. When increased by four (*catur-adhike*): the square of the last root (*antya-pada-kṛtiḥ*), decreased by three (*tri-ūnā*), halved, multiplied by the last root — is the last root. The square of the last root, decreased by one (*vi-ekā*), divided by two, multiplied by the first root — is the first root.

[Recurrence relations for generating new solutions to $Nx^2 + 1 = y^2$ from known solutions. If (x_1, y_1) is a known solution with $k = -1$, then:]

$$y_{n+1} = \frac{(y_n^2 - 3)y_n}{2}$$

$$x_{n+1} = \frac{(y_n^2 - 1)x_n}{2}$$

V.68. When decreased by four: the square of the last root, increased by three and one, the product halved, separately, decreased by one — increased by one, decreased by one, multiplied by the last — the product of the roots (is) the first, the last root.

V.69. In the square of the multiplier (*guṇaka*), the *kṣepa*, by some (number) divided, added, subtracted, halved — the first (root). The last root — the other, divided by the multiplier-root, (gives) the first.

[Given $Nx^2 + k = y^2$, divide by k to reduce the additive. If k divides N or a related quantity, new roots can be found.]

V.70. When the multiplier is a perfect square (*varga-cinne guṇake*): the first (root), divided by its root, becomes (the new root). When the *kṣepa* is a perfect square: multiplied by its root, then (are) the roots.

[If $N = a^2$ is a perfect square, then $a^2x^2 + k = y^2$ gives $y = \sqrt{a^2x^2 + k}$. If also $k = b^2$, then $\sqrt{(ax)^2 + b^2}$ may simplify.]

V.71. The sum of the multipliers, multiplied by eight, divided by the square of the difference of the multipliers — the quantity. The two multipliers, tripled, reversed, increased, divided by the difference — (are) the roots.

V.72. The square, decreased by the sum of the other squares, divided by the half-square of the sum-difference — those multiplied by them, combined and separated — squares; in the product, combined with the *rūpa*.

[The general composition (*bhāvanā*) formula:]

$$\begin{aligned} &\text{If } Nx_1^2 + k_1 = y_1^2 \text{ and } Nx_2^2 + k_2 = y_2^2, \\ &\text{then } N(x_1y_2 \pm x_2y_1)^2 + k_1k_2 = (Nx_1x_2 \pm y_1y_2)^2 \end{aligned}$$

When $k_1 = k_2 = 1$, this generates infinitely many solutions to $Nx^2 + 1 = y^2$ from a single fundamental solution. This is Brahmagupta's great discovery, called the *samāsa-bhāvanā*.

Astronomical Divination Rules

This section gives remarkable “divination” rules: given the planetary residues at a particular day, one can compute the positions of other planets. These were used by calendar-makers (*pañcāṅga-kāras*) to verify their computations.

V.75. “The square of the residue of signs and minutes, multiplied by twenty-two (*dvinavati*), or multiplied by eighty-three (*tryaśīti*), with one added — on a weekday, making the square — the calculator (does this) up to a year.”

V.76. “The residue of the sun’s seconds (*viluptā-śeṣa*), decreased and multiplied by five, likewise by ten — on a Jupiter-day, making the square — the calculator (does this) up to a year.”

V.77. “The square of the residue of revolutions etc., multiplied by three, combined with nine hundred — or the square decreased by eight hundred — the calculator (does this) up to a year.”

V.78. “The square of the residue of revolutions etc., multiplied by four, combined with sixty-five — or the square decreased by sixty — the calculator (does this) up to a year.”

V.79. “The desired residue of revolutions etc., decreased by twenty-two, multiplied by eighty-three, combined with one — making the square — the calculator (does this) up to a year.”

V.80. “The square of the *adhimāsa-śeṣa* (intercalary month residue), multiplied by thirteen, combined with three hundred — or the square decreased by twenty-seven (*tri-ghana*) — the calculator (does this) up to a year.”

V.81. “The residue of the moon’s seconds, multiplied by seventeen, also multiplied by thirteen — separately, combined with one — making the square — the calculator (does this) up to a year.”

V.82. “The square of the *avama-avaśeṣa* (omitted-day residue), multiplied by twelve, combined with one hundred — or the square decreased by three — the calculator (does this) up to a year.”

V.83. “On a weekday, the residue of the sun’s minutes (*kala*), combined with or decreased by the residue of Jupiter-day seconds — the square and product with one — the calculator (does this) up to a year.”

V.84. “The residue of seconds (*vikalā-śeṣa*), combined with ninety-three (*trina-vatī*) and decreased by seventy-seven (*saptaśaṣṭi*) — on the sun’s weekday, making the square — the calculator (does this) up to a year.”

V.85. “On a weekday, the residue of the sun’s minutes, combined with twelve and sixty-three — or decreased by sixty and eight — the calculator (does this) up to a year.”

V.86. “From the residue of the moon’s seconds, the residue of the sun’s sec-

onds, or the residue of degrees — or alternatively the mean (position) desired — the calculator (does this) up to a year.”

V.87. “From the residue of seconds of Venus (*jīva*), Mars (*kuja*) and the Moon; from the residue of Mars’s seconds (*bhauma-liptikā*); from the residue of the moon’s degrees — the Sun — the calculator (does this) up to a year.”

[Verses 75-87 give a remarkable set of “divination” rules. Given a particular planetary residue at a known weekday, these formulas allow one to compute other planetary positions. The constants (22, 83, 5, 10, etc.) encode specific planetary period relations. These verification rules were essential for ensuring the accuracy of pañcāṅga (calendar) computations.]

[Verses 88–99 treat the kuṭṭaka method for solving multiple planetary congruences and refined calendar computations.]

V.100. These problems strike the heart (with joy). One may pose thousands of other problems; given by others — thus one should solve those problems by the stated methods (*karaṇaiḥ*).

V.101. In the assembly of people, among the knowers of fate (*daiva-vidām*), it destroys the radiance (of rivals), like the sun (destroys) the stars — (even) one *kuṭṭaka* problem recited — what to speak of hundreds!

V.102. These problems have been stated for each rule, in the illustrative rules (*sa-uddeśakeṣu sūtreṣu*). By three hundred and three (303) *āryā* (verses), the *kuṭṭaka* and the eighteenth chapter.

*[Thus ends the Eighteenth Chapter, the Kuṭṭaka, in the
Brāhmasphuṭasiddhānta.]*

Chapter XIX: Śaṅku-chāyā-vijñānam

Chapter XIX: On Gnomon and Shadow Knowledge

This chapter contains 20 verses on gnomon (*śaṅku*) and shadow (*chāyā*) computations. These techniques allow the determination of latitude, building heights, water depth, and altitude by mirror reflection.

V.1. Having seen the half-day ghaṭikās, he who knows the sun-knowing (method), knows the degrees of latitude (*akṣa-aṃśakān*). By the ghaṭikās of the interval from rising — from the known, the knowable — he is a knower of the *tantra*.

V.2. By the ghaṭikās of the interval from setting, he who brings the knowable from the known, thence brings the mean motion and the *yuga*-revolutions — he is a knower of the *tantra*.

V.3. He who brings the measures of the darkness-sun, the moon-horn, from the height of the lamp-flame — in the knowledge of the ground between gnomon-base (and object), the shadow — he is a knower of the *tantra*.

V.4. He who knows the desired house-height, knows that interval, is observed in water — shows the top of the house-wall, or in a mirror — he is a knower of the *tantra*.

V.5. By the knowledge of the second-shadow-light-edge interval, he knows the lamp-height — or a knower of gnomon and shadow knows the earth's shadow — he is a knower of the *tantra*.

V.6. Having seen the house-base-interval of the net of light and shadow, having seen the top of the house — a knower of the ground — he knows the house-height, having seen it in oil or water — he is a knower of the *tantra*.

V.7. Having observed the top of the house in water, having spread out the water again, in self-ground-knowledge — he brings from the water the ground or the height of the house — he is a knower of the *tantra*.

V.8. With known shadow-men (measures), the interval of water and wall having been known — in the wall, he who knows the ascent of the sun's light (ray) — he is a knower of the *tantra*.

V.9. The half-day ghaṭikās of the desired day, the prāṇas of the interval of fifteen ghaṭikās — by the day's motion prāṇas, by those he should compute the latitude as before.

V.10. Of the known and knowable planets, by the nāḍikās of the interval from rising — greater or less — by the risings, the known from the known, the knowable on the east-west (is) to be known.

V.11. The known, with its half-duration, by the risings — greater or less by the nāḍikās of the interval from setting — from the known, on the east-west the knowable; when the half-duration is decreased, the knowable.

V.12. Having made the known into the mean, again on another day, that interval is the motion (*bhuktiḥ*). By the rule of three (*trairāśikena*), by the motion — the derivation of the planet’s circle in the *kalpa*.

V.13. The reverse of half the duration (*sthiti*) is the true darkness-measure in an eclipse. From the measure-rise of the sun and moon, by the ghaṭikā-part, from the terrestrial rising.

V.14. The interval of lamp-base and gnomon-base, multiplied by the desired-quantity-gnomon, divided by the lamp-flame-height minus the gnomon — the quotient is the shadow (*chāyā*).

[Similar triangles: If h = lamp height, g = gnomon height, d = distance between them, then shadow $s = \frac{d \cdot g}{h - g}$.]

V.15. The shadow, multiplied by the gnomon-interval, divided by the shadow-interval — (is) the ground (distance). The shadow together with the gnomon, multiplied by the gnomon, divided by the shadow — (is) the lamp-height.

V.16. Having known the gnomon and shadow, from proportion (*anupātāt*) one should compute the elevations (*samucchrayān*) — of houses, temples, trees, and mountains — or having known the height, the shadow.

[This is the standard “rule of three” for similar triangles: $\frac{\text{height}}{\text{shadow}} = \frac{\text{known height}}{\text{known shadow}}$, so unknown quantities can be found by proportion.]

[Verses 17–19 give additional rules on shadow computations and heights/depths using water and mirror reflections. Verse 20 gives the chapter colophon.]

[Thus ends the Nineteenth Chapter on Gnomon and Shadow Knowledge.]

Chapter XX: Chandaś-citty-uttarādhikārah

Chapter XX: On Prosody (Chandaś-citty-uttara)

This chapter contains 19 verses on Sanskrit prosody (*chandah-sāstra*), dealing with combinatorial calculations for metrical patterns — the number of possible arrangements of long (*guru*, ‘G’) and short (*laghu*, ‘L’) syllables. Brahmagupta here presents an early form of combinatorics.

V.1. The *rg*-group (combinations), the permutations (*paryāya*), the aggregate-sums, in the unequal pairs — the *soma* etc., obtained as before, the quadruples, the remainder-additions — the last.

V.2. Decreased by one etc., the first and last, their reversals — in squares etc., of the odd-even (syllables), one should increase the quarters by direct and reverse order.

V.3. One by one, two by two, with additional appendages in those positions — beginning from the square up to the desired — the spread-out (*prastāra*) becomes with barley-middle (*yava-madhya*, i.e. Pascal’s triangle).

[The “Meru of Prosody” (*prastāra*) is Pascal’s triangle: the number of metrical patterns with n syllables and k guru syllables is $\binom{n}{k}$. The “barley-middle” refers to the shape of the triangular arrangement.]

V.4. With one, the last — the two-foot beginning, the three-foot etc., below, separately the numbers. That is to be subtracted, decreased by one, separately from that — combined with one above.

V.5. Up to the quarter, decreasing by one, among the syllables, then increasing by one — in the product beginning from one, the final number of squares etc.

[For n syllables, each of which can be G or L , the total number of possible patterns is 2^n .]

V.6. With one added to the quarter-half — in the uneven (syllables) above, in the even (syllables) at the quarter-half. From the half, doubled, decreased by one — the *yulāni* (pairings) below that, for all.

V.7. By the middle (terms), thus decreased by half, by the quarter-steps in order — equal to the reversed, beginning from the quarter — in the uneven, subtracting one in the middle, from that one should do (the computation).

V.8. With one-plus-equal-to-the-step-beginnings, the setting is excessively subtracted below — the sum of numbers such as it is — the first, decreased by three — in the lost (*naṣṭa*, i.e. finding a missing pattern).

V.9. Made with middle terms and halved — in the equal-number (case), in direct and reverse order — the *kṣepa* (additive). In the uneven, decreased by one — the half, in order, in reverse order, with one added.

V.10. In the equal-number (case), by the staircase in direct and reverse order — thus also in the uneven — to be constructed. When cooked (computed) is seen, the first — the remaining syllables at the end.

[Verses 11-18 continue the treatment of combinatorial prosody, including algorithms for the “Meru of Prosody” (Pascal’s triangle) and rules for lost (*naṣṭa*) and spread-out (*uddhiṣṭa*) metrical patterns. The “staircase” (*sopāna*) is a procedure for listing all possible metrical patterns in systematic order.]

[Thus ends the Twentieth Chapter on Prosody in the **Brāhmasphuṭasiddhānta**.]

Chapter XXI: Golādhyāyaḥ (Jyā-prakaraṇam)

Chapter XXI: The Sphere — Sine Computation

This section includes the Jyā-prakaraṇa (verses 17–23), which gives Brahmagupta’s method for computing the table of sines. The radius is $R = 3270$ minutes. The “segments” (*jyā-khaṇḍa*) are the tabular sine-differences. Brahmagupta uses a second-order interpolation formula, more accurate than the linear interpolation used by earlier authors.

V.17. Having placed marks at the eighth parts of a sign ($3^\circ 45'$ intervals) in direct and reverse order, one should construct chords (*jyā*) for every two-by-two: their halves are the sine-chords (*jyā-ardhāni*).

*[Brahmagupta divides each sign (30°) into 8 parts of $3^\circ 45'$ each, giving 24 intervals in a quadrant (0° to 90°). The *jyā-ardha* (half-chord) is the modern sine: $R \sin \theta$.]*

V.18. The sine-half-differences of sine-halves are the sine-segments (*jyā-khaṇḍa*). Reversed from the last, or: the bow (*dhanus*) is the reverse-sine (*utkrama-jyā*) together with these.

*[The *jyā-khaṇḍa* are the tabular differences: $\Delta_i = R \sin(i+1)\alpha - R \sin i\alpha$ where $\alpha = 3^\circ 45'$.]*

V.19. From the quarter-parts of the half-diameter squared, multiplied by one, two, and three: the roots are the eighth, twelfth, and sixteenth segments; the others (are derived) from these.

[Initial segments computed from $R^2/4$:]

$$\begin{aligned} j_8 &= \sqrt{R^2 \cdot 1/4} = R/2 = R \sin(30^\circ) \\ j_{12} &= \sqrt{R^2 \cdot 2/4} = R/\sqrt{2} = R \sin(45^\circ) \\ j_{16} &= \sqrt{R^2 \cdot 3/4} = R\sqrt{3}/2 = R \sin(60^\circ) \end{aligned}$$

With $R = 3270$: $j_8 = 1635$, $j_{12} = 2312$, $j_{16} = 2832$.

V.20. Having subtracted the fourth part of the sum of squares of equal direct and reverse sines and equal segments from the square of the half-diameter, the square-root of the remainder is the first (sine-chord).

[If j_k and j_{n-k} are known sines, and δ is the common segment-difference, the next sine is:]

$$j_{k+1} = \sqrt{R^2 - \frac{(j_k + j_{n-k})^2 + (2\delta)^2}{4}}$$

V.21. Its half-segments, decreased by those equal to the earth (*jina*) — the second in production.

[This verse contains numerical values encoded as words (saṅketa system). “Jina” = 1 (Lord of the earth). The second-order correction refines the first-order segments.]

V.22. Thus the small or numerous sine-segments — the first segments are the sine-halves of one-sixth, one-fourth, and one-third of the circle’s circumference.

[The reference sines at key angles:]

$$R \sin(30^\circ) = R/2 = 1635$$

$$R \sin(45^\circ) = R/\sqrt{2} \approx 2312$$

$$R \sin(60^\circ) = R\sqrt{3}/2 \approx 2832$$

V.23. From the diameter multiplied by the equal reverse-segment, or from the quarter-part, having done the stated segment operations — the derivation of sine-halves is not difficult from this.

[Summary of the method: Given the tabular differences (*khaṇḍa*), one can compute all sines by second-order interpolation using only multiplication and square roots.]

[

0.5cm][Brahmagupta's sine table uses $R = 3270$ minutes. The twenty-four *jyā-khaṇḍa* (sine-differences) are: 225, 224, 222, 219, 215, 210, 205, 199, 191, 183, 174, 164, 154, 143, 131, 119, 106, 93, 79, 65, 51, 37, 22, 7. The second-order correction formula accounts for the curvature of the sine function, yielding values accurate to within 1 minute of arc.]

Chapter XXII: Yantrādhyāyaḥ

Chapter XXII: On Astronomical Instruments

This chapter contains 43–57 verses on astronomical instruments (*yantra*). It describes the bow instrument (*dhanur-yantra*), staff instrument (*yaṣṭi-yantra*), water instruments, wheel instruments, and gnomon instruments. These were used for measuring celestial altitudes, azimuths, and for timekeeping.

Instruments Described

- **Dhanur-yantra** — A semicircular or arc-shaped instrument for measuring altitudes, resembling a bow.
- **Yaṣṭi-yantra** — A staff or rod instrument for measuring angles and time.
- **Jala-yantra** — Water-based instruments including water clocks and mercury devices.
- **Śaṅku-yantra** — Gnomon instruments for shadow measurements.
- **Cakra-yantra** — Circular instruments with rotating sighting vanes for measuring celestial positions.

[Verses 1–28 describe the bow instrument, staff instrument, and their use in measuring celestial altitudes and azimuths. Verses 29–40 treat water instruments and circle instruments.]

V.41. The *kṣeṭṭa* (object to be observed), by its own light, the *kṣepya* (object aimed at) — equal, thus in mercury (*pārada*), when it becomes — the mixed-time measure from the *śaka* epoch (78 CE) or from the *mudda* (particular epoch).

V.42. The co-altitude (*kota*) going above, held obliquely in the string (*sūtraka*), in small (measures) — having cast upon the eastern fire-place (*prāv-analaka*), the *nāḍikā* (water-clock) flows, falls thus.

[Verses 43–48 contain further rules on the use of instruments for time measurement and planetary observation.]

V.49. A wheel of light-wood, with equal bearing intervals, of lotus-shape — having faith, filled with mercury in the center, well-polished, having performed the *saṃdhyā* (preparation).

V.50. Obliquely in the middle of the co-altitude, the support above — a beautiful semi-circle (*rūpā-radaṃ*) moves. The small wheel with holes — thus constantly in the air it rotates.

[Description of a self-rotating wheel instrument, possibly an early form of anemometer or a water-driven astronomical device. The “holes” regulate the flow of water or air.]

[Verses 51–53: Additional rules on wheel instruments and their construction. The chapter concludes with verses praising the value of observation and proper instruments.]

V.53. The *karaṇa-jyā* (computed sine), the unsteady motion itself — the arrow, the rising, the ascension — by the instrument with confidence, two-fold — from the five-fold of Ari (the sun’s) motion.

*[Thus ends the Twenty-second Chapter on Instruments in the
Brāhmasphuṭasiddhānta.
 End of the section on observation.]*

Appendix: On Variant Readings

Manuscript Sources for Variant Readings

The footnotes in the printed edition record variant readings from four manuscript sources, designated by the following sigla:

The footnotes use a notation such as “ for (*reading*)” to indicate that manuscript has the variant reading instead of the adopted text. Parenthetical notations like “(—)” indicate that the verse number in the source manuscript is different.

Chapter 23: — The Measurement Chapter

The twenty-third chapter of the *Brāhmasphuṭasiddhānta* treats of the measurement of time and the practical application of astronomical calculations. This chapter, like others in the work, is composed in the classical *āryā* metre and presents Brahmagupta's rules for calendrical computation, planetary positions, and the reconciliation of different astronomical systems.

The chapter is divided into two principal sections:

- (i) **Prayogavidhi** — Practical application of the siddhānta
- (ii) **Caturviṃśa** — The twenty-four fold division of time

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### Practical Application (Verses 1-4)

"The solar year, the lunar months and *tithis*, the civil days, days, months, years, the positions of the planets, the *nakṣatras*, and the zodiacal signs—these we call measures (*māna*)."

#### — Verse 1

##### Word Analysis :

(1) *saurairāṇḍa* — the solar year; (2) *māsaistithayaḥ* — lunar months and *tithis*; (3) *sāvanāni divasāni* — civil days; (4) *dinamāsābdāni* — days, months, and years; (5) *grahanakṣatrarāśayaḥ* — planets, nakṣatras, and zodiacal signs—these are called *māna* (measures).

"The measures of solar, lunar, stellar, and civil reckoning, and the computation of planets by these measures separately—the fourfold manner of computation is in use in the world."

#### — Verse 2

*sauracāndrārśasāvanāni* — four measures; *ebhiḥ pṛthak pṛthak* — separately by each; *grahānanayam* — computation of planets; *lokasya* — current in the world.

"A *yuga*-year, abundant and deficient (days), fire (3), subtraction and addition—these (come) from the solar (system). *Tithi*, *karaṇa*, intercalary month, and the changes of zodiacal signs and *parva* (come) from the lunar (system)."

#### — Verse 3

*yugavarṣaṃ vipulābdayanāni* — intercalary and deficient days; *vahniḥ* — the number three; *udviddhi-hānayaḥ* — addition and subtraction—from the solar system. *tithikaraṇādhikamāsāḥ* — from the lunar; *rāśiparvikriyāḥ* — changes of zodiacal signs and lunar half-months.

"For one desirous of performing the *pramāṇa* (measurement), the planetary motion, the rising and setting, and the creation (of almanacs), the civil measure (is useful), as well as the bed (auspicious moment), *prāyaścitta* (expiatory rites), and other ceremonies."

## — Verse 4

*pramāṇam grahagatikṣayodayasūtakaḥ* — for the construction of the pañcāṅga (almanac); *sāvanam mānam* — civil measure; *śayyā* — auspicious moment for lying down; *prāyaścittakriyāḥ* — expiatory rites and ceremonies.

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The Twenty-four Fold System (Verses 5-23)

“Since (computation) is impossible without names (*saṃjñā*) given to each object, the names and numerals well-known in the world (are given).”

— Verse 5

Commentary: The commentator explains that Brahmagupta here gives a systematic nomenclature for the various measures of time and computation. The twenty-four items are enumerated in the following verses.

“A *yugapad*, a *yuga*, beginning from that, or of the sun, of the Yāmya (south), of Bhāskara, or of Kanyā (Virgo), half a sign, the numbers (associated), at the head, from half a day, from two, from Bhrama (revolution).”

— Verse 6

“The sun has been measured by Pulisha, Romaka, Vāraha, and Yavana (Greeks). Therefore, from that (measurement) one must construct a proper *siddhānta*.”

— Verse 7

puliśaḥ romakaḥ vārāhī yavanāḥ — the five *siddhānta* authors; *sūryasya pramāṇam* — measurement of the sun; *racanātmakaḥ siddhāntaḥ* — construction of a scientific treatise.

“If the *siddhāntas* are different, (yet) Bhāskara (the sun) has the same modifications (*saṃskāra*)—the difference (is) abundant and deficient, those having the same (result) for the previous (method).”

— Verse 8

“In it (the *siddhānta*) the five (methods): *pakṣāgati*, *madhya*, *gati*, *uttara*, etc.; *kuṭūākāra* (transformation), *bheda* (division), *chid* (cutting), *caturtha* (fourth proportional), and the sphere (*gola*).”

— Verse 9

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“By Brahmagupta, the son of Jīṣṇu, the knower of mathematics and the sphere, this true *siddhānta* was composed, (which is) Brahma (itself), adorned with the beauty of elegant composition.”

## — Verse 10

*bhaṭṭa brahmācāryaḥ* — the venerable Brahmagupta; *jijñoh tanayaḥ* — son of Jīṣṇu; *gaṇitagolavit* — knower of gaṇita (mathematics) and the celestial sphere; *sphuṭasiddhāntaḥ* — true astronomical treatise; *brahmā* — of the nature of Brahman itself.

“(Various) instruments, conjunction, year-count, diverse multiplication, the *utka* (elevation), the *śakani* (depression), on account of the multitude of operations, (all these) from the diminished (part), the solar *maragraham*...”

— **Verse 11**

“The abundant, the one without method, the *ṛddhicchid* (augmented cut), the *yogketsya* of the standing, the *śaṅkuśye* (gnomon-shadow) declares the year and also the solar year...”

— **Verse 12**

“As much as has not been spoken by me, the remainder of the sphere, thin as a sixth part of a *ksaya* (decrease), this (is) the *prāyaśloka* (resumé verse). This (is) the *saṃjñā-adhyāya* (chapter on nomenclature), the twenty-fourth.”

— **Verse 13**

*mayā yannoktaṃ* — what has not been spoken by me; *golāt agraṃ* — the remainder of the sphere; *kṣayārdhasato ṣaṣṭhāṃśatulyaṃ* — as minute as one-sixth part of the decrease; *prāyaślokaḥ* — resumé verse; *saṃjñādhyaś caturviṃśaḥ* — the twenty-fourth chapter on nomenclature.

**Note on verse numbering:** The original text numbers this as the 23rd *adhyāya* in the internal count, though the final verse calls it the 24th in another reckoning. The manuscript tradition contains variations in chapter enumeration.

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“Multiply the elapsed *sāvana* (civil) days by the solar days; the *asāvana* (non-civil) days—therefore, that attainment is difficult for those of dull intellect.”

— **Verse 5 (alt.)**

“Humanity, the father, the divine, the arm, the eight (*vasu*), the time—the *utkarṣi* (elevation) for the purpose of knowledge, the arm, clearly the ninth (or another).”

— **Verse 6 (alt.)**

“Two-two (make) a zodiacal sign; multiplied by Makara (10, i.e. 60), these (are) the six (seasons); the solar motion (gives) the seasons: *śiśira-vasanta* (spring), *grīṣma* (summer), *varṣā* (rains), *śarad* (autumn), and *hemanta* (winter).”

— **Verse 7 (alt.)**

dvau dvau rāśiṃ — Gemini etc.; *makarāhatāḥ ṣaṭ* — six seasons multiplied by ten (Makara = 10); *sūryagatiḥ* — the sun’s path; *śiśira-vasanta-grīṣma-varṣā-śarad-hemantāḥ* — the six seasons.

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“The principal and the *arha* (worthy), the worshipper and the devotee; by the *vyāsāntara* (diameter difference) for the crow—the sun’s hypotenuse is to be released; two, the decrease is to be shown, the remainder by the moon’s hypotenuse.”

— **Verse 8 (alt.)**

“The six, diminished and increased by fifteen, are the declination-time etc., multiplied by six (*rasa*).”

— **Verse 64 (alt.)**



“The radius and the abundant shadow—their product, divided by that and multiplied by the latitude; the decrease is always in the south (Yāmya); from the sphere, the disc’s declination.”

— Verse 66 (alt.)

“When the sum and difference of declination and latitude are purified by the *akṣarāpada* etc., the half-day; when the square of that is subtracted from the square of the sum—the method when reversed at that position.”

— Verse 67 (alt.)

“The remainder, multiplied by the prescribed (number) and arrived at, divided by the day-number elevated by the *nāḍī*; that (result) multiplied by the half-day hypotenuse, by that radius the result (is) the hypotenuse.”

— Verse 68 (alt.)

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“Thus has been stated in order the section on numerals and the practical application; the diameter multiplied, the long circle, the *ṣaḍāṃśaka* (sixth part), the latitude-decline.”

— Verse 9 (alt.)

“Its diameter, śaki (potency), *karaṇḍū*, that (is) multiplied by the *dvijaya* (twice-conquered), by degrees.”

— Verse 10 (alt.)

“The sun’s hypotenuse, the messengers, by the *trijaya* (triple-conquered), by the *vyāsāntara* (diameter difference) divided, the remainder; the *tijayāmukhyāḥ* (chief of the three-conquered), by the decrease, the operation, the *hutātma* (fire-souled = 3), the diameter.”

— Verse 10 (alt. 2)

“From the diameter, the evil, from the product, the crow—the diameter difference, the release (is) the product; the *prokṣa* (sprinkling), the second middle, thus from the *tithi* (lunar day) multiplied and obtained, the darkness, the diameter.”

— Verse 11 (alt.)

“The intercalary months, the *avama* (omitted) days, the *sambhavana* (production), all the measures; by ten *prāyaśloka*s (resumé verses), the twenty-fourth Measurement Chapter.”

— Verse 12 (alt.)

*Thus ends the twenty-third chapter on Measurement
in the work of Brahmagupta.*

Mean Computation Chapter

The *Madhyamādhikāra* constitutes the first and foundational chapter of the commentary (*vāsanā-vijñāna-bhāṣya*) on the *Brāhmasphuṭasiddhānta*. It treats the computation of mean positions of planets (*madhyama-graha*), beginning with a discussion of the motion of the celestial sphere (*bhacakra*), the definition of time (*kāla-pravṛtti*), and the various *yuga*-systems employed in Indian astronomy.

Introductory Discussion (pp. 752-760)

The commentary opens with a discussion of the *bhacakra* (celestial sphere) and its motion:

“(The celestial sphere), moving eastward (from left to right), was first created by Brahmā (the world-teacher). The sphere, along with the nakṣatras and other celestial bodies having equal *capādas*, along with the meridian staff etc., the circle of nakṣatras and other luminaries that give light—this sphere was fashioned by Brahmā. Even so, the ācāryas (teachers) do not describe their (the planets’) essential nature (*svārūpa*), but have only determined their positions (*sthāna*) for the purpose of computation (*spaṣṭādi*).”

— Commentary v. 1

Hindi Commentary (pp. 752-753): The commentator explains that the celestial sphere (*bhacakra*) was first created by Brahmā (the creator). The sphere, along with the nakṣatras and other celestial bodies, revolves from west to east. The commentator notes that the ācāryas (teachers) do not discuss the nature (*svārūpa*) of these bodies but have only determined their positions (*sthāna*) for the purposes of computation.

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### Key Verses from the Siddhānta (pp. 760-770)

The commentary elucidates the following root text (*mūla*) verses:

“Now (Brahmagupta) declares the commencement of that time whose beginning is unmentioned. Day, month, year, *yuga*, and *kalpa* began from the sun at Laṅkā at the beginning of creation.”

#### — Mūla-śloka 4

**Hindi commentary (pp. 761-762):**

*caityādi kalpa-pratipadyāt laṅkopalakṣitāmbhṛdv ekodayādi-divā pravṛttāḥ* — From the first tithi of Caitra (the first month), at Laṅkā, the days etc. commenced from the sunrise.

*ravidinambaraḥ sṛṣṭyādaḥ kalpādaḥ* — The solar day number at the beginning of creation, at the beginning of the *kalpa*.

The commentator explains the concept of the *kuldiṇa* (civil day) in detail. The number of civil days elapsed since the beginning of the current *kalpa* is 1,58,22,23,78,50,000 according to Brahmagupta’s system.

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On Time Computation and Planetary Motion (pp. 770-790)

The commentary proceeds with detailed mathematical analysis:

“First, the *yugapada* (yuga-epoch method); second, by *parva* (lunisolar method); third, by thirteen months; or fourth, by *sāvana* (civil reckoning).”

— Time-reckoning verse

Exposition: The four types of time-reckoning are discussed:

- (i) *Yugapada* — the yuga-epoch method
- (ii) *Parva* — the lunisolar intercalation method
- (iii) *Trayodaśa-māsa* — the thirteen-month intercalary method
- (iv) *Sāvana* — civil reckoning based on sunrise

Each method is explained with its mathematical basis and astronomical application.

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## Planetary Computations (pp. 790-820)

The commentary provides detailed algorithms for computing mean planetary positions:

### Mean longitude formula :

$$\text{Mean longitude} = \frac{\text{ahargaṇa} \times \text{daily motion}}{\text{civil days in a kalpa}} \mod 360^\circ$$

- *ahargaṇa* = elapsed civil days since kalpa-beginning
- *daily motion* = mean daily motion of the planet
- The result is expressed in degrees of the ecliptic

The commentary discusses the *mandaphala* (equation of centre) and *śīghraphala* (*śīghra* equation) with detailed trigonometric formulas using the *jyā* (sine) function.

*Mandaphala* for the Sun and Moon: the sine of the *manda-kendra* (anomaly) is the multiplier, the radius (*trijyā*) is the divisor.

*Śīghraphala* for Mercury, Jupiter, and Saturn: the sine of the *śīghra-kendra*.

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On the Computation of Weekdays (pp. 820-838)

The final section of this portion treats the computation of weekdays (*vāra-pravṛtti*):

“The sages declare the commencement of the week (to be reckoned) from the sunrise (at Laṅkā); on the earth (at other places), it (is determined) by adding or subtracting the *deśāntara-nāḍikās* (longitude difference in time).”

— Verse 136

“At Laṅkā, the weekday begins with the first *nāḍī* after sunrise; at a place to the west, (it begins) after (sunrise); with the six or eight *nāḍīs* (difference), the weekday is determined when the sun rises. For a place on the eastern or western side of the prime meridian, before or after sunrise (at Laṅkā), by addition or subtraction, the correct time (is found).”

— Expanded v. 136

Hindi Commentary (pp. 836-838):

At Laṅkā, the weekday begins at sunrise in the first *nāḍī*. At a place to the west, after (sunrise) by six or eight *ghaṭīs*.

At a place to the east, the correct time (is found) before sunrise.

The commentator explains that the *vāra* (weekday) begins at sunrise. By this definition, at a place east of the prime meridian (Laṅkā), the weekday begins after subtracting the longitude difference (in *ghaṭīs*); at a place west of the meridian, (the weekday begins) before sunrise by the same amount (longitude difference in *ghaṭīs*).

In the *Siddhānta-śiromaṇi*, Bhāskarācārya states: “*arkodayād dvābhyāṃ bhyas tāniḥ prācyāṃ praticyāṃ dinap-*”—(the computation proceeds from sunrise with modifications for eastern and western positions).

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### **Concluding Verses (p. 838)**

The portion concludes with a verse from the *Manusmṛti*:

“This (universe) existed as darkness, unknowable, without marks, not to be reasoned about, not to be apprehended, as if asleep everywhere.”

### — Manusmṛti 1.5

#### **Commentator’s exposition:**

The commentator explains that Manu has stated that at the time of the deluge (*pralaya*), the perception of weekdays and other time measures is not possible. Therefore, the reckoning of weekdays is only possible during the manifest (*sṛṣṭi*) period. Brahmagupta establishes that the weekday begins from sunrise at Laṅkā and is modified by the longitude difference (*deśāntara*) for other locations.

## Appendix: Structural Overview of the Commentary

The *Vāsanā-vijñāna* commentary on the *Brāhmasphuṭasiddhānta* is organized into the following major chapters (*adhyāyas*):

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*The commentary employs a rich methodology, interweaving:*

- (i) *Mūla-śloka* — The root text verses of Brahmagupta
  - (ii) *Anvaya* — Word-order rearrangement for grammatical clarity
  - (iii) *Bhāṣya* — Sanskrit explanatory commentary
  - (iv) *Hindi Ṭīkā* — Hindi translation and explanation
  - (v) *Upapatti* — Mathematical demonstrations and proofs
  - (vi) *Udāharaṇa* — Numerical examples and worked problems
- 
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*The Brāhmasphuṭasiddhānta  
composed by Śrī Brahmaguptācārya*

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## On the Commencement of the Day

### Commencement of the Weekday

**[Hindi Commentary]** *Prav.rtti* means that which is spoken by Brahmagupta himself. On the question “from when does the weekday commence?” there are many differing opinions among teachers. Āryabha.ta and Siṃhācārya hold that the day begins from the half-risen solar disc. Others hold it begins from sunset. Lāṭadeva and others hold it begins from midnight. The Yavana kings hold that ten *muhūrtas* of night [make the division]. In the *Siddhāntaśekhara*, Śrīpati has refuted the opinions of these teachers.

**[BSS I.36]** When at the beginning of creation the universe was filled with darkness, and the planets had not yet been created, there was no lord of the weekday (*vāreśvara*). Therefore the commencement of the weekday is from sunrise (*udayāt prav.rttiḥ*).

**[Hindi Commentary — Translation]** Before the beginning of creation, the universe was full of darkness. At such a time, discussion of the weekday’s commencement is impossible because the planet-lords of the weekdays did not yet exist. For this reason, the opinion of those who say the weekday commences at midnight is not correct. [Holding that] it begins from midday or from sunset is also not correct. Therefore the opinion of the teachers who hold that the weekday commences at sunrise is indeed correct.

**[Sanskrit Commentary]** In the Sanskrit proof-text (*saṃsk.rtopapatti*) it is known from the verse “The sages state the commencement of the weekday from sunrise at Laṅkā, the city of Rāvaṇa” etc. This is made clear by the verse “From Laṅkā sunrise, south of the prime meridian, east is before, west is after” etc. Thus [ends] verse 36.

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## Longitude Correction

### Rules 37-39: Longitude Difference

**[BSS I.37-39]** The terrestrial circumference (*bhū-paridhi*) is five thousand *yojanas* (*khakha-kha-śara* = 5000). Multiply this by the degrees of difference in latitude (*svāk.sāntarāṃśa*); divide by 360 (*bhaga.nāṃśa*); the square of the result, subtracted from the square of the longitudinal difference (*deśāntara*) [in *yojanas*], gives [the square of] the corrected difference. [Verses 38-39] From the root of the remainder, multiplied by the terrestrial circumference and divided [appropriately], [obtain] minutes (*kalā*); to the east of the Ujjayinī north-south line is positive (*dhana*), to the west is negative (*.rna*). Or from the root divided by the terrestrial circumference and multiplied by sixty, the result is in *gha.tikās*; this is to be applied to the mean or true planet as positive or negative like [the treatment of] a planet.

**[Sanskrit Commentary]** “Terrestrial circumference” (*bhū-paridhi*) means the circumference of the Earth-globe. In answer to “how much?” he says *khakha-kha-śarā* = five thousand *yojanas*. Here the longitudinal difference, latitude, etc., have been explained previously by means of measurements and divisions.

**[Hindi Commentary — Translation]** The terrestrial circumference is five thousand. Multiply this by the latitudinal difference between one’s own place and the standard meridian; divide by 360; the square of that result, subtracted from the square of the longitudinal difference [in *yojanas*]; from the square root of the remainder, multiplying by the planet’s motion and dividing by the terrestrial circumference, one obtains a result in minutes. East of the Ujjayinī north-south line [apply as] positive to the mean or true planet; west as negative. The root multiplied by sixty and divided by the terrestrial circumference gives the result in *gha.tikās*; this is to be applied to the *tithi*-freed *gha.tīs* (i.e., time correction) like a planet.

### Upapatti (Proof/Explanation)

**[Sanskrit Upapatti]** If the diameter measure is 1581, then from the square of the diameter, multiplied by eight, [taking] the square root gives the terrestrial circumference — so it is stated in the *Sūryasiddhānta*. By this rule the terrestrial circumference measure of 5000 is obtained, which is Brahmagupta’s opinion.

**[Upapatti — Translation]** If we assume the diameter measure to be 1581, then by the method stated in the *Sūryasiddhānta*, “the square root of eight times the square of the diameter is the terrestrial circumference,” the terrestrial circumference measure of 5000 is obtained. This same [value] is the terrestrial circumference in Brahmagupta’s system.

## Computation of Days Since Epoch

### Rule 40: Days Elapsed Since Kalpa

**[BSS I.40]** The years elapsed since the beginning of the *kalpa* (*kalpa-gatābdā*), multiplied by 2486 (*rūpā.s.tajina*) and by 7736 (*apavarni-saptanaga*), and divided by 8600 (*kha-svara-sanava*), yield the days and the omitted [tithi]s; the remainders are the fractional parts and defects.

**[Sanskrit Commentary — Vāsanābhā.sya]** At the desired end of a solar year, the years elapsed of the *kalpa* are taken here. The years elapsed of the *kalpa* having been multiplied — he says. By *rūpā.s.tajina* = 2486, and elsewhere by *navārni-saptanaga* = 7736, and then divided by *kha-svara-sanava* = 8600; the results are respectively the days and omitted [lunar days]; the remainders are the fractional parts in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days. There, whatever is the remainder, that is called the *dināṃśa* (fractional part of a day) by the word *dābda*.

**[Hindi Commentary — Translation]** In the desired end of a solar year, the years elapsed of the *kalpa* that are here taken. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days and omitted [days]; the remainders are in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600, the result is days. There, whatever remainder remains, that is called the *dināṃśa* (fractional part of a day).

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## Mean Position of the Sun and Moon

### Rules 24-26: Daily Motion and Latitude

**[BSS I.24]** The minutes of the *manda-phala* (equation of centre), multiplied by the divisor and divided [appropriately], give in order the daily-motion results for the Sun and Moon. In the southern [hemisphere] it is positive, in the northern negative — this is called the *dhruva* (constant). Thus here the staff-head [correction] is reversed; if previously the *vighora-tana-phala* was small, then here [it becomes] solar [correction]. Let astronomers examine the *bhuja* (sine) once, [that of] the delight of all good people; let them examine the *bhū.sa* (ornament) [of calculation].

**[Sanskrit Commentary — Vāsanābhā.sya]** How many revolutions (*bhaga.namāna*) of Prahlāda and the others in a *yuga*, how many elapsed days (*gata-āharga.na*), the computations of weekdays and so on, the computations of mean planets and so forth — in these and other subjects, the first chapter on mean motion (*madhyagati-nāmaka adhyāya*) was composed by me (Brahmagupta) with sixty-*āryā* [verses] (*ṣaṣṭi-pramitā chandasā*); how many subjects there are in the chapter named *Madhyagati*, an enumeration of them has been made. Thus [ends the commentary].

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## Mean Position of Mars and Other Planets

### Rule: Mean Planets Computation

**[BSS — General Rule]** The *manda-phala* (equation of centre) is [obtained] from the *bhuja-jyā* (sine of the anomaly) of the *manda-kendra* and the *pāta-jīvikā* [divisor]; divided by the *ko.ti-jyā* (cosine) and multiplied by the *vyāsārdha* (radius), [it becomes] the *phala-ma.n.dala* (resultant circle/epicycle).

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## Epilogue to the Madhyamādhikāra

### Chapter Conclusion

**[End of Chapter 1]** Thus the first chapter, the *Madhyamādhikāra* (Chapter on Mean Motions), in the *Brāhmasphuṭasiddhānta* [of Brahmagupta].

**[Hindi Commentary — Translation]** The revolutions of the planets and others in a *yuga*, the elapsed *aharga.na*, the computations of weekdays and so forth, and the computations of mean planets and so on — in these and other subjects, the first chapter named *Madhyagati* was composed by means of sixty-three *āryā* [verses]; that is, in the first chapter named *Madhyagati*, however many subjects there are, an enumeration of them has been made. Thus [ends the commentary on Chapter 1].

*[End of the Madhyamādhikāra. This first section, pages 839–893, has been completely explained. It covers mean motions of all planets, day computation, deśāntara (longitude correction), and yugādi calculations.]*

## — Spa.s.tādhikāra

The *Spa.s.tādhikāra* (Chapter on True Positions) is the second chapter of the *Brāhmasphuṭasiddhānta*. Here the purpose of *spa.s.tīkara.na* (making true), the twenty-four sine table (*jyā-pañcaviṃśatikā*), versed sines (*utkrama-jyā*), *mandaphala* (equation of centre), *śīghra-phala* (equation of conjunction), and true daily motion (*spa.s.ta-bhukti*) are presented. This section spans pages 896–978.

## Purpose of Spa.s.tīkara.na

### Rule 1: The Need for Correction

**[BSS II.1]** Since the mean planet is not observed in the zodiac (*bhaga.na*) every day equal to [its mean computation], therefore I shall state the procedure for making the mean [planet] true — the *madhyasphu.tīkara.na*.

**[Sanskrit Commentary — Vāsanābhā.sya]** First, the chapter on true motion (*sphu.ta-gaty-adhyāya*) is being explained. There the purpose of the commencement is stated. Since a planet equal to the mean planet is not observed in practice in the zodiac (*bhaga.na*, the nak.satra circle) day by day, therefore I shall state the *sphu.tīkara.na* (procedure for making true). In answer to “what is the defect of the mean?” he says: the mean planet is hypothetically located (*parikalpita*) on the *kak.sā-ma.n.dala* (deferent circle). And the real planet is not on the *kak.sā-ma.n.dala*; revolving on the *prati-ma.n.dala* (epicycle) with mean motion, it is observed on the *kak.sā-ma.n.dala* with true motion. Therefore, by whatever computation the planet situated on the *prati-ma.n.dala* may become equal to [the observed position] on the *kak.sā-ma.n.dala*, that *sphu.tīkara.na* I shall state. Thus [ends commentary on] verse 1.

**[Hindi Commentary — Translation]** For the reason that the true planet in the *krānti-v.rtta* (ecliptic) is not seen every day equal to the mean planet, for that reason I (Brahmagupta) call that procedure which makes [it] equal to the observed [position] the *sphu.tīkara.na*. The mean planet is hypothetically [placed] on the *kak.sā-v.rtta* (deferent); the planet, revolving with true motion on the *prati-v.rtta* (epicycle), is observed on the *kak.sā-v.rtta*. Therefore, by whatever computation the planet situated on the *prati-v.rtta* becomes equal [to the observed position] on the *kak.sā-v.rtta*, that *sphu.tīkara.na* I state. Thus [ends commentary on] verse 1.

## The Twenty-Four Sines

### Rules 2-9: Sine Table in 24 Steps

#### Introduction to the Jyā-pañcaviṃśatikā

Brahmagupta gives twenty-four *krama-jyās* (sines in sequence), each arc being  $3\frac{3}{4}^\circ$  ( $= 225'$ ), filling the complete quadrant of  $90^\circ$ . These sines are expressed poetically through *bhūta-saṅkhyās* (word-numerals), displaying Brahmagupta's literary talent alongside his mathematical genius.

#### [BSS II.2-9] The 24 Sines

The twenty-four sines are encoded in *bhūta-saṅkhyā* (word-numeral) verses. Each word corresponds to a numerical value:

**Verses 2-9:** *ṣaḍ-bud-agni* ( $6+14=20$ ) = 214; *suta-agni* ( $7+3=10$ ) = 427; *rasa-dvādya* ( $6+12=18$ ) = 638; *munīndu* ( $7+1=8$ ) = 846; *vasu-candra* ( $8+1=9$ ) = 1051; *indunava-nava-candra* ( $1+9+9$ ) = 1251; *rasa-tithi-yamala* ( $6+15+1$ ) = 1446; *raviśriyamā* ( $12+1$ ) = 1624.

**Verses 3-9** continue the sequence up to the 24th sine = 3270 (the full radius).

#### Krama-jyā Table (Sines in Sequence)

**[Sanskrit Commentary]** Their diameter-measure is the radius. “These are the *aghe-jyā*” (sines) — thus. *Sanu-yamalā* = 214; this is the form of the first sine. *Suniya-mavedā* = 427, the second. *Vasu-jvalana-gha.tikāḥ* = 638, the third. *Rasa-k.rta-vasavaḥ* = 846, the fourth. *Śaśi-pañcha-khendavaḥ* = 1051, the fifth. *Candra-dara-sūryāḥ* = 1251, the sixth.

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## Utkrama-jyā (Versed Sine) Values

### Utkrama-jyā Table (Versed Sines)

The *utkrama-jyās* (versed sines) are constructed by the reversed order of the half-chords. In modern trigonometry these are known as “ $1 - \cos$ ” (versed sine = radius – cosine).

**[Hindi Commentary — Translation]** The twenty-four half-chords (*ardhajyā*), [named] *muniyamalā*, *muniyamavedā*, etc., are of twenty-four numbers, as written below. The relation of *vyastā* (reversed, i.e., versed sine) is given after. The twenty-four *ardhajyās* (*krama-jyās*) are: 214, 427, 638, 846, 1051, 1251, 1446, 1624, 1817, 1991, 2145, 2312, 2459, 2594, 2832, 2933, 3021, 3096, 3145(?), 3207, etc., 3263, 3270.

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## Arc-Jyā Conversion

### Rule: Finding Arc from Jyā

**[BSS — Arc-Jyā Rule]** From the *jyārdha* (half-chord, sine), divided by the *trijyā* (radius), the square root gives [the first approximation], slightly different for odd and even [intervals]; from that the arc is obtained. After that, again [taking] the square root of [the sine] combined with its square [correction], [the more precise arc is obtained].

**[Sanskrit Commentary]** The twenty-four versed sines (*utkrama-jyā*), [named] *muno's.tayamā* etc., are of twenty-four numbers, as written below: 7, rad (29), 62, 111; 174, rau (292), 337, 435, rabha1 (521), 676, 811, rau (985), 1114, 1296, 1453, 1635, 1824, 2019, 2214, 2424, 2632, 2543(?), 3056, rera70 (3270).

**[Hindi Commentary — Translation]** The *jyārdha* (half-chord), divided by the *trijyā* (radius), from whatever root (*pada*) is obtained, slightly differing for odd and even [intervals], that becomes the arc. After that, again, having combined [the correction] with the square of that same root, the next arc is obtained.

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## Manda and Śīghra Equations

### Rules 41-44: True Motion Computation

#### Introduction to Manda-phala and Śīghra-phala

The *manda-phala* (equation of centre) and *śīghra-phala* (equation of conjunction) are the two principal corrections for making the planet true. The *manda-phala* is computed from the *manda-kendra* (anomaly of the epicycle); the *śīghra-phala* from the *śīghra-kendra* (anomaly of conjunction).

#### [BSS II.41-44]

**41:** The *bhuja-jyā* (sine of anomaly) of the *manda-kendra*, multiplied by the *pāta-jīvikā* and divided by the *ko.ti-jyā* (cosine), [then] multiplied by the *vyāsārdha* (radius), gives the *manda-phala* (equation of centre) and the motion.

**42:** From that, first the small or second [epicycle], or from the *manda-saṃsk.rta* (mean corrected by manda), from the mean, one should give the *manda-phala*, or again by the *śīghra-bhukti*.

**43:** Taking the *manda-saṃsk.rta*, or again by the *śīghra-bhukti*, one should then compute the *śīghra-phala*, combined by proportion (*trairāśika*).

**44:** Having established the *śīghra-phala*, the true velocities (*sphu.s.t.a-gatikaḥ*) of the planet are to be brought together; then the wise [astronomer] should compute the *vakra-bhukti* (retrograde motion).

**[Sanskrit Commentary]** Multiplied by the *manda-kendra-gati-jyāntara* (*bhogyakha.n.da*, the tabular difference), divided by the *prathama-jīvā* (first sine); whatever result is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* etc., [when this is] less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and the other planets, the mean motion minus the *manda-gati-phala* is called *manda-sphu.ta-gati* by the teachers.

**[Hindi Commentary — Translation]** The *manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā* (*prathama-jyā*); the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and other planets, the mean motion minus the *manda-gati-phala* is the *manda-sphu.ta-gati*. The *śīghrocca-gati* minus its own *manda-sphu.ta-gati* is the *śīghra-kendra-gati* of Mars and other planets.



## The Manda Paridhi (Epicycle) Values

### Rules 20-21: Manda Epicycle Corrections

#### [BSS II.20-21]

**20:** The difference between that [quantity] and the half-diameter of the epicycle (*dyudala-paridhi*), multiplied by the *trijyā* (radius) and by the *nata-jīvā* (sine of altitude), [when the planet is] in decrease (*ūne*) is positive (*dhana*), in increase (*adhike*) is negative (*.rna*); this is the *sphu.ta-paridhi* (true epicycle) of the half-day.

**21:** For the Sun, that is the *manda-paridhi* in the *.rna-manda-phala* or *dhana-manda-phala* at midday. For the Moon, the same, and by those same [quantities] in *gha.tikās*, the Moon also [is corrected] at that same *kapāla* (division); thus the *nata* (altitude) is obtained.

**[Sanskrit Commentary]** Fourteen degrees (*catudaṃśāṃśāḥ*), in two places, diminished by two-thirds of a degree (*bhāgadwayaṃśena rahitāḥ*) — then in the *.rna* or *dhana manda-phala* of the Sun; at midday (*dinamadhye*) the *manda-paridhi-aṃśa* are obtained. In the *.rna* or *dhana manda-phala*, before the half-day *kapāla*, for the Sun separated by fifteen *gha.tikās*, the half-day *paridhi* measure is to be made successively increased or decreased by two-thirds of a degree. For the Moon, in the *.rna* or *dhana manda-phala*, twelve [degrees] diminished by two *jina-lipata* (?) — that is, in two places, diminished by thirty-two degrees plus twenty-four minutes (*dvātriṃśadaṃśākhyacaturviṃśatikalā rahitāḥ*) — then at midday those *manda-paridhi-aṃśa* are obtained.

**[Hindi Commentary — Translation]** From fourteen degrees, in two places, diminishing by one-third of a degree (20 *kalās*), in the *.rna-manda-phala* or *dhana-manda-phala* of the Sun, at midday the *manda-paridhi-aṃśa* is obtained. In the *.rna-manda-phala* or *dhana-manda-phala*, in the *kapāla* before half-day, having subtracted fifteen (14?) *gha.tikās* from the *nata* Sun, in the midday *manda-paridhi* measure, successively one-third of a degree (20 *kalās*) is to be added and subtracted.

## Summary of Manda Paridhi Values

### Manda-paridhi Summary Table

Below is presented a numerical summary of the *manda-paridhi* (epicycle) values. For each planet the *manda-paridhi* varies up to twenty-four degrees.

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## True Daily Motion

### Rules: Daily Motion After Correction

**[BSS — True Daily Motion]** The difference between the *śīghra-kendra-sphu.ta-bhukti* and [the mean motion], multiplied by the [tabular difference] and corrected from the stated quotient — this is the *śīghra-gati*; [if negative] it is *vakra-gati* (retrograde motion). Thus [goes] the proof-verse of Sinhavarman.

**[Hindi Commentary — Translation]** The *śīghra-kendra-sphu.ta-bhukti-antara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā*; the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

### Geometric Explanation (Upapatti)

**[Upapatti — Geometric Proof]** By the difference between mean and true, motion becomes threefold. That motion which is different every moment is *spa.s.tā*; another is *madhyā* (mean). The true motion (*sphu.tā gati*) is also twofold by the division into daily and instantaneous: thus the daily *manda-sphu.tā-gati* and the instantaneous *manda-sphu.tā-gati*. The daily *spa.s.tā-gati*, the instantaneous *spa.s.tā-gati* — by the teacher the daily *manda-sphu.ta-gati* and the daily *sphu.ta-gati* are computed.

The *graha-manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied by the *prathama-jyā* and divided [by the radius]; whatever quotient is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* (or *kkayādi-kendra*), whether less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

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## Special Corrections for Different Planets

### Planet-Specific Corrections

#### Mercury and Venus

**[Sanskrit Commentary]** For Mercury and Venus, the *śīghrocca* (apsis of the *śīghra* epicycle) is [derived] from the Sun; from the seizure of the Sun [it is different], otherwise [it would be the same]. Their *śīghrocca* is connected to the Sun; the *mandocca* (apsis of the *manda* epicycle) is independent — this is the special feature.

**[Hindi Commentary — Translation]** The *śīghrocca* of Mercury and Venus is related to the Sun itself; their *śīghrocca* is Sun-related, the *mandocca* is independent. Therefore for these two planets a special correction procedure is necessary.

#### Mars , Jupiter , and Saturn

**[Sanskrit Commentary]** For Mars and the others, the *manda-sphu.ta-gati* and the *sphu.ta-gati*. From the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti* [is obtained]; the *śīghra-phala-ādha-bhogya-jīvā*, multiplied by the *madhya-jivyā* (mean sine), is to be divided. Having brought together the *sphu.ta-bhukti* by the stated quotient; if together with that, by the *manda-phala-ādham-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

**[Hindi Commentary — Translation]** For Mars and other planets, the *manda-sphu.ta-gati* and *sphu.ta-gati* — from the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti*; the *śīghra-phala-adha-bhogya-jīvā*, multiplied and divided by the *madhya-jīva*. By the resulting [quotient], the *sphu.ta-bhukti* is to be brought together as stated; if together with that, by the *manda-phala-adha-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

## The Chapter Conclusion

### Epilogue to the Spa.s.tādhikāra

**[End of Chapter 2]** Thus the second chapter, the *Spa.s.tādhikāra* (Chapter on True Positions), in the *Brāhmasphuṭasiddhānta*.

**[Hindi Commentary — Translation]** In this manner, the explanation of the *Madhyamādhikāra* (first chapter) and the *Spa.s.tādhikāra* (second chapter) in the *Brāhmasphuṭasiddhānta*, composed by Ācārya Brahmagupta, has been completed. In this, the mean motions of planets, *sphu.tīkara.na*, sine mathematics (*jyā-ga.nita*), *manda-phala*, *śīghra-phala*, *deśāntara*, *nata-karma*, and other subjects have been described in detail.

*[The Spa.s.tādhikāra continues with detailed calculations for each planet's manda and śīghra corrections, tables of paridhi values, geometric upapattis (proofs), and special rules for Mercury and Venus across pages 896-978 of the original edition.]*

*Spaṣṭādhikāraḥ — Chapter on the Determination of True Positions*

**English Translation**

This chapter deals with the methods for computing the true (corrected) positions of planets from their mean (average) positions. The corrections account for the eccentricity of planetary orbits and the epicyclic models described in Indian astronomy.

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**English Translation**

**Verses 48-49. Śīghra-spaṣṭikaraṇa (True Position via Śīghra Correction):**

Having taken half the śīghra-kendra (anomaly from the śīghrocca) of the traversed arc, and half of the retrograde-direct kendra of that same planet, knowing its retrograde-direct period, the true planet should be computed. The construction is: the square-root of the sum of the squares of the śīghra-kendra retrograde-direct quantities is the divisor. The vertical line (koṭi) from the centre of the śīghra-nīcocca-vṛta (epicycle) up to the mean position equals the parama-phala-jyā (maximum equation sine); the horizontal (bhuja) at the centre of the śīghra-nīcocca-vṛta is the product of the east-west components, whose square-root is the triñṇaka-karṇa. The parama-phala-jyā is declared to be proportional to the śīghra-nīcocca-vṛta-śālākā (radius) on the circumference of the perimandala. That also, when the planet is at the highest point on that perimandala, causes parallax (paracchāda). Therefore, in an iron globe one should demonstrate everything.

**Commentary Translation**

[Commentary of Varāhamihira:]

For Mars and Jupiter, the retrograde motion begins at specific śīghra-kendra añśa values. The manda-spaṣṭa (true position via manda correction) and the śīghra-spaṣṭa (true position via śīghra correction) are computed. The śīghra-kendra-gati is the velocity at the śīghra-kendra. The days traversed and the remaining time are computed from these kendra values. By the śīghra-kendra divisions, the ucca positions remain stable.

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**English Translation**

**Verses 12-13. Planetary Mean Motion (Madhyamāgama):**

First, having multiplied the zodiacal signs (bhacakra-bhāga, 43,200 minutes of arc) by the given days, divide by the Kalpa or Yuga days [1,577,917,828,000]. The quotient obtained is the pūraka-kṣepa (accumulator of revolutions).

That same pūraka-kṣepa, multiplied by the elapsed days and divided by the solar years, gives the mean daily motion (madhya-bhukti). That, added to or sub-

tracted from the previous mean longitude, gives the mean position (madhyama) of the planet.

### Commentary Translation

*[Varāhamihira's Commentary:]*

The zodiacal signs (bhacakra-bhāga), that is 43,200 minutes of arc, are first multiplied by the given days and divided by 1,577,917,828,000 solar years. The quotient obtained is the pūraka-kṣepa. That same pūraka-kṣepa, multiplied by elapsed days and divided by solar years, gives the mean daily motion (madhya-bhukti). That madhya-bhukti, added to or subtracted from the previous mean position, gives the mean (madhyama) of the planet.

### English Translation

#### Verses 14-15. Mean Apogee and Śighrocca:

The mandocca (apogee of manda epicycle) and the śighrocca (apex of śighra epicycle), divided by the yuga-pūraka-kṣepa and multiplied by elapsed days, gives the mandocca or śighrocca.

That mandocca or śighrocca, when added to or subtracted from the previous mandocca or śighrocca of the yuga, gives the current manda-śighrocca.

The mandocca and śighrocca, multiplied by elapsed days and divided by the yuga-pūraka-kṣepa, gives the result. Adding or subtracting this from the previous mandocca or śighrocca gives the current values. In this manner, knowing the planet's madhyama, mandocca, and śighrocca, the foundation for computing the true position is established.

### English Translation

#### Verses 16-17. Manda-spaṣṭikaraṇa (Equation of Centre):

The maximum manda-phala (equation of centre) of the planet, diminished by the mandaka (eccentric anomaly divisor), divided by the half-sine (jyā-ardha), gives that maximum phala of the planet.

Multiplying the manda-phala-jyā by the parama-phala-jyā, the result obtained is the parama-phala (maximum equation). By this, the planet becomes manda-spaṣṭa (corrected via the equation of centre).

The mandaka is the square-root of the sum of squares of the doḥ (sine) and koṭi (cosine). The manda-jyā multiplied by the trijyā and divided gives the manda-phala. That parama-phala, when subtracted from or added to the mean position, gives the manda-spaṣṭa (true) position of the planet.

### English Translation

#### Verses 50-51. Planetary Path and Visibility:

Having fixed the path regulation to the epicycles and instruments, from that the installation in the kakṣa-maṇḍalas (orbits) is established. These rising and setting portions, when undivided [in the orbit], present the characteristics of the maṇḍala divisions; that difficulty in rising and setting will arise when the planet is divided [by epicyclic motion].

The śīghra-kendra values for Mercury and Venus are described. When Mercury's śīghra-kendra equals 50, it rises after the Sun; when it equals 24, it rises before [specific values for visibility]. The rising and setting are reversed relative to the orbital path.

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### *Tripraśnādhikāraḥ — Chapter on the Three Questions of Diurnal Motion*

This important chapter deals with the “three questions” (*tripraśna*) of spherical astronomy: determining the *lambana* (parallax in longitude), the *nāḍī* difference (time difference), and related quantities needed for computing solar and lunar phenomena.

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### English Translation

#### **Verses 54-55. Rising and Setting Śīghra-kendra Values:**

Now, the śīghra-kendra añśa values for the rising and setting of Mars and the others: When Mercury's śīghra-kendra equals 28 [bhauma], 14 [cakre], 17 [muni-nagendra], then the rising of Mars, Jupiter, and Mercury occurs; the setting is opposite to these.

[Values: 50 in Svātī, 24 in Jina, 155 in the desired tithi, 177 in Muni-nagendra] The rising and setting are reversed relative to the orbital path, the forward one being retrograde.

The śīghra-kendra determines the heliacal rising. At 28 degrees, Mars rises before the Sun; at 14, [a value for] Jupiter; at 17, Saturn. The heliacal settings are opposite. These kendra values are well-known; for Mars, the setting śīghra-kendra is 332; for Jupiter, 385; for Saturn, 353.

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When the elapsed [days' values] of the *bhā* (signs) and minutes of arc, joined with the manda-phala-spaṣṭa and śīghra corrections, are divided by the days — thus is the derivation by this method. Still, since all [planets] are corrected by the solar orbit, the planet revolves in the nīcocca-vṛta (epicycle) in that time.

When the planet is situated at the highest point of the perimandala, then there is maximum perturbation; at that time, since the Sun is nearby, the planet is not observed. The rising and setting are first seen when the śīghra-kendra is appropriate; then, even at the Sun's setting, the epicyclic motion having changed, the planet gradually attains reduction [in elongation]. Therefore, the planet is observed at rising and setting.

---

#### **Verses 58-61. Declination and Shadow Calculations:**

The *jina-māga-jyā* (24-degree sine) multiplied by the solar *jyā* and divided by the semi-diameter gives the result. The desired ascensional difference (*kranti-jīvā*) from the equinox is for the Sun.



At the desired arc, subtracting from the square of the trijyā, the remainder's root, from the equinox semi-duration, the day-duration is obtained.

The declination-sine (krānti-jyā), multiplied by the equinoctial shadow and divided by 12, gives the *sthiti-jā*. The day-and-night prāṇa, multiplied by the semi-diameter, are considered.

The day-and-night half combined with the ascensional difference gives the arc-measure. Those [prāṇas], separated, are the divided [portions] of the nāgikā [serpent of time].

---

### Verses 62-63. Twelve-Degree Sines and Ascensional Differences:

The sines at twelve-degree intervals are to be computed; for the Earth etc. 1325 [values apply]. The difficult [computations] yield 1863; for Mithuna [Gemini], the vomiting-sines are 3030. For the elapsed [periods], the feeble [sines] are obtained. The *jina-māga-jyā* multiplied by the solar jyā gives the *pumānu-kṣoṇi-jyā* (terrestrial sine). The Mithuna values at 66°30', Mercury's [value] at 19°59', and the nine-fold cycle at 13°30'. By these, the equinoctial shadow is computed. These, combined with the ascensional difference for one's own location, give the prāṇas.

---

### Verses 64-67. Aries-to-Gemini Sine Computation:

Now, the computation of *cara-khaṇḍa* (ascensional difference segments) for Aries etc.:

For Aries, Taurus, and Gemini, the same fields apply; for Cancer, Leo, and Virgo, the same fields; and for Libra, Scorpio, and Sagittarius, and similarly for Capricorn, Aquarius, and Pisces.

The day-and-night durations should be shown for each; in the Spāṣṭādhikāra itself, I have demonstrated the cara-dala computation for the learned. When, at a location, the prāṇas between the terrestrial sine and shadow are computed, this interval is the time-difference inquired for the lagna (ascendant) computation.

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### Verses 28-31. Shadow and Gnomon Formulas (Chāyākaraṇī):

The altitude-sine (hr-jyā), multiplied by 12 and divided by the śaṅku (gnomon), gives the shadow (chāyā). The square of the semi-diameter divided by the equinoctial hypotenuse gives the hypotenuse (karṇa).

The product of the equinoctial hypotenuse and the koṭi, divided by the square of the semi-diameter, gives the result. In the north (sāumya), diminished by the sun's diameter-sine; in the south (yāmy), added — this is the arc-measure of the bow.

In the north, added; in the south, diminished — these are the east-west prāṇas. The elapsed specific portions; the remainder of the excess portion is the difference. Half the day-sine is the east-west *nata-prāṇa* (hour-angle in time units).

Having computed the chāyā-karṇa (hypotenuse of gnomon and shadow), multiply by the day-and-night. Then, divide by the day-and-night half and the chāyā-karṇa; the resulting quantity, compared with the square of the semi-diameter, gives the product of the equinoctial hypotenuse and the koṭi as a sine.

In the northern hemisphere, subtract the *chāyā-buddhi-jyā*; in the southern, add similarly. The arc of this result is computed, then with the *cara-dala-prāṇa*, in the northern sphere added and in the southern subtracted, giving the east-west *prāṇas*.

---

### Verses 32-33. Alternative Shadow Computation:

The square of the semi-diameter divided by the equinoctial hypotenuse gives the hypotenuse (*karṇa*). Or, the product of the *lambaka* (co-latitude) and the quantity, divided by the square-root of the sum of the squares of the gnomon and its semi-diameter.

Or the product by the gnomon-multiplication, divided by the equinoctial hypotenuse, gives the *śaṅku*. The square of the hypotenuse, subtracted [from the appropriate quantity], its root divided by 12 gives the shadow.

---

### Verses 34-36. Computation of Nata-kāla (Hour Angle):

With the day-and-night half and the *chāyā-karṇa* hypotenuse as multipliers, the equinoctial hypotenuse times the *koṭi*, divided by the square of the semi-diameter, gives the result in the north.

Diminished by the sun's diameter-sine in the north, added in the south — this is the arc-measure. Added in the north, diminished in the south — these are the east-west *prāṇas*.

The elapsed specific portions; the remainder of the excess portion. Half the day-sine is the east-west *nata-prāṇa* (hour-angle).

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### Verses 57-62. Latitude-related Sine Computation (Lambhākṣa-jyā):

The *śaṅku* (gnomon), known as the *lambaka-chāyā*, multiplied by the northward [quantity], gives the result. The equinoctial hypotenuse [gives] the purity; the highest latitude-*śaṅku* is the *aksa-ucca-natā*.

The elevated sine's *koṭi-jyā*, the altitude-sine is the *bhuja* or the *nata-jyā*. The semi-diameter [applies to] the *karṇa*-field-circle.

The *śaṅku* and the *kṣaya* quantities, with the aggregate of eight *puruṣas*, become the *lambhākṣa-jyā*; its degrees and minutes are the subsequent [measures].

Divided by the equinoctial hypotenuse, or the *śaṅku* divided by the *kṣaya* separately with the *trijyā*, in the latitude field, the *lambhākṣa-jyā* is elevated.

Having subtracted the *nata-lambhākṣa*, the sines [yield] difficulty. The *śaṅku* multiplied by *kṣaya*, divided by twice the *kṣaya*, remains.

The *lambhākṣa* degrees and minutes' remainder, the root of the square of the *trijyā*, is to be divided. The *bhuja-jyā* of all elevated differences is single.

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**Verse 91. Alternative Lambhākṣa-jyā Computation:**

For computing the lambhākṣa-jyā (latitude sine), having divided from 90 degrees, the sine is to be computed. Or alternatively: the jyā of the difference between 90 degrees and the latitude degrees is the lambaka-jyā. The aksa-lambaka-jyā [is obtained from] the product of the śaṅku and chāyā (gnomon and shadow), multiplied by 12 and divided by the shadow. Thus the lambhākā-jyā is obtained by an alternative method.

*Derivation:*

The sine of (90 - latitude) = lambaka-jyā, the sine of (90 - latitude) = madhya-jyā. Or:

$$\frac{\text{aksā-jyā} \times 12}{\text{palamā}} = \text{lambaka-jyā}, \quad \frac{\text{palamā} \times \text{lambaka-jyā}}{12} = \text{madhya-jyā}$$

This is the derivation of the aksa and lambaka.

**[91]**

**Verses 63-66. Computation of Nata-kāla (Hour Angle):**

Having multiplied the udayāstamaya (rising-setting) by the twelve akṣepa (declination), from the yāmyodaya (southern rising), the result is the *cala* and the true difference of the doḥ-jyā (altitude sine difference).

Proclaiming that maṇḍala, multiplying the result by the śaṅku or lambaka, divided by the dyu-jyā (day-radius), gives the nata-kāla (hour angle).

The doḥ-jyā-utkānti-jyā (altitude sine minus declination sine), diminished by the arka-kṣepa, divided by the nata-kāla, multiplied by the trijyā and divided by the dyu-jyā, gives the desired śaṅku.

The product of the nata and utkānta-jyā, divided by the square of the dyu-jyā, that portion is the śaṅku. The shadow multiplied by 12, its root divided [appropriately] gives the kṣepa.

Having multiplied the udayāstamaya by the twelve akṣepa from the yāmyodaya, the result is the *cala* and the true doḥ-jyā-vivara. Proclaiming that maṇḍala, multiplying the result by the śaṅku or lambaka and dividing by the dyu-jyā, gives the nata-kāla. The doḥ-jyā-utkānti-jyā, diminished by the arka-kṣepa and divided by the nata-kāla, multiplied by the trijyā and divided by the dyu-jyā, gives the śaṅku. The product of the nata and utkānta-jyā, divided by the square of the dyu-jyā, gives the śaṅku; the shadow multiplied by 12, its root gives the kṣepa.

**Verses 137-139. Construction of the Armillary Sphere (Gola-yantra):**

The celestial sphere (khagola) has seven rays [circles]; in its centre is situated the terrestrial sphere (bhūgola). Joined by rods (daṇḍaiḥ), this khagola is in the form of an instrument (yantra-rūpaka).

The ecliptic (krānti-vṛtta), the equator (sama-maṇḍala), the north-south circles (dakṣiṇottara-vṛtta), and the prime vertical (sadyat-kṣitija) should be constructed in the sphere by those who know.

By this instrument alone, the true position of the planets is known. Therefore, the instrument should always be constructed by those who know the sphere (gola-vid-bhiḥ).

The celestial sphere with seven circles, having the terrestrial sphere at its centre, joined by rods, is this armillary sphere. The ecliptic, equator, meridian circles, and prime vertical

should be constructed in the sphere by those versed in spherics. By this instrument alone, the true positions of the planets are known; therefore, astronomers should always construct this instrument.

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### Verses 128-130. Time Unit (Prāṇa) Computation:

Sixty ghaṭikās make one nāḍī; sixty nāḍīs are called prāṇas; sixty vipalas make one pala — by these units the learned compute time.

Having multiplied the cara-jyā (ascensional difference sine) and divided by the trijyā, the result gives the prāṇas. Their sum is the sthiti-kāla (duration of visibility); the ayanakāla is similarly computed.

Having divided the tanu-jyā (sine of zenith distance) and dividing by the trijyā, the result is the pramāṇa-kāla (measure-time). In the sthiti (position), dividing by the cara-jyā, those prāṇas are the true ones.

Sixty ghaṭikās equal one nāḍī; sixty nāḍīs are prāṇas; sixty vipalas equal one pala — these are the time units used by the learned. Multiplying the cara-jyā by the trijyā and dividing gives the prāḍnas. Their sum is the sthiti-kāla; the ayanakāla is similarly computed. Dividing the tanu-jyā by the trijyā gives the pramāṇa-kāla; in the position, dividing by the cara-jyā gives the true prāṇas.

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### Verse 148. Computation of Midday Shadow:

By the proportion of the chāyā-field:

$$\frac{\text{tri} \times 12}{\text{śāṅku}} = \text{chāyā-karṇa}$$

$$\sqrt{(\text{chāyā-karṇa})^2 - 12^2} = \text{midday shadow}$$

[148]

[Hindi commentary:] Multiplying the trijyā by 12 and dividing by the midday śāṅku gives the midday chāyā-karṇa. The square-root of the difference of squares of the chāyā-karṇa and 12 gives the midday shadow by an alternative method. [148]

*Derivation:* From the similarity of the chāyā-fields demonstrated in the previous verse's derivation, by proportion:

$$\frac{\text{tri} \times 12}{\text{midday śāṅku}} = \text{midday karṇa}$$

Therefore  $\sqrt{(\text{chaya-karṇa})^2 - 12^2} = \text{midday shadow}$ .

[148]

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### Verses 140-142. The Mahāpāta and Lunar Month:

The mahāpāta (great conjunction) occurs at the meeting of the Moon and Sun; then their difference becomes zero when they are at the same ecliptic point.

The lunar month (cāndra-māsa), by the aggregation of tithis (lunar days), is also a sāvana month, traversed by the Sun alone.

The combination of these two months is called a saṃ

The mahāpāta (great conjunction) occurs when the Moon and Sun meet; then their difference is zero when they are at the same ecliptic longitude. The lunar month, by the collection of tithis, is also a sāvana month, traversed by the Sun. The year (saṃ

### Verses 131-133. Declination (Krānti) and Ayanāṃśa:

When the Sun is at the beginning of Meṣa (Aries), multiply by 66 [the kāla value]. When the Sun is at the equinox (viṣuvat), at the beginning of Meṣa, the declination-sine (krānti-jyā) there is considered as the equinoctial sine (viṣuva-jjyā).

That, diminished by its kṣepa, is to be known as the true declination by those who compute. In the uttarāyaṇa (northward course), it is southern; by those kṣepas, it becomes northern.

The ayanāṃśa (precession of equinoxes) is that from which the declination arises by differentiation. Therefore, knowing the ayanacalana (precessional motion), the true [position] should be computed by the prescribed method.

When the Sun is at the beginning of Aries, multiplied by 66; at the equinox, the declination at the beginning of Aries is considered the equinoctial sine. That, diminished by its kṣepa, is the true declination. In the northward course, the declination is southern; in the southward, northern. The ayanāṃśa is that from which the declination differs; knowing this precessional motion, the true [position] is computed properly.

### Kṣitijyā (Terrestrial Sine):

Well-known: In the southern [hemisphere], the southern bhuja's elevation, the product of the lambaka and the appropriate quantity, divided by the chāyā-karṇa, gives the result. From this, the krānti-jyā (declination-sine) is obtained.

### Verses 134-136. Computation of the Lagna (Rising Sign):

From the rising-time (rāśi-udaya) of the Sun's sign, the ghaṭikās of the currently rising sign, having subtracted those from the given time, until the sign-risings are completed.

Those [signs], joined with the previous signs, constitute the lagna. Dividing the remainder by 30, the result is in degrees and minutes.

That should be added to the sign; its lagna is thus computed by those who compute true positions. Thus the lagna should always be computed by the wise who know planetary mathematics.

From the rising-time of the Sun's sign, the ghaṭikās of the currently rising sign, subtracted from the given time until the sign-risings are accounted for. Those signs, joined with the previous ones, give the lagna. Dividing the remainder by 30 gives the degrees and

minutes, which are added to the sign; this gives the lagna for true-position computation. Thus the lagna is always to be computed by wise planetary calculators.

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### *Key Mathematical Formulas from the Spaṣṭādhikāra and Tripraśnādhikāra*

This section presents the key mathematical formulas from the Spaṣṭādhikāra and Tripraśnādhikāra chapters, covering sine table construction, shadow calculations, hour angle computation, and arithmetic operations.

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#### **Śighra-spaṣṭikaraṇa Formula:**

The śighra-phala-jyā (anomaly equation sine) multiplied by the śighra-karṇa gives the true position correction.

The kendra-kardṇa (eccentric distance) is computed as the square-root of the sum of the square of the trijyā and the square of the manda-phala-jyā, minus twice the product of the manda-phala-jyā and the kalpa-jyā.

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#### **English Translation**

##### **Verses 100-101. Construction of the Sine Table (Jyotpatti):**

From the diameter (vyāsa), having subtracted three from the diameter-parts [3438-3 = 3435], made into parts and divided by six, the remainder is taken as the trijyā (radius sine,  $R \approx 3438'$ ).

Having multiplied the previous previous-jyā by the previous previous-difference-segments, and dividing by the trijyā, the other jyās are obtained; they are considered to originate from the previous one.

From the diameter, subtracting three from the diameter-parts gives 3435 parts. Dividing by six, the remainder is taken as the trijyā. With this trijyā, multiplying the previous previous-jyā by the previous difference-segments and dividing by the trijyā, the other jyās are obtained; they are considered to originate from the previous jyā.

#### **English Translation**

##### **Verses 102-103. Sine Computation Rule:**

The first jyā is a trijyā-segment; the second from half of the remainder; the third also likewise — this is the rule of Brahmagupta's method.

The sum of their difference-segments is remembered as the jyā-āvalaya (sine-circle). The six-degree interval jyās are well-known among accurate calculators.

The first jyā is the 7.5-degree sine, a trijyā-segment; the second is the 15-degree sine; the third similarly in sequence. The sum of their difference-segments is the jyā-āvalaya. The six-degree interval sines are well-known among accurate calculators.

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**Chāyākaraṇī Formulas:**

**Shadow:** The shadow equals 12 (gnomon height) multiplied by the altitude-sine, divided by the śaṅku (gnomon's effective height).

**Hypotenuse:** The hypotenuse equals the square of the semi-diameter divided by the equinoctial hypotenuse.

**Shadow-hypotenuse:** The hypotenuse of gnomon and shadow equals the square-root of the sum of squares of the shadow and the gnomon.

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**English Translation****Verses 104-106. Second-Difference Interpolation (Gaurī-guptā):**

Having subtracted the square of the bhuja-jyā (sine of arc) from the square of the jyā, divided by the square of the trijyā, and having subtracted the result from the first jyā, the next jyā is obtained.

The difference between the previous and subsequent jyā is the first segment (khaṇḍa); from that, the next segment; from that, having subtracted the previous segment, the third segment is obtained.

Thus, in sequence, all jyās should be computed from the divided jyā. The difference-jyās separately yield a more accurate result.

**English Translation****Verses 107-108. Difference Segments for Sine Computation:**

For odd [positions], having subtracted the square of the dvi-jyā (double sine) from the square of the dvi-jyā, the root from the remainder is the second segment. Then, having subtracted the previous segment, [the third is obtained].

The third segment, then having subtracted from the previous two, the fourth; then having subtracted from the previous three, thus in sequence all segments are obtained.

Having subtracted the square of the bhuja-jyā from the square of the jyā, divided by the square of the trijyā, and subtracting the result from the first jyā, the next jyā is obtained. The difference between previous and subsequent jyā is the first segment; the next segment from that; having subtracted the previous segment from that, the third segment is obtained. Thus, in sequence, all jyās are computed from the divided jyā; the difference-jyās separately give a more accurate result.

For odd positions, having subtracted the square of the dvi-jyā from the square of the dvi-jyā, the root of the remainder is the second segment. Then, having subtracted the previous segment, the third; having subtracted from the previous two, the fourth; from the previous three, the fifth; thus in sequence all segments are obtained.

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**Lambhākṣa-jyā Formulas:**

**Lambaka-jyā (co-latitude sine):** The aksa-jyā (latitude sine) multiplied by 12, divided by the palamā (equinoctial shadow), gives the lambaka-jyā.

**Madhya-jyā (meridian sine):** The palamā multiplied by the lambaka-jyā, divided by 12, gives the madhya-jyā.

**Nata-kāla Formulas:**

**Hour angle:** The nata-kāla equals the taj-jya multiplied by the śaṅku or lambaka, divided by the dyu-jyā (day-radius).

**Śaṅku:** The śaṅku equals the taj-jya multiplied by the trijyā, divided by the dyu-jyā.

**Krānti-jyā (Declination Sine) Formulas:**

The krānti-jyā equals the lambaka-jyā multiplied by the krama and divided by the viśuva. Alternatively, it equals the lambaka-jyā multiplied by the karṇoccātā and vi, divided by the chāyā-karṇa and vi.

The kṣetra-jyā equals the krānti-jyā multiplied by the vi, divided by the chāyā-kardna.

**English Translation****Verses 117-119. The Twenty-Four Operations of Pāṭigaṇita:**

Addition (saṃkalita), subtraction (vyavakalita), multiplication (guṇana), division (bhājana), squaring (varga), square-root (varga-mūla), cubing (ghana), and cube-root (ghana-mūla) — these five and three [make eight basic operations].

Five special [operations make] thirteen [in total]; these operations are remembered as gaṇita (mathematics). Proclaimed by the ācāryas, by these all mathematics is accomplished.

A number joined with a unit part is called a vyakta-rāśi (manifest number). That which is a number joined with the unit part is [called] avyakta (unmanifest).

Saṃkalita is addition; vyavakalita is subtraction; guṇana is multiplication; bhājana is division; varga is the product of two equals; varga-mūla is the root of that equal quantity; ghana is the product of three equals; ghana-mūla is the root of that quantity. These operations are called gaṇita (mathematics), proclaimed by the teachers; by these all mathematics is accomplished.

**English Translation****Verses 120-121. Square, Cube, and Their Roots:**

That which is the triple portion of a number, or joined with a unit part and halved, that becomes the square (varga). That which is the triple portion of a number.

Or joined with a unit part, that becomes the cube (ghana). Its root is called the ghana-mūla (cube-root); that becomes the measure of the equal-root quantity.

That which is the triple portion of a number, or joined with a unit part and halved, is the square. That which is the triple portion of a number, or joined with a unit part, is the cube. That which is its root is called the cube-root; that is the measure of the equal-root quantity.



**English Translation****Verses 109-111. Tithi, Yoga, and Nakṣatra Computation:**

Having subtracted the Sun from the true Moon, the remainder that results, divided by twelve parts, gives the tithis (lunar days); from half of that, the true [tithis].

The tithis are twelve digits (aṅgulis) from the difference of the Sun and Moon. Joined with fifteen tithis, the yoga is formed.

The yoga is the sum of the signs, degrees, and minutes of the Sun and Moon. That, joined with fifteen and divided by twenty-seven, gives the nakṣatra.

Having subtracted the Sun from the true Moon, dividing the remainder by twelve gives the tithis. The tithis are twelve digits from the difference of Sun and Moon. Joined with fifteen, the yoga is formed. The sum of the signs, degrees, and minutes of Sun and Moon, divided by twenty-seven, gives the nakṣatra.

**English Translation****Verses 112-113. Nakṣatra and Daily Motion Difference:**

Having subtracted the Sun from the true Moon, dividing the remainder by thirty, that nakṣatra becomes the true sign-measure.

The true motion of the Sun and the true motion of the Moon — their difference is called the tithy-ardha-bhukti (half-tithi daily motion).

Having subtracted the Sun from the true Moon, dividing by thirty gives the nakṣatra as the true sign-measure. The true motion of the Sun and the true motion of the Moon — their difference is the tithy-ardha-bhukti.

**English Translation****Verses 114-116. Spherical Astronomy — Indian and Yavana Methods:**

In the opinion of the Yavanas (Greeks), the Earth's sphere is situated at the equatorial region; its highest point is at its centre; there indeed is the kṣitija-vṛta (horizon circle).

In the Indian method, the akṣa-vṛta (latitude circle) is the madhyamaṇḍala (prime vertical); the kṣitija is that great circle; the sama-rekhā (equinoctial line) is also there.

The elevation-sine's koṭi-jyā, the altitude-sine is the bhuja or nata-jyā. The semi-diameter [applies to] the karṇa-field-circle in both methods (ūbhayamato-dyanna).

In the Yavana (Greek) view, the Earth's globe is at the equator, its highest point is at the centre, and there the horizon circle lies. In the Indian method, the latitude circle is the prime vertical, the horizon is that great circle, and the equinoctial line is also there. The common spherical knowledge of both methods is called "ubhayamato-dyanna" (accepted by both schools).

*Candra-grahaṇādhikāraḥ — Chapter on the Calculation of Lunar Eclipses*

This chapter deals with the computation of lunar eclipses, including the determination of the time, magnitude, and duration of the eclipse. Brahmagupta gives detailed rules for

finding the *parva* (full moon day), the *sparsā* (first contact), the *madhya* (middle), and the *mokṣa* (last contact) of a lunar eclipse.

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### English Translation

#### Verses 70-71. Sparsā-piṇḍa (First Contact):

At the true moonrise on the day of double motion, the piṇḍa (magnitude) that is the sparsā-piṇḍa (first contact magnitude), from that the computation of the middle of the eclipse.

That sparsā-piṇḍa, from that the computation of the eclipse middle. Or by its half, the time of the eclipse is proclaimed.

At the true moonrise on the day of double motion, the piṇḍa is the sparsā-piṇḍa (first contact magnitude), from which the middle of the eclipse is computed. Or by half of that, the time of the eclipse is proclaimed.

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### English Translation

#### Verses 72-73. Eclipse Limits by Declination:

The difference between the Moon and Sun's southern declination, that is the decrease or increase. The difference between their northern and southern declinations, that is the decrease or increase.

The difference between the southern and northern declinations, that decrease or increase. That bhuja-jyā is the karaṇa (semi-duration); that shows the sparsā and mokṣa (first and last contacts).

The difference between Moon and Sun's southern declination gives the decrease or increase. The difference between their north-south declinations gives the decrease or increase. That bhuja-jyā is the karaṇa, which determines the sparsā and mokṣa contacts.

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#### Verses 74-77. Eclipse Contacts and Duration:

The difference between southern and northern declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa contacts.

That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa. Or by its half, the time of the eclipse is proclaimed.

When the sparsā and mokṣa occur, then the eclipse is proclaimed. Or by its half, the time of the eclipse is proclaimed.

Or by its half, the time of the eclipse is proclaimed. That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa.

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**Verses 78-80. Summary of Eclipse Rules:**

The sparśa-piṇḍa, from that the computation of the eclipse middle. Or by its half, the time of the eclipse is proclaimed.

The difference between Moon and Sun, southern motion, that decrease or increase. The difference between their north-south declinations, that decrease or increase.

The difference between south-north declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparśa and mokṣa.

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**Verses 81-84. Eclipse Computation Summary:**

At the true moonrise on the day of double motion, the piṇḍa is the sparśa-piṇḍa (first contact magnitude), from which the middle of the eclipse is computed.

That sparśa-piṇḍa, from that the eclipse middle is computed. Or by its half, the time of the eclipse is proclaimed.

The difference between Moon and Sun, the southern motion, that decrease or increase. The difference between their north-south declinations, that decrease or increase.

The difference between south-north declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparśa and mokṣa.

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**Verses 85-87. Middle of the Eclipse:**

When the sparśa and mokṣa occur, then the eclipse is proclaimed. Or by its half, the time of the eclipse is proclaimed.

Or by its half, the time of the eclipse is proclaimed. That bhuja-jyā is the karaṇa; that shows the sparśa and mokṣa.

The sparśa-piṇḍa, from that the eclipse middle is computed. Or by its half, the time of the eclipse is proclaimed.

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**English Translation****Verses 122-124. Solar Eclipse Computation:**

In a solar eclipse, the lambana (parallax in longitude) should be subtracted from the difference between the Moon and Sun; that is the pure [value] at the sparśa (first contact), and similarly at the mokṣa (last contact).

The sparśa-kāla (time of first contact), increased by the lambana at the sparśa, is pure at the mokṣa. The middle of the eclipse is accomplished by half the sum of the two.

The pariveśa (diameter) of the Moon, diminished by that of the Sun, that becomes the grāhaṇa-māna (magnitude of the eclipse), as proclaimed by the ācāryas.

In a solar eclipse, the lambana is subtracted from the Moon-Sun difference; that is pure at first contact and at last contact. The first contact time, increased by lambana at first contact,

is pure at last contact. The eclipse middle is by half the sum of the two. The Moon's diameter, diminished by the Sun's, is the eclipse magnitude, as taught by the masters.

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### English Translation

#### Verses 125-127. Lambana (Parallax in Longitude):

The difference between the lambaka-jyā of one's own place and the dyu-jyā, divided by the trijyā, the result obtained is called the lambana (parallax in longitude).

The lambaka-jyā divided by the dyu-jyā, multiplied by the trijyā, the result obtained is the lambana, as spoken by the ancient masters.

That lambana-produced quantity, divided by time, becomes the true [value]. That, converted into nāḍīs, becomes the form — this is the lambana computation.

The difference between the lambaka-jyā of one's place and the dyu-jyā, divided by the trijyā, gives the lambana. The lambaka-jyā divided by the dyu-jyā, multiplied by the trijyā, gives the lambana as spoken by the ancients. That lambana-produced quantity, time-divided, becomes true; converted into nāḍīs, this is the lambana computation.

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#### Verses 88-90. Eclipse Contacts (Sparsā and Mokṣa):

The difference between Moon and Sun, the southern motion, that decrease or increase. The difference between their north-south declinations, that decrease or increase.

The difference between south-north declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa.

That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa. Or by its half, the time of the eclipse is proclaimed.

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## Candragrahaṇavikārah (Lunar Eclipse Variations)

### Introduction

This chapter deals with the computation of lunar eclipses, following Brahmagupta's exposition in the . The commentary by Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī provides elaborate explanations in Hindi, with mathematical demonstrations .

### Verses 11-12: Computation of (Desired Eclipse Magnitude)

#### Sūtra — Verses 11-12:

**Translation:** “The half-duration (*sthitidala*) diminished by the desired quantity, multiplied by the daily motion (*bhukti*) and divided by the deflection (*villepa*), gives the *bhuja* (sine). The *koṭi* (cosine) is computed by the rule of the desired quantity (*iṣṭeṇu*). From the half-sum of the diameters (*mānaikyārdha*), the *karṇa* (hypotenuse) being subtracted, the result gives the true obscuration (*channa-māna*).”

**Literal Meaning:** The verse gives the method for computing the *iṣṭagrāsa* (desired eclipse magnitude). From the *sthitidala* (half-duration) diminished by the desired quantity, multiplied by the *bhukti* (daily motion) and divided by the *villepa* (deflection), one gets the *bhuja* (sine). The *koṭi* (cosine) is computed by the rule of *iṣṭeṇu* (desired quantity). From the *mānaikyārdha* (half-sum of diameters), the *karṇa* (hypotenuse) being subtracted, the result gives the true obscuration.

**English Translation of Hindi Commentary:** “Subtracting the desired (*iṣṭa*) from the *sthitidala*, multiplying by the *bhukti* and dividing by the *villepa*, one obtains the *bhuja*. By applying this result to the motions of the Sun, Moon, and node (*pāta*), one computes the instantaneous lunar latitude (*tātkālika candra-śara*), which is the *koṭi*. The square root of the sum of their squares is the *karṇa*. From the *mānaikyārdha* (sum of the radii of the eclipsed and eclipsing bodies), subtracting the *karṇa*, the remainder is the desired magnitude (*iṣṭa-grāsa*).”

**English Translation of Proof:** “After the contact (*sparśī*), for whatever desired number of ghaṭīs the knowledge of the *grāsa* is required, that desired number of ghaṭīs should be subtracted from the *sthityardha*. The remainder is used in proportion. If in sixty ghaṭīs the difference in motion between the Sun and Moon gives a certain number of minutes (*kalā*), then [by proportion] in the *iṣṭa*-diminished *sthityardha* ghaṭīs, whatever result is obtained is the *bhuja*. From the result obtained by proportion, by applying the motions of the Sun, Moon, and node, the instantaneous lunar latitude (*tātkālika candra-vāra*) is computed — that is the *koṭi*. The square root of the sum of the squares of these two (*bhuja* and *koṭi*) is the *karṇa*. Subtracting the *karṇa* from the *mānaikyārdha* gives the desired *grāsa*.”

### Verses 13-14: Computation of (Time from Desired Magnitude)

#### Sūtra — Verses 13-14:

**Translation — Verse 13:** “From the square of the half-sum of the measure diminished by the *grāsakalā*, having subtracted the square of the instantaneous latitude (*tātkālika-vikṣepa*), the square root of the remainder — made like a *tithi* [i.e., multiplied by 60 and divided by the difference of motions] — [gives a time].”

**Translation — Verse 14:** “Having subtracted [this time] from the *pragrahaṇa-sthit yardha*, the time from the *pragrahaṇa* [i.e., from contact] is obtained. Having subtracted from the *mokṣa-sthit yardha*, [one obtains the time] before the *mokṣa* [release].”

**Literal Meaning:** “From the square of the half-sum of the measure [of diameters] diminished by the *grāsakālā*, having subtracted the square of the instantaneous *śara* (latitude), the square root of the remainder — made *tithi*-like (*tithivat*) [by multiplying by 60 and dividing by the difference of motions of Sun and Moon] — is the result. That is, having multiplied the square root of the remainder by 60, divided by the difference of motions of Sun and Moon, and having applied the motions of Sun, Moon, and node by that resulting time, the instantaneous latitude (*tātkālika-vikṣepa*) is computed. From that, again by the rule ‘*grāsakālona-pramāṇa*’, etc., iteratively (*asakṛt*), the square-root-time becomes stable. That [time], subtracted from the *pragrahaṇa-sthit yardha* (contact half-duration), the time from the *pragrahaṇa* (from contact) is obtained.”

**English Translation:** “From the square of the *grāsa*-diminished *mānaikyārdha*, having subtracted the square of the instantaneous *śara*, the square root of the remainder — made *tithi*-like iteratively — that is, having multiplied the square root of the remainder by 60, divided by the combined difference of motions of Sun and Moon, and having applied the motions of Sun, Moon, and node by that resulting time, the instantaneous lunar latitude is computed. From that, again by the rule beginning ‘*grāsakālona-pramāṇa*...’, iteratively (*asakṛt*), the square-root-time becomes stable. That, subtracted from the contact half-duration (*pragrahaṇa-sthit yardha*), gives the time from contact, meaning that after contact, in that desired time, the desired *grāsa* is obtained.”

**English Translation of Proof:** “Due to the ignorance of the latitude at that time (*tātkāla*), the computation has been done using the middle-time latitude. Therefore, it is proper to determine the accurate time by an iterative method (*asakṛt-ghātinā*). The remaining proof [follows] the method of Bhāskara: ‘From the square of half the sum of diameters diminished by the *grāsa*, having subtracted the square of the *vikṣepa*...’ etc.”

## Verse 9: Sphuṭikaraṇa of (Refinement of Half-Duration of Totality)

### Sūtra — Verse 9:

**Translation:** “The own motion (*svagati*) — [i.e.,] the motions of the Sun, Moon, and node — multiplied by the *sthitivimardārdha* *nāḍikās* and divided by sixty; both [motions] are applied [to the Sun, Moon, and node] repeatedly in the eclipsed body (*chādyā*), [and] the result is additive or subtractive.”

**Literal Meaning:** “The own motion — the motion of the Sun, the motion of the Moon, and the motion of the node — multiplied by the *sthitivimardārdha* *nāḍikās* (the half-duration of totality *ghaṭis* multiplied separately) and divided by sixty; applied to the Sun, Moon, and node in the eclipsed body, both [additive and subtractive] repeatedly, the result being additive or subtractive.”

**English Translation:** “The [method is] made accurate by the method of Bhāskara: ‘*sthit yardha-nāḍī-guṇitā sva-mukti*’ etc., and ‘*vimardārdha-phala-ona-yukta*’ etc.”

## Sūryagrahaṇādhikārah (Solar Eclipse Chapter)

This chapter (, the fifth chapter of the text) deals with the computation of solar eclipses. The chapter begins with the statement of purpose and proceeds through the computation of the five (sines) required for eclipse calculation: , , , , and .

### Verse 1: Statement of Purpose

#### Sūtra — Verse 1:

**Introduction:** “Now the chapter on solar eclipse is begun. First, the purpose of its commencement is stated —”

**Translation:** “Because the inter-planetary [distance] computed by the *grahabhāskara* [method] previously is an obstruction in the inter-planetary [computation], therefore I shall state the computation of *dyuti* (parallax) with its own degrees, so that there may be agreement.”

**Technical note:** The *grahabhāskara* refers to the method given by earlier astronomers (like Bhāskara I) for computing the distance between planets, which Brahmagupta finds to be defective for solar eclipse computation.

**Literal Meaning:** “Previously, in the eastern horizon (*pūrve kṣitije*), by the degrees of the *grahabhāskara* method, the planets become visible or invisible (*haśya-āhaśya*) in their own time-degrees. Later, in the western horizon (*paścima-kṣitije*), by the degrees of the planetary distances, [the same] according to number. Therefore, I shall state that computation — the computation of the *praharāntara*. That is: in the eastern horizon, on whatever day after the rising of the planet, the rising of the Sun [occurs] after a certain number of time-ghaṭis, on that day the planet becomes invisible (*haśya*) in the remaining night. Similarly, in the western horizon, after the setting of the Sun, when the setting [occurs] after a certain number of time-ghaṭis, on that day in the evening the planet’s invisibility (*haśyatva*) [is determined].”

**English Translation:** “For whatever reason, in solar eclipses, the five *vyās* (*udaya-jyā*, *madhyajyā*, *ravidayākārṇa*, *viśiṃśaṅku*, *hr̥kṣepa*) — by which earlier teachers like Bhāskara computed the *prāṇayana* — from that, the *gaṇitaikya* (agreement between observed and computed values) is not obtained. For that reason, so that their agreement may exist, I state the *prāṇayana* [method]. The *tithi-anta* [refers to] the relation with the *graha*: *udaya-jyā* [refers to] parallax (*bhramā*), *madhyajyā* [to] the sine of the zenith-distance of the tenth degree (*daśama-lagna-natāṃśajyā*), *viśiṃśaṅku* [to] the ascensional difference (*ha-gati*), *hr̥kṣepa* [to] the sine of the zenith-distance of the nonagesimal (*viśiṃśa-natāṃśajyā*). In the *Sūryasiddhānta* too, the solar eclipse *prāṇayana* is by the aforesaid five *vyās*. In the *Paulīśasiddhānta*, in solar eclipses, separate five *vyās* of the Moon are computed; therefore there the solar eclipse is accomplished by the two-*vyā* method.”

### Verses 2-3: Conditions for (Deflection) and (Depression)

#### Sūtra — Verses 2-3:

**Translation — Verse 2:** “When one [planet] is equal to the *viśiṃśa-lagna* (nonagesimal ascendant), there is no *lambana* (deflection in longitude). When it is greater or less than that, *lambana* occurs. When its declination-sine (*krāntijyā*) is equal to the latitude-sine (*akṣajyā*), then there is no *avanati* (depression).”

**Translation — Verse 3:** “Otherwise, *avanati* occurs. When both *lambana* and *avanati* are possible, then the rising [point] equal to the *lagna*, the *ghaṭikās* elapsed since its rising, its *śaṅku* (gnomon), and the *prāṇas* of its motion — [are computed] thereafter.”

**Literal Meaning:** “At the end of the *tithi*, at the time of computed conjunction (*gaṇitāgata-darsāntikāle*), having computed the *lagna* by the previous method stated in the *Tripraśna*, the *viśrīma* [nonagesimal] is to be computed. When the Sun is equal to one [point] with that *viśrīmalagna*, there is no *lambana*. When the Sun is greater or less than that *viśrīma*, *lambana* occurs — this meaning is self-evident. Similarly, when the northern declination-sine (*krāntijyā*) of that *viśrīma* is equal to the latitude-sine (*akṣajyā*) of one’s own place, then there is no *avanati*; otherwise *avanati* occurs. When both *lambana* and *avanati* are possible, then having made the rising equal to the *lagna* — the *rāśyudaya-eṣi-lagna* of one’s own place — by the method stated in the *Tripraśna*: ‘the first diminished by the *bhoga*, increased by the *bhukti*, [gives] the middle *udaya*’, etc., the *ghaṭikās* elapsed since the rising of that *viśrīma* are to be computed.”

## Verse 7: Computation (Computation of True Conjunction Time)

### Sūtra ( ) — Verse 7:

**Translation:** “The *ghaṭikā-miṣa* [of *lambana*] multiplied by five and divided [by some number] yields the *labdha-kalā* called *lambana*. Then the Sun, Moon, and node at the computed conjunction time (*gaṇitāgata-darsānta-kālīka*) are to be diminished [or increased].”

**Literal Meaning:** “The *ghaṭikā-miṣa* multiplied by five, divided by five [? *pañcayyā*], the result is the *labdha-kalā* called *lambana*. Then the Sun, Moon, and node at the computed conjunction time are to be diminished (*ūnāḥ*). If the *lambana* is positive (*dhana*), then the Sun, Moon, and node at the computed conjunction time are to be increased by the *labdha-kalā*; then those become the true Sun, Moon, and node at the true conjunction time (*spaṣṭa-darsānta-kālīka*).”

**English Translation of Proof:** “From the Sun, Moon, and node at the computed conjunction time, all [quantities] beginning with the *śara* [latitude] are to be computed. When the *lambana* is negative (*ṛṇātmaka*): ‘The minutes of motion of the Sun, Moon, and node are separately multiplied by the *lambana-ghaṭikā-miṣa* of sixty-five — by proportion: ‘then by the *lambana-ghaṭikā*, what [result]?’ — by the *labdha-kalās*, the Sun, Moon, and node at the computed conjunction time are to be diminished. When the *lambana* is positive (*dhanātmaka*), they are increased by the *labdha-kalās*; then they become the true [quantities] at the true conjunction time. In the *Siddhāntaśekhara*: ‘Having divided the motion-multiplied *lambana-nāḍī-miṣa* by five, the resulting minutes... The Sun, Moon, and node are added in positive [*lambana*] and subtracted in negative *lambana*’ — by this, what was stated by Śrīpati-nācārya is here spoken in accordance.”



## Candracchāyādhikārah (Moon's Shadow Chapter)

This chapter deals with the computation of the Moon's shadow, which is essential for understanding lunar phenomena. The shadow of the Earth falls on the Moon during a lunar eclipse, and its computation involves calculating the (shadow), (middle shadow), and related quantities.

### Verse 1: Beginning of the Chapter

#### Sūtra — Verse 1:

**Introduction:** “Now the chapter on the Moon's shadow is begun. First, the purpose of its commencement is stated —”

**Translation:** “Previously, the difference between the Moon and the *lagna*, and afterwards, that of the *saptalagna* etc., because from that the shadow of the Moon (*indu-ccchāyā*) is daily [computed], therefore I shall state the computation (*anayana*) of that.”

**Literal Meaning:** “Previously (in the eastern horizon), the difference between the Moon and the *lagna*, and afterwards (in the western horizon), the difference between the *saptalagna* and the Moon, and daily by effort — by which cause the Moon's shadow (*indu-ccchāyā*) is known, and from the Moon's shadow that difference is [also] known — therefore I shall state that computation (*anayana*), the shadow-computation (*chāyā-anayana*). In the *Siddhāntaśekhara*: ‘Usually the difference between the diminished-Sun-*lagna* and the *lagna-nāśin* in the west’ etc. — by this, what was spoken by Śrīpati-nācārya is here spoken in the same manner.”

**English Translation:** “Now the chapter on the Moon's shadow is begun. First, its purpose is stated. The difference between the Moon and the *lagna* in the eastern horizon, and the difference between the *saptalagna* and the Moon in the western horizon — because from these, daily, the Moon's shadow is known (first, the knowledge of the Moon's shadow is obtained), and from the Moon's shadow the knowledge of that difference is obtained — therefore I state the shadow-computation.”

### Verse 2: Shadow Computation

#### Sūtra — Verse 2:

**Translation:** “The parts of the upright-time-difference (*ūrdhvakālīka-udbala-naṣṭa-samasthiti-pāta*) [having] the form of a *prabhahṛṇḍa* [spherical triangle], the semi-diameter (*vyāsa*) of the shadow at the horizon (*tala-lagna*), the proximity to its own *agra* — there is no abbreviation (*saṅkṣepaḥ*).”

**Note:** This is a technical verse whose exact interpretation requires the accompanying commentary. The *prabhahṛṇḍa* is a type of spherical triangle used in Indian astronomy.

### Verses 3-4: Detailed Shadow Calculation

#### Sūtra — Verses 3-4:

**Introduction to Verse 3:** “Now [the author] states what is to be done —”

**Translation — Verse 3:** [Same structure as verse 2, establishing the method.]

**Translation — Verse 4:** “As the root (*mūla*) diminished by the root, or the elapsed [time], having been extracted, another [quantity] comes therefrom. Divided by the own motion (*svabhakti*), that is the desired shadow-motion (*abhīṣṭa-ccchāyā-gati*). That same [quantity], extracted from the root (*mūlodḍhṛtā*).”

*Note:* This verse describes an iterative root-extraction method for computing the rate of change of the shadow. The *mūla* here refers to a base quantity from which differences are taken.

## Grahodayāstādhikārah (Planetary Rising and Setting)

This chapter deals with the computation of the rising and setting times of planets. The chapter explains how to determine the times when planets become visible (, heliacal rising) and invisible (, heliacal setting) based on their elongation from the Sun.

### Verse 1: Pratyaya (Introduction)

#### Sūtra — Verse 1:

**Introduction:** “Now the chapter on planetary rising and setting is begun. First, the purpose of its commencement is stated —”

**Translation:** “Because previously, in the inter-planetary computation, the *grahabhāskara* distance is an obstruction (*bādhaka*), therefore, with its own degrees, so that there may be agreement — I state the computation of *dyuti*.”

*Note:* This verse serves as a parallel to the solar eclipse chapter’s opening, establishing that Brahmagupta’s method supersedes earlier approaches that failed to achieve agreement (*aiṃya*) between computation and observation.

**Literal Meaning:** “Previously, in the eastern horizon, by the *grahabhāskara* distances in their own degrees, in their own time-degrees, later in the western horizon, by the planetary distances in their own time-degrees, according to number, the planets become visible or invisible (*haśya-āhaśya*). Therefore, I shall state that computation — the computation of the *praharāntara*. That is: in the eastern horizon, on whatever day after the rising of the planet, the rising of the Sun occurs after a certain number of time-ghaṭīs, on that day the planet becomes invisible (*haśya*) in the remaining night. Similarly, in the western horizon, after the setting of the Sun, when the setting occurs after a certain number of time-ghaṭīs, on that day in the evening the planet’s invisibility (*haśyatva*) [is established].”

### Verses 8-9: Daily Rising and Setting

#### Sūtra — Verses 8-9:

**Translation — Verse 8:** “By iterative computation (*asakṛt-karma*) from the instantaneous (*tātkālika*) planet-*lagna*, the daily rising and setting are to be computed. First, at whatever time the *grahodita-ghaṭī* [elapsed ghaṭīs since planetary rising] is to be known, the *lagna* at that time (*tātkālika lagna*) and the planetary rising-*lagna* are to be computed. Computing the difference-ghaṭīs of both, by those (computed ghaṭīs) moving the planet, again computing its rising-*lagna*, from the instantaneous fixed *lagna* again computing the desired ghaṭīs — in this manner, iteratively (*asakṛt*, repeatedly) doing [this].”

*Note:* This verse describes the iterative procedure (*asakṛt-karma*) for refining the computation of a planet’s rising time. The method involves successive approximation until the computed value stabilizes.

### Verse 9: Moon’s Rising and Setting Computation

#### Sūtra — Verse 9:

**Translation:** “The rising and setting of the Moon (*indu*) are to be computed by time-degrees (*kālāmśa*) equal to digits (*aṅka-saṃmitaiḥ*). If there is diminution (*hīnatva*) or excess (*adhikatva*), in that interval the *yoga-kāla* [conjunction time] would be [obtained].”

*Note:* The *aṅka* (digit) is a unit equal to 12 degrees. The verse states that the Moon’s rising and setting are computed using time-degrees measured in these units.

**Literal Meaning:** “By time-degrees equal to digits (*aṅka*), that is, by twelve time-degrees (*dvādaśa-kālāmśa*), the rising and setting of the Moon (*indu* = *candra*) are to be computed. Here, if the computed time-degree (*paṭhita-kālāmśa*) has diminution or excess from the desired time-degree (*iṣṭa-kālāmśa*), then in that interval, between those two computed time-degrees, the *yoga-kāla* (conjunction time), or *yogavat-kāla* (time like conjunction), would be [obtained], as clearly stated in the seventh verse.”

**English Translation:** “By time-degrees equal to digits (*aṅka*), that is, by time-degrees equal to twelve, the rising and setting are to be computed. If the computed time-degree has diminution (*hīnatva*, being smaller) or excess (*adhikatva*) from the desired time-degree, then in that interval, between those two time-degrees, the *yoga-kāla* (conjunction time) would be [obtained].”

## Śuklonnatyadhikārah (The Brightness of the Moon)

This chapter deals with the computation of the illuminated portion of the Moon's disk — the (bright) portion. The chapter explains how to determine the amount of the Moon's surface that is illuminated by the Sun at any given time.

### Verses 1-3: Sphuṭaśuklonnati Computation

#### Sūtra — Verse 4:

**Introduction:** “Now the chapter on the Moon's bright elevation (*śuklonnati*) is begun.”

**Translation — Verse 4:** “The ghaṭikās [which are] the *khacarārdha* (half of the celestial sphere's motion) — by the Sun, Moon, node, and *lagna*, and from the rising of the own declination (*sva-krānty-udaya*), [which are] the remaining [ghaṭikās] elapsed from the own *lagna* — having made [the computation] from the setting of the Sun, [these are] to be computed.”

*Note:* The *khacarārdha* refers to half the diurnal motion of the celestial sphere. The verse establishes the parameters needed to compute the Moon's illumination.

**Literal Meaning:** “The *khacarārdha* ghaṭikās [which are] from the setting of the Sun (*ravyastāt*), [which are] the remaining [ghaṭikās] elapsed from the own-*lagna* of the Moon — or the elapsed/deficient ghaṭikās of the Moon's rising-*lagna* — are to be computed. By what? By the Sun, Moon, node, *lagna*, and from the rising of the own declination. Here, this is stated: at the time of the Sun's setting, the Sun, Moon, node, and *lagna*. Having computed these, by the own declination (*sva-krānti*), by the Moon's declination (*candra-krānti*), from the rising, [computing] the elapsed or deficient ghaṭikās of the Moon's rising-*lagna* from the *udaya-lagna* by the method ‘diminished by the *bhoga*, increased by the *bhukti*, [gives] the middle *udaya*’, etc. After the Sun's setting, the ghaṭikās by which the Moon sets — those are the elapsed ghaṭikās. Before the Sun's setting, the ghaṭikās by which the Moon sets — those are the deficient ghaṭikās. Similarly, in the eastern horizon, from the Sun's rising, the elapsed or deficient ghaṭikās of the Moon's rising are to be computed — this is to be understood from the meaning.”

**English Translation:** “By iterative computation (*asakṛt-karma*) from the instantaneous planet-*lagna*, the daily rising and setting are to be computed. First, at whatever time the planetary elapsed ghaṭis are to be known, the *lagna* at that time (*tātkālīka lagna*) and the planetary rising-*lagna* are to be computed. Computing the difference-ghaṭis of both, by those (computed ghaṭis) moving the planet, again computing its rising-*lagna*, from the instantaneous fixed *lagna* again computing the desired ghaṭis — in this manner, iteratively (*asakṛt*, repeatedly) doing [this]. And for the knowledge of the *sthita-kāla*, computation from the *sthita-lagna* (*sapta-bhastala-lagna*) should be done. In this way, from the first day of lunar visibility (*candra darśana*) up to the full moon (*pūrṇimā*), on the desired day, the setting time from the Sun's setting and the rising time from the Sun's rising are to be computed.”

### Verse 5: The (Bright Elevation)

#### Sūtra ( ) — Verse 5:

**Translation — Verse 5:** [Recapitulates the parameters for computing the bright elevation.]

*Technical summary:* The verse reiterates that by the Sun, Moon, node, *lagna*, and from the rising of the declination, the *khacarārdha* ghaṭikās from the Sun's setting are to be used to compute the Moon's bright elevation.

**English Translation of Proof:** “On whatever day, after the Sun's setting, the Moon sets after an additional number of time-ghaṭis. Before the Sun's rising, the Moon rises

[after a certain number of ghaṭīs]. On that very day, when the Moon is visible, the two bright elevations (*śva-jy-unnati*) are to be computed. The bright elevation of the bright [half] (*sita-śuklonnati*) is half the disk (*bimbārdha*) of the Moon's bright [portion]. Thus Bhāskara computed the *śuklonnati* in the first *pāda* of the month-end. In the second and third *pādas*, half the disk is dark (*kṛṣṇa*). Therefore there the dark *śuklonnati* occurs. Thus by the teacher (*ācārya*), two *śuklonnatis* are computed. But here, the *pāda* discussion is not made. Bhāskara, having considered in his mind that the dark *śuklonnati* is not clearly observable, computed only the bright *śuklonnati*. And it is stated: 'Brahmagupta and others have produced the dark *śuklonnati* also for the second and third *caraṇas*; that is not approved by me. For neither is the dark *śuklonnati* clearly perceived by men.'

## Verses 11-12: (True Bright Portion) Computation

### Sūtra ( ) — Verses 11-12:

**Translation — Verse 11:** "Or when the middle motion (*madhyagati*) is multiplied by its own shadow (*tacchāyā*) and by the semi-diameter (*sva-kara*) multiplied by the *tithi* — the results are additive or subtractive. In equality (*sāmya*), the difference is [taken]; in direction-difference (*dig-anyatva*), [it is] the *avanati* [depression] expressed in digits (*aṅkya*)."

*Note:* This verse gives the method for computing the true bright portion (*spaṣṭa-śukla*) of the Moon. The *madhyagati* is the mean daily motion, and the *tacchāyā* is the shadow corresponding to the planet's altitude.

**Literal Meaning:** "In the *hṛjya* [radius = 3438'] of the Sun and Moon, the *kṣepī* [sine of latitude], having been multiplied by the respective middle motion and by the semi-diameter (*vyāsadala*) multiplied by the *tithi* and by fifteen [i.e., by the *tithi*-multiplied semi-diameter multiplied by 15], divided by the *triṇyā* [radius = 3438'], separately, separately, the *natī* [depressions] of the Sun and Moon are obtained. Or, their middle motion, their *kṣepa*-equal *hṛjya*, the shadow being multiplied by that, multiplied by the *tithi*-multiplied own shadow-*karman*, their difference is [taken]."

**English Translation of Proof:** "The first computation [is made] accurate by the method of Bhāskara: 'Having multiplied by the *sahakṣepa* [combined latitude] of the Moon, by the own middle motion-*tithi*-aṁśa-multiplied [quantity]' etc."

## Chandraśuklonnatyadhikārah (Moon's Bright Elevation)

This is the seventh (chapter) of the , dealing with the Moon's bright elevation . It comprises eighteen verses and covers the computation of the arc of illumination on the Moon's disk.

### Verses 1-2: Introduction

#### Sūtra — Verses 1-2:

**Introduction:** “Now the beginning of the chapter on the Moon's bright elevation (*candra-śukla-unnati*).”

**Translation — Verse 1:** “The parts of the upright-time-difference (*ūrdhvakālika-udbala-naṣṭa-sama-sthiti-pāta*) [having] the form of a *prāgrahuṇḍa* [spherical triangle], the semi-diameter (*vyāsa*) of the shadow at the horizon (*tala-lagna*), the proximity to its own *agra* — there is no abbreviation (*saṅkṣepaḥ*).”

**Translation — Verse 2:** “As the root (*mūla*) diminished by the root, or the elapsed [time], having been extracted, another [quantity] comes therefrom. Divided by the own motion (*svabhakti*), that is the desired shadow-motion (*abhīṣṭa-ccchāyā-gati*). That same [quantity], extracted from the root (*mūlodḍhṛtā*).”

**Note:** Verse 2 establishes an iterative root-extraction algorithm for computing the rate of change of the shadow, which is essential for determining the Moon's illumination over time.

### Verse 18: Chapter Conclusion

#### Sūtra — Verse 18:

**Introduction:** “Now [the author] states the conclusion of the chapter —”

**Translation — Verse 18:** “Thus, in [the topics of] *bāhu* (base), *koṭi* (perpendicular), *karṇa* (hypotenuse), *spāṣṭa* (true [longitude]), *sita* (bright [part]), *parilekha* (outline), *sūtra* (line), and *kardama* [mud] — by eighteen doors (*dvāra*), [this is] the seventh chapter [on] the Moon's bright elevation (*candra-śuklonnati*).”

**Note:** The chapter lists eight topics (*bāhu*, *koṭi*, *karṇa*, *spāṣṭa*, *sita*, *parilekha*, *sūtra*, *kardama*) covered in 18 verses. The *kardama* refers to the “mud” or penumbral shadow region.

**Literal Meaning:** “The *parilekha-sūtra* itself [is] the *karṇa*... the *parilekha-sūtra-karṇaḥ*. The meaning is clear.”

**English Translation:** “In this way, in [the topics of] *bāhu*, *koṭi*, *karṇa*, *spāṣṭa*, *sitāṅgula*, *parilekha-sūtra* (*karṇa*), by eighteen meaning-verses, is the seventh chapter named *Chandraśuklonnati*.”

**Note:** The *sitāṅgula* (bright digit) is a unit for measuring the illuminated portion of the Moon's disk. One *āṅgula* (digit) = 1/12 of the Moon's diameter.

### Concluding Verse ( )

**Translation — Concluding Verse:** “That *tilaka* (ornament) which was spoken by the son (*sūnu*) of Madhusūdāna, which was spoken here by Śrīpathi for the conqueror (*jīṣṇu*) — having placed that in [my] heart, this new [work] *Sūryagati* has been composed by Sudhākara.”

**Colophon:** “Thus, in the *Brāhmasphuṭasiddhānta-nutana-tilaka* composed by Śrī Kṛpāludatta-sūnu Sudhākara Dvivedī, the seventh chapter — the *Candraśuklonnatyadhikārah* — [is concluded].”

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**[Thus ends the Seventh Chapter — the Moon’s Bright Elevation — in the *Brāhmasphuṭasiddhānta* composed by Śrī Brahmagupta.]**

## Appendix: Notes on the Text and Commentary

### About the

The (*Brāhmasphuṭasiddhānta*, “Correctly Established Doctrine of Brahma”) is the magnum opus of the great Indian mathematician and astronomer **Brahmagupta** (598–668 CE). Composed in 628 CE at Bhinmal (in present-day Rajasthan, India), it consists of 24 chapters ( ) containing a total of 1008 verses ( ) in and metres.

1. — Lunar Eclipse Variations  
Deals with the computation of eclipse magnitudes, durations, and the *iṣṭagrāsa* (desired obscuration) method.
2. — Solar Eclipse Chapter  
Covers the computation of solar eclipses, including the five *jyā* (sines) required: *udaya-jyā*, *madhyajyā*, *ravidayākarṇa*, *viśiṃśaṅku*, and *hṛkṣepa*.
3. — Moon’s Shadow Chapter  
Treats the computation of the Earth’s shadow as seen on the Moon’s disk.
4. — Planetary Rising and Setting  
Explains the methods for computing the heliacal rising (*udaya*) and setting (*asta*) of planets.
5. — The Brightness of the Moon  
Computes the illuminated portion of the Moon’s disk.
6. — Moon’s Bright Elevation  
The seventh chapter, with 18 verses, covering the arc of illumination.

### The Commentary ( )

The Hindi commentary ( ) in this edition was composed by **Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī**. The commentary follows the traditional structure of Indian mathematical commentaries:

- (*Sūtra*) — The original Sanskrit verse
- (*Sūtrārtha*) — Literal meaning of the verse (marked . .)
- (*Vivṛti*) — Detailed explanation (marked )
- (*Hindi Bhāṣya*) — Hindi explanation (marked )
- (*Upapatti*) — Mathematical proof or demonstration

### Mathematical Notation

Brahmagupta’s work represents one of the high points of Indian mathematics. Key contributions in these chapters include:

- Rules for computing eclipse magnitudes using the (*bhuja*, sine) and (*koṭi*, cosine)
- The (*karṇa*) formula:  $\text{karṇa} = \sqrt{\text{bhuja}^2 + \text{koṭi}^2}$
- Iterative methods (, *asakṛt*) for refining planetary positions
- Proportional reasoning (, *trairāśika*) for time conversions
- Geometric constructions using (*kṣetra*, diagrams)



**[p. 1259]**

Sū. Bhā.—The true longitude of the Moon: having applied the corrections for equation of centre (manda-phala) and evection (śīghra-phala) to the mean longitude, the true longitude is obtained. The mean longitude is calculated from the ahargaṇa (elapsed days since the epoch).

**[p. 1260-1264]**

Sū. Bhā.—The Moon's equations (manda-phala) are tabulated for each 15° of anomaly. The tabular differences are: 39, 36, 31, 24, 15, 5. These are applied successively to obtain the true longitude.

Vi. Bhā.—When the anomaly lies between two tabulated values, multiply the residual arc by half the difference of the passed and upcoming tabular differences, divide by 900', and add to half the sum of the two differences.

**[p. 1265-1269]**

Sū. Bhā.—The śīghra corrections for the Moon are computed from the śīghra anomaly, which is the difference between the Moon's longitude and the Sun's longitude. The maximum śīghra equation is 80' (80 arc-minutes).

**[p. 1270-1274]**

Sū. Bhā.—To find the lagna (ascendant): multiply the time elapsed since sunrise by the Sun's daily motion, add to the Sun's longitude, and apply the ayanāṃśa correction.

**[p. 1275-1279]**

Sū. Bhā.—The time of conjunction is found by dividing the difference in longitudes by the difference in daily motions. The result gives the number of days until conjunction.

**[p. 1280-1284]**

Sū. Bhā.—A prāṇa equals 4 seconds of time. Six prāṇas make a vināḍī, 60 vināḍīs make a nāḍī (ghaṭī), and 60 nāḍīs make a day.

**[p. 1285-1287]**

Here end the verses on lunar computation; next follows the section on shadow calculations (chāyā-vyavahāra).

**[p. 1288-1292]**

Proof.—Let  $j_1$  and  $j_2$  be two successive tabular sines, with arc difference  $h = 15^\circ$ . For an intermediate arc at distance  $\theta$  from  $j_1$ , the sine is:

$$\sin(\theta) = j_1 + \frac{\theta}{h}(j_2 - j_1) - \frac{\theta}{h}\left(1 - \frac{\theta}{h}\right)\frac{j_2 - j_1}{2}$$

This is Brahmagupta's second-order interpolation formula.

**[p. 1293-1297]**

Proof.—Applying the kuṭṭaka (pulverizer) to solve  $137x + 15 = 251y$ : divide 251 by 137, quotient 1, remainder 114. Divide 137 by 114, quotient 1, remainder 23. Divide 114 by 23, quotient 4, remainder 22. Divide 23 by 22, quotient 1, remainder 1. Working backwards yields the multiplier and quotient.

**[p. 1298-1302]**

Sū. Bhā.—The shadow (chāyā) for a gnomon of 12 aṅgulas:

$$\text{Shadow} = \frac{12 \times \cos(\text{zenith distance})}{\sin(\text{altitude})}$$

When the Sun is at the prime vertical, special rules apply using the nara (sine of latitude) and śaṅku (gnomon).

**[p. 1303-1307]**

Proof.—The lagna is found from the oblique ascension. Let  $\phi$  be the latitude,  $\varepsilon$  the obliquity of the ecliptic. For elapsed time  $t$  (in degrees from sunrise), the lagna  $\lambda$  satisfies:

$$\tan(\lambda) = \frac{\sin(\text{R.A.})}{\cos(\text{R.A.}) \cos(\varepsilon) - \tan(\phi) \sin(\varepsilon)}$$

**[p. 1308-1312]**

Sū. Bhā.—For Mars, with manda anomaly  $\mu$  and epicycle radius  $r = 39$  (where deferent radius  $R = 60$ ):

$$\text{manda-phala} = \arcsin\left(\frac{39 \sin(\mu)}{\sqrt{60^2 + 39^2 + 2 \cdot 60 \cdot 39 \cos(\mu)}}\right)$$

This gives the equation of centre for Mars.

**[p. 1313-1317]**

Proof.—First apply manda correction to obtain manda-sphuṭa:  $\lambda_{\text{ms}} = \bar{\lambda} + \text{manda-phala}$ . Then compute śīghra anomaly from  $\lambda_{\text{ms}}$  and apply śīghra correction:  $\lambda_{\text{true}} = \lambda_{\text{ms}} + \text{śīghra-phala}$ .

**[p. 1318-1322]**

Sū. Bhā.—The semi-diameter of the Earth's shadow at the Moon's distance:

$$R_{\text{shadow}} = \frac{R_{\text{Earth}} \cdot (D_{\text{Moon}} - D_{\text{Sun}})}{D_{\text{Sun}}}$$

where  $D$  denotes distance. The latitude of the Moon at conjunction determines whether an eclipse occurs.

**[p. 1323-1327]**

Sū. Bhā.—A planet becomes visible when it is far enough from the Sun. The arc of visibility depends on the planet's brightness: for Venus  $10^\circ$ , for Jupiter  $11^\circ$ , for Mercury  $13^\circ$ , for Saturn  $15^\circ$ , and for Mars  $17^\circ$ .

**[p. 1328-1337]**

Proof.—Ṛjuvṛtta (straight epicycle) corrections and planetary phase computations. These final derivations summarise the algebraic methods used throughout the *Brāhmasphuṭasiddhānta* for astronomical computations.

**[p. 1339-1343]**

Sū. Bhā.—The sum of two positive quantities is positive; the sum of two negative quantities is negative. The sum of a positive and a negative is their difference; if they are equal, the

sum is zero.

**[p. 1344-1348]**

Sū. Bhā.—Zero multiplied by any quantity is zero. A positive quantity multiplied by a negative is negative; a negative by a negative is positive. Zero divided by any quantity is zero.

Sū. Bhā.—A quantity divided by zero becomes that fraction of which the denominator is zero. This *khacheda* (zero-divisor, or “zero-seed”) was later corrected by Bhāskara II.

**[p. 1349-1353]**

Sū. Bhā.—To solve  $Nx^2 + 1 = y^2$  (Pell’s equation): first find the least solution  $(x_1, y_1)$  by inspection or by the cakravāla (cyclic) method. From this, infinite solutions can be generated by composition (bhāvanā).

**[p. 1354-1358]**

Proof.—For  $N = 61$ : start with  $a_0 = 8$  (since  $8^2 = 64 \approx 61$ ). The interpolator  $k_0 = 3$ . The cyclic process generates successive approximations:

$$a_1 = \frac{a_0 + m_0}{k_0}, \quad k_1 = \frac{m_0^2 - N}{k_0}$$

where  $m_0$  is chosen so that  $a_0 + m_0$  is divisible by  $k_0$  and  $m_0^2$  is close to  $N$ .

**[p. 1359-1363]**

Sū. Bhā.—The sum of an arithmetic series:  $S_n = \frac{n}{2}(2a + (n-1)d)$ . The sum of squares:  $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ . The sum of cubes:  $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$ .

**[p. 1364-1368]**

Sū. Bhā.—In compound interest, if principal  $P$  grows at rate  $r$  per period for  $n$  periods, the amount is  $A = P(1 + r)^n$ .

**[p. 1369-1370]**

Here conclude the verses of the algebra section, with praise of the siddhānta and transition to the planetary motion chapter.

**[p. 1371-1375]**

Sū. Bhā.—Having computed the mean longitude (madhyama) from the ahargaṇa, apply the manda and śīghra corrections in order to obtain the true longitude (sphuṭa).

**[p. 1376-1380]**

**Manda epicycles:** Mars 39, Mercury 22, Jupiter 32, Venus 14, Saturn 60. **Śīghra epicycles:** Mars 73, Mercury 121, Jupiter 71, Venus 132, Saturn 39 (in units where  $R = 60$ ).

Table 2: \*

Epicycle dimensions and maximum equations (degrees where  $R = 60$ )

**[p. 1381-1385]**

Sū. Bhā.—The rising time of a planet is computed from its longitude, the oblique ascension of the ecliptic, and the latitude of the place. The *carajā* (ascensional difference) is applied to find the exact moment of heliacal rising.

**[p. 1386-1390]**

Sū. Bhā.—Thus have been explained the motions of the planets, the Sun and the Moon, with their corrections. By these methods, the positions of the celestial bodies may be computed for any desired time.

**[p. 1391-1398]**

**Colophon.**—This *Brāhmasphuṭasiddhānta* was composed by Brahmagupta, son of Jṣṇughāṭa, in the year 628 of the Śaka era (550 CE), at the age of 30. May it bring clarity to all who study it.

## Colophon

**Key Mathematical Contents:** The *Brāhmasphuṭasiddhānta* is notable for: (1) algebraic rules for arithmetic with positive and negative numbers, zero, and surds; (2) the kuṭṭaka (pulverizer) method for linear Diophantine equations; (3) the varga-prakṛti method for Pell's equation; (4) astronomical calculations for planetary positions, eclipses, and conjunctions; (5) trigonometric interpolation formulas.

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## — Planetary Shadow Calculations

**Text (v. 41)** — When a shadow passes over the karna-ucchyāsa-ardha-gola (half-sine sphere of the hypotenuse), the shadow turning from the hypotenuse becomes vyasta-golā (reversed/inverted). On the karna-gola (hypotenuse sphere), the palabhā (equinoctial shadow) equals the base of the gnomon (śaṅku-tala), and the palabhā is always directed northward. Of these two, applying saṃskāra (correction), the difference between the eastern-western threads at the face of the shadow becomes the bāhu (base), and the root of the gnomon is to be understood by means of the bandha (construction) at the shadow-face.

**Commentary (Ṭikā)** — When the shadow, having gone around the sphere of the hypotenuse and its altitude, becomes inverted through the hypotenuse, then on the hypotenuse-sphere the palabhā equals the base of the gnomon; the palabhā is always northward. Applying correction to them, the difference between the east-west threads at the shadow-mouth becomes the base (bhuja); by the bandha at the shadow-mouth the root of the gnomon is to be known.

**Proof (Upapatti)** — In the Siddhāntaśekhara it is thus stated by Śrīpati Nārāyaṇācārya: “That which is the difference between the east-west gnomon-bases is the bāhu; that itself becomes the north-south (koṭi).” Thus the demonstration is established.

**Text (v. 42)** — Now he states the method for finding the base (bhuja):

*prācyaparāyāḥ śaṅkūstalaṃ tadapraṃ grahaṃ me ca*

The base of the gnomon extends east-west; that astra (side) and the planet—these become the measure.

**Commentary** — *prācyaparāyāḥ* (east-west): the base of the gnomon extends to the root of the gnomon. The astra (flank) of the gnomon, or the planet, becomes the measure.

**Proof** — Now the proof: from the east-west limit at the root of the gnomon, the demonstration of the base is clear according to the siddhānta-yukti.

“The side of the gnomon and the planet”—this is clear.

**Commentary** — Of the same directions (ekādiśa), the sum of the east-west gnomon-bases; and of two directions (dvyādiśa), the difference of the east-west gnomon-bases—from the east-west line, the base extending to the root of the gnomon becomes the bhuja (base). The astra (side) of the gnomon and the star become the constituents.

**Text (v. 43)** — The base extending east-west from the gnomon-base (śaṅku-tala) becomes the bhuja; the karna (hypotenuse is) the north-south. By the dvijaya (second victory), the root of the inverse from the middle of the dvikṣepa (double-latitude) becomes the koṭi.

**Commentary** — *śaṅkutalaṃ prācyaparāntaṃ* means the difference between the east-west threads at the root of the gnomon; the north-south becomes the bhuja. By the second application, the root of the inverse of the hypotenuse and the base—from the middle of the double-latitude (dvikṣepa-madhya)—becomes the koṭi (perpendicular).

**Proof** — Of the two meeting at the double-latitude, that which is the perpendicular dropped from the east-west threads is the bhuja. Also, the perpendicular dropped from the east-west threads at the root of the gnomon is the same bhuja. By the sum of the gnomon-bases of one direction, and by the difference of the two directions, extending to the east-west threads at the root of the gnomon, the perpendicular from the east-west threads also becomes the bhuja.

## — Characteristics of Yukti (Proofs)

**Text (v. 69)** — Now he states the characteristics of yukti (proof):

*ubhe mānakhyākhaṇḍi grahayor madhyāntaram yukti-grhyoḥ*

When the difference between two planets of equal minutes (māna) is divided by the sum of the vi-ākhaṇḍa (discs), at the time of eclipse there is a yukti (conjunction).

**Commentary** — The dvigu compound *mānakhyākhaṇḍi* is not clearly explained in this statement; its meaning requires further elucidation.

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**Expanded Text** — When the difference between two planets of equal minutes (*samakala*)—i.e., the latitude-difference (*kendra-antara*)—if the difference between the two planets is less than the sum of their discs (*vi-bhakta-yogākhaṇḍita*), then there is a yukti (conjunction), that is, an ācchādana (eclipse) like a grahaṇa; thus, the upper planet becomes the grāhya (the eclipsed) and the lower planet becomes the grāhaka (the eclipser). If, when divided by the sum of the discs at the time of eclipse, their yukti (meeting) does not occur. In the Siddhāntaśekhara: “*mānakhyākhaṇḍi dhruvara-vikare syān medo-dvike tu nyūne medo grahaṇa-vād-vihac-chādako 'stanaḥ syāt*”—this statement of Śrīpati and the statement of Bhāskara, “*mānakhyākhaṇḍi dhruvara vikare 'sye bhaved yogaḥ*” etc., are in agreement with the Ācārya’s teaching.

**Proof** — Now, the definition of yukti is stated: If the difference between two planets of equal minutes is less than the sum of their *vi-bhakta-yogākhaṇḍa* (discs), then there is a yukti of the two planets, that is, an ācchādana (covering) like an eclipse (the upper planet is the eclipsed, the lower planet is the eclipser). If the difference between the two planets is greater than the sum of the *sphuṭa-vibhakta-yogākhaṇḍa*, then there is no yukti of the planets. In the Siddhāntaśekhara, from the verse beginning “*mānakhyākhaṇḍi dhruvara-vikare syān medoḥ dvike tū*” etc., in the Vijñāna-bhāṣya—Śrīpati in the Siddhānta-śiromaṇi, from “*mānakhyākhaṇḍi dhruvara vikare 'sye bhaved yogaḥ*” etc., Bhāskarācārya too has spoken in accordance with the Ācārya.

## — Planetary Conjunctions

**Text (v.57)** — Of the bhuja, koṭi, and karṇa, whatever bhuja-half is to be transformed, making it into gunakas (multipliers) divided by the semi-diameter (vyāsārdha). Thus, the motion of the two planets—according to their construction the difference between them becomes the ku-bra-antara; the difference between the gnomon-sides (śaṅku-kuvra) is due to the equality of the roots of the gnomon etc. By rotation, the difference between their two gnomons comes about. The rest is clear.

**Commentary** — The teacher explains: Of the base, perpendicular, and hypotenuse, the half-base for transformation—those half-bases divided by the semi-diameter should be made into multipliers. Thus the motion of the two planets—their difference as constructed—the difference of the gnomon-sides arises from the equality of the gnomon-roots. By the revolution, the difference of their gnomons occurs. The remainder is self-evident.

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**Proof** — Now the upapatti: From the sphere arise the bhuja, koṭi, and karṇa; from the position of the gnomon-tortoise (śaṅku-kūrma), the true bhuja—in this manner, with its application, the proof demonstrated by the Ācārya is established.

**Commentary** — When by two observation-nāḍīs they are made unequal, then there is a yoga (conjunction). In the yoga (mahā-yoga), when there is a difference between the two planets (*grahayor-madhyāntare*, i.e., at the divided kendra), like the moon-sixth-day-increase (*ṣaṭ-jñonitivat*), of the same directions (*samyad-śoḥ*, i.e., one-and-different-directions), the difference of the east-west gnomon-flanks (*prācya-śaṅku-vaprayoḥ*)—the saṃyogāntara (union-difference)—the bāhu (base)—the meaning is: from the middle of the one-to-be-increased, the yogāntara (difference of union) of each gnomon-base being applied, the bhuja of them is produced.

**Commentary (contd.)** — In the sum of the two planets, the karṇa is produced. Their own karṇa—at the root of the difference of the squares of the bhuja and karṇa—the koṭi is produced. Thus, by the koṭi-bhuja-karṇa of the two planets, by the gnomon-construction, whatever half-bhuja is to be transformed (*yad-bhujyārdha-pariṇāta*), having made them divided by the semi-diameter (*vyāsārdha-hṛtān*), they become transformed.

**Commentary (contd.)** — Of the east-west (transformed-half-bhuja-east-west-line-threads), in the east (eastern direction), in the west (western direction), extending the koṭi separately; the bāhu (base) extended by the koṭi; dividing, from the middle of the directions (from the divided middle), the bhuja-extent (bhuja-prā-viddha-lekha); dividing the karṇa too, of the bāhu-extents (their own bhuja-extents), their own gnomons—two; from the middle to their astra-extent (at the drawing of their own gnomon-sides), as before, two previously assumed.

**Commentary (contd.)** — The dvikṣepa-madhya-stha-cchāyā (shadow situated at the middle of the double-latitude, i.e., by the eye situated at the middle point) shows separately its own gnomon-side and the planet. In the yoga (union of the two planets), their difference of gnomon-kuvra occurs because of the equality of the gnomon, bhuja, etc.; otherwise (if the planetary union is not at the same point), their difference of gnomons alone occurs. In the revolution (in the yoga, only the difference of gnomon-kuvra-remaining, otherwise of them), in the great yoga, the difference of gnomon-kuvra should not be taken as the difference between the planets; this should be known in the planetary-union-minimum-kalpa.



## — Examination of Astronomical Systems

**Editorial Note** — In this chapter Brahmagupta subjects the astronomical system of Āryabhaṭa to rigorous examination. The refutation of Āryabhaṭa I's (476 CE) planetary theory is here demonstrated in detail.

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**Text (v. 3)** — Now he refutes the yuga stated by Āryabhaṭa:

*āryabhaṭo yugapādāmśāne yātānāha kaliyugādaṁ yat*

Āryabhaṭa stated that two quarter-yugas had elapsed at the beginning of Kali-yuga. From what cause? Therefore that is not true; hence his own yuga-aṣṭa (eighth of a yuga) is not valid.

**Commentary** — Āryabhaṭa stated (*āha*) that at the beginning of Kali-yuga, two yuga-pādas (quarter-yugas) had elapsed (*yātān*). From what cause?—from that reason, therefore his own yuga-aṣṭa—of one yuga, the beginning established, thus two in the Kṛta-yuga—from that, that is not true (*na sat*).

**Proof** — According to Āryabhaṭa's system, from the commencement of one yuga to the beginning of Kali-yuga, three yuga-pādas =  $\frac{3 \times 4320000}{4} = 3240000$ . And according to the Ācārya's (Brahmagupta's) system, Kṛta + Tretā + etc.  $\frac{4320000 \times 9}{10} = 3888000$ . The difference between the two = 648000 years. These, by the excess of number in the Ācārya's system, even in the middle of Kṛta-yuga, Āryabhaṭa's eighth-yuga reaches the end of Kṛta-yuga.

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**Text (v. 11)** — Now again he refutes Āryabhaṭa's stated weekday:

*śrīkṣuro dinavāro gurur ādviko 'sya bhavati kalpādaṁ*

The first weekday accepted by him at the beginning of the kalpa is Thursday (guru). *na bhavaty akam yasmād occuro vistaratas tasmāt*—Since this is not established (*na bhavati*), because his extension (*vistara*) is baseless, therefore (it is invalid).

**Commentary** — By Āryabhaṭa in his own treatise, “from Thursday before the Bhārata”—by this, Thursday is accepted at the beginning of the kalpa. By this the meaning is: From what cause?—His (Āryabhaṭa's) acceptance (*śrīkṣuraḥ* = *svikāraḥ*), the first weekday at the beginning of the kalpa being Thursday becomes established; therefore his acceptance is *vistara* (unsupported and unauthoritative).

**Proof** — According to Āryabhaṭa's system, before the commencement of Kali-yuga, in the present kalpa, six manvantaras and three quarters of a yuga-catur (yuga-quadruple) had elapsed. Thus from that, from one manu of seventy-two yugas, the elapsed yugas at the beginning of the kalpa =  $72 \times 6 + \frac{3}{4} = 432\frac{3}{4}$ . Multiplying by the saura-civil days of the yuga-pañcita: =  $432 \times 1577917500 + \frac{3 \times 1577917500}{4}$ . The weekday at the beginning of the kalpa by this: the remainder when divided by seven =  $34 = 6$  (Friday); but if Thursday is taken as the starting weekday, then the elapsed weekday at Kali-yuga-beginning = 0. The present Thursday itself would be established. Thus by Āryabhaṭa's system, Thursday at the kalpa-beginning is arrived at.

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**Text (v. 13)** — Now again he refutes Āryabhaṭa's stated weekday:

*kṣadhikaiḥ śataiḥ caturbhir varṣaiḥ sahasraiḥ caturbhir yāmirekaḥ*

With four hundred years increased, and with four thousand, one yāmi. Between the day and weekday, in the aṁśa of the sun and the two nights...

**Commentary** — By Āryabhaṭa, two systems were constructed. In one, the civil days in a yuga = 1577917500; at the kalpa-day of short motion, creation. In the other, the civil days in a yuga = 1577917800; at the short-motion night, creation. In both, the number of yuga-years 4320000 is the same. The three days of yuga-pañcita-sāvana = 300. Therefore from the two

systems, by weekday computation, there is a difference of three hundred yuga-year-days; and proportionally, the difference of one day-weekday =  $\frac{4320000}{300} = 14400$  yuga-years.

**Commentary (contd.)** — By this the meaning is: According to Āryabhaṭa's system, between the day and weekday in the aṃśa of the sun and nights, with fourteen years, thousands, seasons, increased by hundreds of years, the difference between day and weekday falls by one. Thus the weekday computation is not stable.

## — Hypotenuse and Diagonal

**Proof** — In the proof of the 13th rule of the Sphuṭādikāra it is shown that in the first term:

$$krama-jyā \times \text{circumference} / bhojyā = bhuja-phala$$

In the second term, the maximum bhuja-phala occurs; in the third term:

$krama-jyā \times \text{circumference} / bhojyā = bhuja-phala$ . In the second term, the maximum *bhuja-phala* becomes zero.

In the fourth term, the maximum  $utkrama-jyā \times \text{circumference} / bhojyā = bhuja-phala$ . Although at the end of even terms and at the beginning of odd terms, due to the conjunction of both terms, the *paridhi* (circumference) mentioned in even and odd terms, when taken, does not produce *phala-nāma* (result); there is no harm from it because, by the difference of circumference readings, by the proportion in the difference of terms, the taking of circumference should be done—thus Āryabhaṭa has indicated. Therefore the Ācārya's refutation is not correct.

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**Text (v. 20)** — Now, if by proportion according to Āryabhaṭa's system the circumference is made precise, even then it is not correct—thus he refutes:

$$vyāsārdha-hṛto bāhuḥ \text{ paridhi-vibheṣāhṛtaḥ phalonayutaḥ}$$

The *bāhu* (side) divided by the semi-diameter, divided by the difference of circumferences, with the result added;

*prathamād vikonakaś ca yat tad asat padayoḥ paridhi-pātāt* — and the *vikonaka* (angular deviation) from the first, that is invalid because of the *paridhi-pāta* (falling of circumference) in the terms.

**Commentary** — The *bāhu* (side) by the *bhuja*, divided by the difference of circumferences, diminished by the end-circumference, divided by the semi-diameter by *vijaya*—the first circumference, and the *vikonaka* from the second: then in order, the first circumference with the result added becomes the true circumference—this well-known method of computing the true circumference also becomes invalid by Āryabhaṭa's system.

Why? Because of the reading of circumference in the terms. In meaning, in that system, there is the taking of circumference in the terms. There, sometimes the first is of even-term type, sometimes of odd-term type. Therefore it varies in correction procedures. This is invalid.

And the *vikonaka* from the first—because of the correctness of the instantaneous correction—this is to be considered by the wise.

## Celestial Latitude

**Text (v. 135)** — Of the Moon (Indu), the *utkrama-jyā* (reversed sine, i.e., the *bhuja-utkrama-jyā*), multiplied by the *viśleṣa-apakrama* (by the arrow of separation, by the *apakrama*, and by the maximum brightness-sine), divided by the square of the *vijyā* (*viśama-jyā*), the result is the *dvikṣepa* (double-latitude) in *lamba-māna*. That was stated by Āryabhaṭa as proper (*yuktam*); therefore (*tataḥ*) it is invalid (*asat*). Since (*yataḥ*) at the end of the *kṣaya* (at the *gola-saṅkhyā*), the result produced by that calculation—in his system, the *śrāddhāyana-saṅkhyā* (number of *śrāddhāyanas*) is produced from it—therefore that is invalid.

**Commentary** — Of the Moon (*indoḥ* = *candra-sya*), the *utkrama-jyā* (the *bhuja-utkrama-jyā*), multiplied by the *viśleṣa-apakrama* (by the arrow of separation, by the *apakrama*, and by the maximum brightness-sine), divided by the square of the *vijyā*—the *dvikṣepa* in *lamba-māna* is obtained. Āryabhaṭa stated this as proper; but from that cause it is invalid. Since at the end of the *kṣaya* (at the *gola-saṅkhyā*), the result produced by that calculation—in his view, the *śrāddhāyana-saṅkhyā* is produced from it—therefore that is invalid.

**Proof** — Here the Caturveda-ācārya: “Indeed this is twofold: this Āryabhaṭīya statement is *vrātya* (heterodox):”

*viśleṣa-pakrama-guṇām utkramaṃ vistara-dhaṃ kṣati-bhaktam*

Here, in the *Maṭadīpikā*, Parameśvara: “If the *sāyana-candra-utkrama* be by the *koṭi*, by the *utkrama-jyā*—if thus, then this of Āryabhaṭa is not correct.”

This operation—the *krama-jyā*, the two *kṣepas*, the *utkrama-jyā* according to Āryabhaṭa’s system—whatever is done there is real; thus Bhāskara’s refutation, actually based on spherical demonstration, is true. And the plane-figures and field-figures are clear.

## Cosmic Cycles (Yugas)

**Hindi Commentary (v. 55)** — For the reason by which, in the computation of planets and the positions of pāta, mandocca, and śīghrocca at the beginning of Meṣa (Aries), in whatever time they again come to the position in Meṣa—that time is called Mahā-yuga. The yuga stated by Śrīpathyavidyā-cūdra and other ācāryas is correct, because according to their view, at the beginning of Mahā-yuga, the planets are produced at the beginning of Meṣa (in a different place). That is, according to their view, if planetary computation is done in the Mahā-yuga approved by the smṛti-texts composed by smṛti-kāras like Manu, then the position of the planets in Meṣa etc. is not established. In the Romaka composed by Śrīpaittha, nowhere is there discussion of Mahā-yuga-beginning. Therefore, by the word Mahā-yuga, here the Mahā-yuga stated by the Ācārya should be taken. When all the planets and the positions of pāta, mandocca, and śīghrocca are at the beginning of Meṣa—that alone is named Mahā-yuga-ādi. In the Mahā-yuga-ādi stated by Śrīpaittha, this position does not occur; therefore according to their view, the Mahā-yuga-ādi is not correct. The Ācārya's refutation here is correct.

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**Text (v. 56)** — Now again, the very same is refuted at night:

*kadinādaṁ smṛti-buktaṁ grahamokṣapatir dina-kṣaye pralayaḥ*

At the beginning of the *kadina* (many days), as stated in the Smṛti, the liberation of planets, at the destruction of the day, the dissolution.

*tātyatibahūni yasmāt mahāyugeṣṭo 'prasiddhim avam* — Since there are very many of those, the *eṣṭa* (desired) of Mahā-yuga is of unestablished, low (nature).

**Commentary** — At the beginning of *kadina* (many days), the creation of planets and stars; at the destruction of the day, at the end of many days, the dissolution of the planetary measures—thus it is in the Smṛti-texts. Thus in the Mahā-yuga of many days, since there are sixty-times-hundred yugas, therefore this stated by him is unestablished, not true from any source. What is the use of the gravity-ritual by the taking of many yugas? This is invalid. For even by taking many yugas, the planetary computations are not in this, not at any time obtained from the position—thus the computation is clear to those who know.

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**Expanded Commentary** — At the beginning of *kadina* (*bahu-dinādaṁ*), the *mokṣa-patiḥ* (creation of planets and stars); at the *dina-kṣaye* (end of many days), the *pralayaḥ* (destruction) of the planets and stars—this is stated in the *smṛti* texts. In the Siddhāntaśekhara, Śrīpati has stated: “The creation and dissolution of the luminous planets occurs at the beginning and end of those days.”

In the Siddhānta-śiromaṇi, Bhāskarācārya: “Since their creation is at day-beginning, and dissolution at day-end, the intelligent should contemplate this.” By this, the creation of planets at day-beginning is stated. But the Sūryasiddhānta-kāra states that “from the beginning of many days, from the Śāta-yuga, in the seven Vedas, Brahmā having created, was appointed at the end.” For the refutation of the time of happiness etc. stated by Brahmagupta-Śrīpati-Bhāskarācārya, and for the refutation of the Sūryasiddhānta-kāra's view, in the Siddhānta-tattva-viveka, there is much elaboration by Kamalākara—there is no substance without name.

## Chapter on Mathematics

**Editorial Note** — In this chapter Brahmagupta's algebra (*bīja-gaṇita*), arithmetic (*pāṭi-gaṇita*), geometry, and series summation are presented. This *Dvyāśīti-koṇy-adhyāya* (12th chapter) is of supreme importance in the history of world mathematics.

### Algebra: Multiplication and Powers — Verse 1

**Text (v. 1)** — *dānaṃ ca dattam ardhaṃ pañca tena māna-syārdheṣa-pañcanavamāṃśaṃ tadā kim*

A donation was given: half, five; of that measure, half less one-fifth and one-ninth—then what?

**Commentary** — *Nyāsa* (statement):  $\frac{1}{2}$ . The answer is stated as having become  $1\frac{5}{8}\frac{3}{4}$ .

Here due to confusion, the actual written answer is  $\frac{3}{4}\frac{5}{8}$ . (See his translation p. 283.)

Here the Caturveda-ācārya: "This is to be acknowledged."

"Thus here the Ācārya, one of the *pañca-jātya*, spoke."

"Because he reads, therefore, having made the meaning gone, he did not speak."

"By Skandasena and others, his name was made *Bhāga-māta*."

### Geometric Progression — Verse 5

**Text (v. 5)** — The sum of fractions with equal denominators (*sahasra-cchedāṃśa-yuktiḥ = tulya-harāṃśānāṃ yogaḥ*), divided by the denominator (*cheda-vibhaktā*), becomes the result of the first class for all quantities.

Those multiplied by the second-class measures; denominator (*hara*) multiplied by denominator—the fraction thereof, with the denominator, becomes the result for all quantities.

For the quantities of the first class, "*yogontara tulya-harāṃśakānām*" etc.—as stated by Bhāskara; and for the second *prabhāga-jāti* combination, as in the *Līlāvati*: "*lavāla-vadhana-abhyaṃ harā haranāḥ*" etc.—thus stated by Bhāskara.

### Division and Geometric Problems — Verse 6

**Text (v. 6)** — Now he states the *karana-sūtra* for division (*bhāgāhāra*):

*parivartyaṃ bhāgahara-cchedāṃśāṃ cheda-saṃguṇāc chedaḥ*

Having interchanged, the denominator-fraction of the divisor (*bhāgahara*) multiplied by the denominator;

*śeṣo 'mśaguṇo bhājasya bhāgāhāraḥ sarveṣāṃ tayoḥ* — The remainder multiplied by the numerator of the dividend; the division of both of all.

**Commentary** — The denominator-fraction of the divisor (*bhāgahara-chedāṃśī*)—the denominator-fraction of the divisor is interchanged with the denominator of the dividend; the interchanged-denominator becomes the divisor-denominator.

Thus the numerator of the dividend becomes the numerator-multiplied-by-interchanged-numerator.

Thus for both, for mixed sums, the division is produced.

For all, by the rule "*rūpāricchedaguṇāni*" etc.

Bhāskara's division method is the same.

The example of Caturveda:

**Example** — Where the area is six *haṣṭa* (square cubits), with one and a half, with one *yama* and a half. Ten sides and a half, and the *koṭi*—let it be of that measure.

This is a geometric problem: given area  $324 \frac{3}{8}$ , side  $90 \frac{1}{2}$ , find the *koṭi*.

**Commentary** — *Nyāsa*: Area =  $324 \frac{3}{8}$ ; *bhuja* =  $90 \frac{1}{2}$ ; the *utkrama-jāta koṭi* =  $11 \frac{3}{5}$ .

Here again the Caturveda-ācārya.

“Others here also give an example of proportion-calculation, as:”

## Series Summation — Verse 9

**Text (v. 9)** — Alternatively, calculation by the *laghutama-apavartana* (smallest reduction):

Statement:  $\frac{1}{2} + \frac{1}{3} + \frac{3}{4} + \frac{1}{5}$ .

The denominators 2, 6, 12, 4: by reduction with 2, 3:

From this reduction, produced:  $2 \times 2 \times 3 = 12$ .

This multiplied: 12, or by the sum of denominator-numerator,

$$\begin{aligned} \frac{1}{2} \times \frac{6}{1} + \frac{1}{3} \times \frac{12}{2} + \frac{3}{4} \times \frac{12}{3} + \frac{1}{5} \times \frac{12}{4} \\ = \frac{6}{2} + \frac{12}{6} + \frac{12}{4} + \frac{3}{5} = 9 = \text{sum result.} \end{aligned}$$

## Binomial Expansion — Geometric Demonstration

**Editorial Note** — At page 734 of this section, the proof of the rule “*sthāpyo-ntyaghana*” is demonstrated by geometric diagrams. This geometric demonstration of the expansion of  $(a + b)^3$  was accomplished in India centuries before similar developments in Western mathematics.

**Geometric Demonstration** — Side of the square =  $a + b$ .

Breadth =  $a + b$ .

Area of the square =  $(a + b) \times (a + b) = a^2 + 2ab + b^2$ .

The area of the field multiplied by the length etc.—the *ghanatamaka* (solid-cube) field is related; the examples shown are not correct, and are not to be examined by the wise.

**Proof** — For making the cube of  $(a + k)$ , by the definition of cube as equal triple-product:

$$(a + k)(a + k)(a + k) = (a + k)^3$$

By multiplication:  $(a^2 + a \cdot k + k^2)(a + k) = (a^2 + 2a \cdot k + k^2)(a + k)$

$$= a^3 + 2a^2 \cdot k + a \cdot k^2 + a^2 \cdot k + 2a \cdot k^2 + k^3$$

$$= a^3 + 3a^2 \cdot k + 3a \cdot k^2 + k^3 = (a + k)^3$$

This agrees with the Ācārya's “*sthāpyo-ntyaghana*” (place-the-last-cube).

“*sthāpyo-ntyavargaṃ dviguṇā-ntyaniḥnaṃ*” etc.—as stated by Bhāskara in the *Līlāvātī*, this follows the same method.

By the ancients, only the investigation of the sum of squares and cubes was done.

For making the cube of the sum of two-plus-two, the first is called *ākṣepitva*; the second two is called *uttara*; then the last-ghana is placed; from that the last-square (*antya-kṛti*) is doubled, called *uttara*; the cube of the sum of two-plus-two becomes.

Here the Caturveda-ācārya gave this example: “A square pond, equal to the extent of hastas. Its depth also; the result is the *ghanatamaka* (solid cube).”

**Binomial Table** — As the best *rūpa* (form): this itself is the twofold *pada-siddhānta*:

$$\frac{\frac{a \times n \times k}{1}}{n=1} \quad \left| \quad \frac{\frac{a \times n(n-1)k}{1 \times 2}}{n=2} \quad \left| \quad \frac{\frac{a \times n(n-1)(n-2)k}{1 \times 2 \times 3}}{2} \quad \left| \quad + \dots \right. \right.$$

Here if  $n = 2$ :  $(a + k)^2 = a^2 + a \times 2 \times k + k^2$

If  $n = 3$ :  $(a + k)^3 = a^3 + a^2 \times 3 \times k + a \times 3 \times k^2 + k^3$

**Proof** — By this much, “*sthāpyo-ntyavargam dviguṇā-ntyanighnam*” etc.—the proof of the square, and “*sthāpyo ntyaghana*” etc.—the proof of the cube become clear.

## Square Root Extraction — Verse 10

**Text (v. 10)** — One should divide the denominator; having obtained (the result) of the two, one should divide them separately, separately.

From the square, the root from the divided term—that becomes the square-root.

**Commentary** — For making the cube of the sum of two-plus-two, the first is called *ākṣepitva*; the second two is called *uttara*; then the last-ghana is placed; from that the last-square (*antya-kṛti*) is doubled, called *uttara*; the cube of the sum of two-plus-two becomes.

Here the Caturveda-ācārya gave this example: “A square pond, equal to the extent of hastas. Its depth also; the result is the *ghanatamaka* (solid cube).” *Ghanatamakam*.



## Colophon

### Colophon —

Thus, in the *Śrī-Brāhmasphuṭasiddhānta*, the *Dvyāśīti-koṇy-adhyāya* (12th chapter), named the eleventh chapter. || 11 ||

Thus, in the *Śrī-Brāhmasphuṭasiddhānta*, the *Tantraparīkṣādhyāya* (chapter on examination of systems), named the eleventh chapter. || 11 ||

Thus, in the *Śrī-Brāhmasphuṭasiddhānta*, the *Duṣpradādhyāya* (chapter on difficult gifts), named the eleventh chapter. || 11 ||

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## The Third Class of Partnerships

### Circumference and Proportion

(BSS 12.37 Vol. II, p. 891)

This verse states that the circumference is obtained by multiplying the diameter successively by the prescribed factors, as taught by previous teachers including Śrīpati. Brahmagupta, Bhāskarācārya, and Līlāvātī have all taught this in their respective works.

**171.** The true circumference is that obtained by the rules taught by the two Dvivedis; the circumference otherwise (from crude approximation) is erroneous.

### Mathematical Proof

#### End of Arithmetic Chapter

Here ends the *Rāśi Vyavahāra* (Arithmetic Operations). Now, under division, the quantities of grains corresponding to the inclined angle and base angle are stated for the derivation of measures.

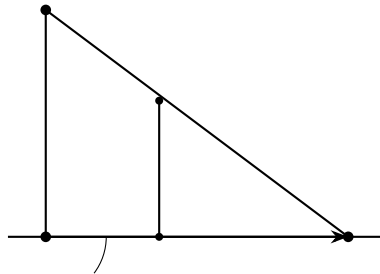
### Shadow Calculations

*Sūtra-bhāṣya*: The *nara* is the *śaṅku* multiplied by eight — the *nara* of the shadow divided by the *śaṅku* is that in the *saṅkrama*; this is the *śuddha*; divided by the eastern or western semi-diurnal arc, it becomes the corrected value.

The *Chāyā Vyavahāra* (shadow computations) section continues with detailed derivations. The *nara* is the sine of latitude, and the *śaṅku* is the gnomon (12 aṅgulas). The shadow is computed from the relationship between these quantities.

### Geometric Construction of Shadows

The accompanying diagrams illustrate the geometric relationships: the gnomon (vertical pole) casts a shadow on the ground, forming an angle with the sun's ray. The *karna* (hypotenuse) equals the distance from the lamp's height. Similar triangles yield the proportions.



*Shadow* (Chāyā), *śāna* = base. *Karna* = lamp-height. From similarity of triangles, the *karna*-base proportion gives: base = *chāyā-prāntara*; then by proportional reasoning,  $12 \times \text{base/shadow} = \text{lamp-height}$ . Similarly for the other triangle pair, the same result follows. Thus the teacher's statement is proved. (BSS 12.41–42 commentary)

## Algebraic Operations —

(BSS 12.46)

Where a quantity has both integer (*kala*) and fractional (*vikala*) parts, the integer part is called *sakala* and the fractional part *vikala*; these are then processed by the rule to obtain the square root.

### Upapatti (Mathematical Proof)

*Upapatti*: Let the integer part be  $k$  and the *vikala* part  $vi$ . Then the quantity is  $k + \frac{vi}{60}$ . The square is:

$$= k^2 + \frac{2k \cdot vi}{60} + \frac{vi^2}{60^2}$$

*V̄tti-bhāya*: When a quantity has both *kala* and *vikala* parts, to find its square, the integer part is called *sakala* and the fractional part *vikala*; then the rule states the method. Note: "" means 'square' in Sanskrit. (BSS 12.46 commentary)

(BSS 12.47)

This verse gives the method for division with fractions. The dividend, combined with the divisor (in the manner of adding fractions), is divided by half the divisor multiplied by the denominator of the divisor; the result, added to the *prakṛti*, gives the quotient. (BSS 12.47 commentary)

(BSS 12.49)

The section on division with fractions (*bhāgahāra*) is explained. The rule involves the manipulation of fractional quantities through their denominators. (BSS 12.49 commentary)

## Astronomical Problems —

The following sections deal with astronomical computations, including the determination of time, planetary positions, and the computation of various astronomical quantities.

### Mixed Astronomical Problems

(BSS 13.6)

He who knows the *vyatīpāta*, *vaidhṛti*, *bṛhaspati-varṣa*, and the motion of the nodes in a *yuga* — he is the knower of the essence of time. (BSS 13.6 commentary)

(BSS 13.10)

He who knows the sum of the *avama* (omitted tithis) and the *praghara* (excess days), combined with the mean (longitudes) at the desired time — he is the knower of the *grahyaka* (the quantity to be known). (BSS 13.10 commentary)

### Third Set of Problems

(BSS 13.40)

From the product of the elapsed days of the *kalpa* and seven, divided by seven — the remainder gives the day of the week (counting from Sunday).

*Upapatti*: From the *kalpa* elapsed days multiplied by seven, the remainder from division by seven gives the day of the week starting from Sunday. (BSS 13.40 commentary)

### Lunar Calculations

## Planetary Theory — &

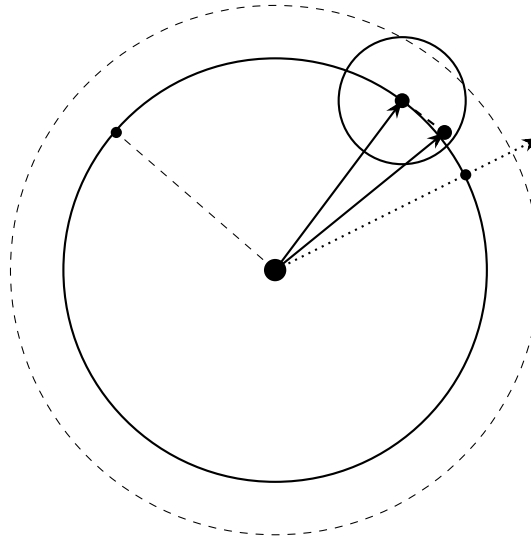
### Epicyclic Model

The computation of planetary positions involves the *mandasphuṭa* (equation of centre) and *śighraphala* (equation of anomaly, or second inequality). Brahmagupta gives the following method:

*Vṛtti-bhāṣya*: The *phala* obtained by multiplying by the *bhuja* (sine of anomaly) and dividing by the *karna* — its arc gives the difference between the first and second true positions; that is the true position. (BSS 378-581 commentary)

### Proof of Epicyclic Correction

*Upapatti*: *Ja* = geocentric planet; *Ma* = mean planet via manda; *Ma.3* = *śighrakendra*; *Mar* = *śighrakendra-jyā* = geocentric sine; *Sp.* = true planet; *Sp.3* = true *kendra*. (BSS *upapatti*)



*Bhum* = geocentric planet. *Spaṣṭa* = true *kendra-jyā*. *Ma.3*. *Ma.* = mean *kendra*. *Mū.spaṣṭa* = *śighraphala*. From similarity of triangles: geocentric sine/*śighrakarna* = true *kendra-jyā*. Alternatively, true *kendra* = difference between true planet and apogee. The manda-true *kendra-jyā* meets the epicycle; by arc-correspondence, manda-true *kendra-rāśi* = difference between manda-true and equation. (BSS *upapatti*)

The *mandasphuṭa* (true planet via equation of centre) is computed as:

1. Compute the *mandakendra* (mean anomaly): difference between mean longitude and apogee
2. Find the *mandaphala* (equation of centre) from the sine table
3. Apply to the mean longitude to get *mandasphuṭa*
4. Compute *śighrakendra* from *mandasphuṭa*
5. Find *śighraphala* and apply to get the true longitude

## Advanced Astronomical Computations — &

(BSS 21.25-26)

The *iṣṭajyā* (desired sine) is multiplied by the results of the five operations and divided by the radius; the remainder, when 90 is subtracted from the *nava* position, gives the *śeṣa*; the *manoratha* (desired arc) is obtained. The arc is thus found from the sine without the *vyārdha* (half-chord).

*Sūtra-bhāya*: The desired sine is divided by the semi-diameter, together with the two-foot arc-difference; the remainder is clear. (BSS 21.25-26 commentary)

### Upapatti — Detailed Proof

*Upapatti*: From previous proofs:  $jya = \frac{(180-ca) \cdot ca \cdot jra}{10125 - \frac{(180-ca)ca}{4}}$ . Let  $ya = (180 - ca) \cdot ca$ . Then  $jya = \frac{jya \cdot ya}{10125 - ya/4}$ . (BSS 21.25-26 *upapatti*)

(BSS 21.50-52)

These verses compute the *bhogakalā* (daily motion in minutes of arc). For the Sun, Moon, and planets, the daily motion is found by taking the difference in longitude between successive days and converting to *kalās*.

*Vṛti-bhāya*: The mean daily motion in *kalās* of the Sun (or Moon), multiplied by the respective factor, gives the *bhogakalās* of the *nakṣatras*. (BSS 21.50-52 commentary)

The *bhogakalā* is the daily motion of a planet in terms of *kalās* (arc-minutes). The computation involves:

- Finding the difference in longitude between successive days
- Converting this difference into minutes (*kalās*)
- For the Sun: about 1° per day (approximately 60 *kalās*)
- For the Moon: about 13° per day

## Final Chapters — Eclipses and Colophons

The concluding portion of the *Brāhmasphuṭasiddhānta* contains extensive rules for eclipse calculations, including the computation of the *sparsā* (first contact) and *mokṣa* (last contact), the magnitude of the eclipse, and the *akṣṣavalana* (deflection due to latitude).

## Colophons —

*Here ends the Brāhmasphuṭasiddhānta*

Thus concludes the *Brāhmasphuṭasiddhānta*. This treatise contains 24 chapters in all, comprising 12 chapters of mathematics (*Gaṇitādhyāya*) and 12 chapters of astronomy (*Golādhyāya*). The teacher Brahmagupta composed this work in Śaṃvat 628 (CE 666).

## Appendix: Summary of Mathematical Rules

**Shadow rule :** The shadow is proportional to the *nara* (sine of latitude) and inversely proportional to the *śaṅku* (gnomon height).

**Square root of non-squares :** The method of successive approximation:  $\sqrt{N} = a + \frac{r}{2a+1}$  where  $a$  is the largest integer such that  $a^2 \leq N$  and  $r = N - a^2$ .

**Equation of centre :** The correction for the eccentricity of planetary orbits, computed using the *vyā* (sine) of the *kendra* (anomaly).

**Second inequality :** The correction for the planet's position relative to the Sun-Earth line, computed from the *śīghrakendra* (anomaly of conjunction).

**Kuṭṭaka :** The pulverizer method for solving linear Diophantine equations  $ax + c = by$ , essential for calendar calculations.

**Bhogakalā :** The daily motion of a planet in arc-minutes, computed from the difference in longitude between successive days.

**Arc from sine :** The inverse problem of finding the arc from its sine, solved by Brahmagupta's algebraic formula without using the *vyārdha* (half-chord) table.

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**[Introductory Note.]** Now [the Ācārya] speaks of the computation of the motion (bhoga) of the starry nakṣatra and the computation of the nakṣatra as freed from the planets (graha-mukta-nakṣatrānayaṇa).

**[Mathematical Rule.]** The fifteenth [operation]: The sum of the degrees of conjunction-motion (yoga-bhoga-kalā) =  $\text{Caṃ} \times 15$ . The sum of all =  $1 \text{ Caṃ} + 3 \text{ Caṃ} + 15 \text{ Caṃ} = 27 \text{ Caṃ}$  = the total motion of all the nakṣatras (sarvāna-kṣatra-bhogaḥ). Here  $\text{Caṃ}$  (cakra) denotes a complete revolution.

**[Dvivedī's Commentary.]** The pure revolutions (cakkala) are the numbers [representing] the total motion of all the nakṣatras; then the degrees of motion of the starry nakṣatra (śrambijñud bhoga-kalā) [are obtained]; its daily motion = revolution —  $27 \text{ Caṃ}$ . From that, the kalpa-born revolutions [are obtained] as: (revolution —  $27 \text{ Caṃ}$ ) civil days (kudina) = cakkala.

In 27 civil days, the kalpa revolutions [are obtained] by this Ācārya; [this is] well established (tumūpapaṇṇam). In the *Siddhāntaśekhara*: “cakkāriyā vā śāsvata-sativadhana-āhatāni śodhayin sudina-laya-dvaroṣa-cakra”. [The meaning is that] the daily motion of the kalpa (kalikā-gatiḥ) which is the Vaivasvata [value] at mid-[period] is to be known (mr̥tiḥ). The degrees of nakṣatra motion (nakṣatra-bhoga-kalikā), the times of the prahara, etc., the remaining degrees to be traversed (bhogya-kalikā) — from the excess, the gatagamya [values] become the days traversed and remaining to be traversed (gata-gamya-dināni). Thus [it is said] by Śrīpati in the fifteenth [chapter].

“The sum of the motions of all [nakṣatras] multiplied by the revolutions [gives] the Vaivasvata [value]; this is the yoga of the starry [nakṣatra],” and so on — thus spoke Bhāskara; and all teachers [hold] this same meaning.

**[Verses 50-51]**

### | Upapatti (Proof/Demonstration)

**[Mathematical Derivation.]** The sum of the six doubled (ṣaṭ-śraddha) motion-degrees of the nakṣatras =  $\frac{3 \text{ Caṃ}}{2} \times 6 = 3 \text{ Caṃ} \times 3 = 6 \text{ Caṃ}$ . The sum of the six [times] single-motion degrees of the nakṣatras =  $\frac{\text{Caṃ}}{2} \times 6 = \text{Caṃ} \times 3$ . The sum of the fifteen [times] single motion (sama-bhoga) degrees of the nakṣatras =  $\text{Caṃ} \times 15$ . The sum of all = total nakṣatra motion degrees =  $6 \text{ Caṃ} + 3 \text{ Caṃ} + [15] \text{ Caṃ} = 27 \text{ Caṃ}$ .

**[Hindi Bhāṣya — English Summary.]** When the mean daily motion of the Moon (candra-madhyagati-kalā) is multiplied by [certain] quantities, the motion degrees of the six, the śarva, and the equal nakṣatras are obtained. The remaining [degrees], when subtracted from the revolution degrees (bhagaṇa-kalā), give the motion of the starry nakṣatra (śrambijñuta-bhoga).

Alternatively: The kalpa revolutions of the starry nakṣatra are obtained by multiplying the kalpa revolutions of the Moon by eighty-seven and subtracting from the kalpa civil days. From these revolutions, the daily motion of the starry nakṣatra is obtained by proportion (anupāta): if in kalpa civil days such-and-such kalpa revolutions [are obtained], then in one day [what]? — thus the daily motion in time-units.

When the nakṣatra motion degrees are subtracted from the planet-time aggregate (grahakāla-samūha), the nakṣatras are obtained. That is, in the planet-freed degrees (graha-



mukta-kalā), as many pure motion degrees of the nakṣatras as there are, so many are the traversed nakṣatras (gata-nakṣatra). The remaining degrees are the current degrees of the present nakṣatra; subtracting these from the nakṣatra motion degrees yields the remaining degrees to be traversed (gamya-kalā). Then, if in one day [a certain value] is obtained in planetary motion degrees, by proportion from the gata-gamya degrees, the days traversed and days remaining are obtained.

**[Verses 50-52]**

| **Upapatti (Proof).** Here the Ācārya has postulated (kalpita) the previously stated time-quantity (kāla-jñā) as the desired [value] for the second [process]. Then śrada, sama-śāim, kudūdit — these three quantities (muñjā) produce one śramba field (kṣetra); and śrada-jñā, labdha-jñā, tijñā — these three quantities produce a second śramba field. These two śramba fields are homogeneous (sajātiya); therefore, proportion (anupāta) is applied [between them].

$$\frac{\text{labdhajñā} \times \text{iṣṭa}}{\text{tijñā}} = \text{samaśāim-kud}$$

And the time-quantity (kāla-jñā) = [the result of this proportion]. The three quantities (trayaṃ muñjānām) generate the proportional relationship between the two śramba fields.

| **Atrôpapattiḥ (Demonstration thereof)**

| **[Śloka on Shadow Computation.]** In the shadow-hypotenuse circle (châyā-karṇe gole), how the equinoctial shadow (palabhā) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) — this is demonstrated thus: if the palabhā is obtained as the base (bhuja) in the twelve-aṅgula sign (dvāda-śāṅgula-rāśi), then what is the gnomon-like [quantity]? The gnomon-equivalent (śaṅku-tulya) =  $\frac{\text{pama} \times \text{śaṅku}}{12}$ ; but śaṅku =  $\frac{12 \times \text{tri}}{\text{chāka}}$ . By elevation (utthāpanena), the gnomon-equivalent =  $\frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}}$ .

| **[Dvivedī's Explanation.]** Here, for computing the palabhā at a time different from mid-day (dvāda/dinārk), [the Ācārya] speaks.

| **[Hindi Bhāṣya — Summary.]** At a time different from midday, multiplying the Sun's shadow by the shadow-hypotenuse (châyā-karṇa) and dividing by the tijñā yields the circular shadow of the hypotenuse-circle. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the base (bhuja). In the northern hemisphere (uttara-gola), subtracting the northern bhuja from the karṇa-dvitiya and adding the southern bhuja yields the palabhā. In the southern hemisphere (dakṣiṇa-gola), merely subtracting the second shadow from the bhuja yields the palabhā.

**[Verses 46-50]**

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**[Introduction.]** Now [the Ācārya] speaks of the questions of the śrambya [type].

**[Hindi Bhāṣya — Summary.]** Those who know the sama-śaku [from] the declination (krānti), who know the sama-śutta [from] the latitude (lamba-āṃśa), who know the koṇa-śaku, who know the koṇa-śaku-chāyā, and who know the ghaṭi [from] the koṇa-dvitāja entrance — they are the scholars (paṇḍita) of jyotiṣa-siddhānta. Here there are five questions.

**[Verse 7]**

| *Idānīm anyān praśnān āha* (Now he states other questions.)

**[Sūtra-bhāṣya.]** The difference of the east-west [line] from the gnomon-base (śaṅku-tala-prācyā-parāntara) is the base (bhuja). He who, having understood the two bhuja-values, knows the equinoctial shadow (viśuvac-chāyā) [i.e.,] the palabhā, or who knows one bhuja [together with] the equinoctial shadow and the sun — he is a gaṇyaka. Thus here there are three questions.

**[Verse 8]**

**[Vivaraṇa-bhāṣya.]** The difference of the east-west line [measured] from the gnomon-base (śaṅku-tala-pūrvāpara-rekha) is declared to be the bhuja. He who, having known the two bhuja-values, knows the palabhā, or having known one bhuja, knows the palabhā and the Sun (ravi) — he is a gaṇyaka. Thus here there are three questions.

**[Verse 7]**

**[Dvivedī's Introduction.]** Now [the Ācārya] speaks of the śrambya questions.

**[Hindi Bhāṣya — Summary.]** The difference between the gnomon-base (śaṅku-tala) and the east-west line is the bhuja. The person who, having known the two bhuja-values, knows the palabhā, and likewise [he who], having known one bhuja, knows the palabhā and the Sun (ravi) — they are the paṇḍitas of jyotiḥ-śāstra. Here there are three questions.

**[Verse 7]**

| *Idānīm anyān praśnān āha* (Now he states other questions.)

**[Sūtra-bhāṣya.]** He who knows the pātāla-śaṅku of the Sun [and] the kumatha-śaṅku, or who knows the Sun having known the uddheśa, or [who knows] the iṣṭa-śaṅku [and] the kudivo-iṣṭa-śaṅku [as] the pātāla-gaṇaḥ and pātāla-śaṅku [respectively], the ravi-ratha-śaṅku — of these two, separately, each at its own base (prthak tale) or at the gnomon-base (śaṅku-tale), he who knows — he alone is the tantra-jña. Thus here there are four questions.

**[Verse 9]**

**[Dvivedī's Introduction.]** Now [the Ācārya] speaks of the śrambya questions.

**[Hindi Bhāṣya — Summary.]** The person who knows the Rāhu-path (bhūmā-mārga) in an eclipse, who knows that path, and who knows the koṇa-śaku-chāyā — here there are five questions.

**[Verse 7]**

**[Eclipse Triangle Computation.]**

*koṇaduta-stha-ravi-gataṃ bhuva-protkuṭaṃ bhuvādri-viparyaṃ bhuvyacāpāṃśaṃ ekaṃ bhujaḥ*

*bhuvālpasatikāvadhi lamba-aṃśa dvitīyo bhujaḥ | koṇadutaṃ va svainikādri-viparyaṃ nanāṃ-śāstu-tīyo bhujaḥ*

In this triangle: the Sun's passage at the koṇaduta, the earth's protuberance (protkuṭa), the earth-mountain reversal — the degrees of the earth-curve (bhuvyacāpāṃśa) — these form one bhuja. The latitude-degrees (lamba-aṃśa) up to the earth's alpaka-satika [form] the second bhuja. The koṇaduta and the svainika-mountain reversal [form] the third bhuja.

In this triangle, the Sun's passage being [known] as the earth's protuberance (bhuva-protkuṭa), the angle [is] the natakala (nata-kālah). The southern koṇaduta-nata-amṛtyanna-koṇa = 85; by the rule of proportion (anupātaḥ): if by the natakāla-jyā the hasta-jyā is obtained, then by the arc-jyā of 85 (jyā 85), what is obtained?

The bhuja-jyā =  $\frac{\text{hasta-jyā} \times \text{jyā } 85}{\text{natakāla-jyā}} = \frac{\text{hasta-jyā} \times \text{tri} \times \text{jyā } 85}{\text{natakāla-jyā} \times \text{tri}} = \frac{\text{hasta-jyā} \times \text{tri} \times 285}{2495 \times \text{natakāla-jyā}}.$

### | Upapattiḥ (Proof/Demonstration)

**[Dvivedi's Upapatti — Summary.]** Here, since the longitude difference (deśāntara) is not known, one should compute the conjunction-time (saṃmīlana-kāla) and disjunction-time (unmīlana-kāla) of the total eclipse by the method of lunar eclipse, from the true Sun (sphuṭa-ravi) and true Moon (sphuṭa-candra), devoid of deśāntara correction. On that day, the second saṃmīlana-kāla should [also] be computed; if it exceeds the computational saṃmīlana-kāla, then one should know whether [the place] is situated in the eastern or western part from the second line.

Because to the east of the line, midday (madhyāhna-kāla) occurs first in one's own country and later in the line-country; therefore, the saṃmīlana-kāla of one's own country exceeds the saṃmīlana-kāla of the line-country. In the west, the opposite occurs. Thus, from the unmīlana-kāla, the iṣṭa-prāsa-kāla, the sparśa-kāla, or the mokṣa-kāla should be examined.

**[Sūtra-bhāṣya.]** The first half of the earlier [section] of the Tri-praśna-adhyāya [is to be] placed as the fourth subsidiary section (upayupardha-saṃṃ nyasyāt). All else of the Tri-praśna [verses] 58-60, the extent (āyāmi) [is] clear.

### [Verses 49-50]

**[Vivarāṇa-bhāṣya.]** At the half-day time (dinārdha-dina-kāle), the shadow multiplied by the shadow-hypotenuse (chāyā-karṇa-guṇā), divided by the tijñā, then the circular shadow of the hypotenuse-circle (chāyā-karṇa-golīyā) would be [obtained] as the ravi-value. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the bhuja. By that bhuja, diminished by the karṇa-dvitiya-prācya, [or] joined with it — in the northern hemisphere, the equinoctial shadow (viśuvac-chāyā), i.e., the palabhā, results. In the south (i.e., the southern hemisphere), when that is diminished by the difference from the karṇa-dvitiya-prācya (bhuja-māna), then the palabhā results. Thus, [the shadow] joined with or diminished from the difference by the bhuja, or diminished by the southern bhuja, then the palabhā results.

### [Verses 49-50]

### | Atrôpapattiḥ (Demonstration thereof)

#### [Proof of Gnomon-Equivalence.]

How the palabhā in the shadow-hypotenuse circle (chāyā-karṇa-gola) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) is demonstrated thus: if the palabhā is obtained as the bhuja in the twelve-aṅgula sign, then what is the gnomon-equivalent (śaṅku-tulya)?

$$\begin{aligned} \text{śaṅku-tulya} &= \frac{\text{pama} \times \text{śaṅku}}{12}, \quad \text{but } \text{śaṅku} = \frac{12 \times \text{tri}}{\text{chāka}}, \text{ by elevation} \\ \text{śaṅku-tulya} &= \frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}} = \text{pama} \end{aligned}$$

Therefore, it is proved (siddham) that in the karṇa-gola, the gnomon-equivalent (śaṅku-tulya) equals the palabhā. The karṇa-dvitiya-jyā becomes the inverse circle (vyasta-golā). The palabhā is the north-south [quantity]. Thus, by the correction of the chāyā-pravi-parasūna, the bhuja is spoken of; by the addition-subtraction (kṣepa-riyena) with the karṇa-dvitiya-jyā, by the correction of the bhuja, the palabhā results.

In the *Siddhāntaśekhara*: “Now, the difference of the natapūrva-pāścima (east-west natal difference), the śakibuta-jñā-makā. The dakṣiṇōttara-bhujā, joined with and diminished, the Sun dwelling in the north (saumya-vartin) [gives] the palabhā. In the southern [hemisphere], again, at the inanāyām, thus it is always diminished by that difference. Thus, joined with and diminished by the difference, the shadow becomes fixed [in] aṅgula-ścakā.” This [is] said by Śrīpati; [it is] in accordance with the Ācārya alone.

### [Verses 49-50]

*valanasya muce taimyaḥ pṛthag grahak khaṇḍakena | vṛttiḥ kṛtiḥ sparśavimarśanimadhyagratā krameṇôr bimha-vagamayaḥ |*

Of the deflection (valana), the release (muca), the taimya, separately; by the section of the planet (graha-khaṇḍakena); the vṛtti, the kṛti, the contact and release (sparśa-vimarśa), the centrality of the middle (nimadhyagratā) — in sequence, the understanding of the eclipse-disc (ur-bimha) [is obtained].

By Bhāskarācārya, the very method of Śrīpati [was] propounded (pratipāditāḥ) in the *Viśvadarpaṇa*.

**[Verses 14-15]**

### **[Dvivedī's Commentary on Eclipse Drawing.]**

The answers to the two questions beginning “viśvadarpa-maṇḍala-dikā” etc., have been explained in the eclipse chapter (grahaṇādhikāra). The latter half of that [is called] śrambe. Now the answers to the question beginning “samparkaṃ maṇḍalai” etc. are spoken.

**[Hindi Bhāṣya — Summary.]** In the prāhlādūta, the arc of the instantaneous declination (tātkālīka krānti) is of a simple form with slight difference; [this is] the balana-sūtra. From the balana-jyā, the divya-jyā should be computed. In the māna-cakra-bāhṛta [circle], above the balana-sūtra, the contact-sphāra (sparśa-kālīka sāra) and release-sphāra (mokṣa-kālīka sāra) of the Sun should be given in the direction of their [respective] sāra. In the madhya-balana-sūtra, the madhya should be given from the centre of the prāhla-disc. From the moon's horns (śarāghra), by the prāhla-pramāṇa-vyāsa [diameter], having drawn quickly, the contact, release, and mid-line prāsa should be computed. And by drawing the parilekha [projection] on the earth, the contact-release śrīra-prāsa is obtained.

### **| Upapattiḥ (Proof/Demonstration)**

**[Dvivedī's Proof — Summary.]** When in the māna-cakra-bāhṛta circle there is the centre of the prāhlā-kṛtā, then the prāhla-bimbagrānta and prāhla-bimagrānta remain contiguous (saṃlagna). Therefore, in the known māna-cakra-bāhṛta circle, the direction should be fixed (śrīṣṭa). The common endpoint of the sama-maṇḍala and the māna-cakra-bāhṛta circle is the eastern mark (prācī cihna) in the māna-cakra-bāhṛta circle.

From there, giving the balana-jyā, the line from the centre [through] the balana-jyā-pragata is the krānti-tṛt-prācī. One should give the jyāvat-khādāna from the balana-sūtra. Because the sāra remains south and north from the krānti-vṛt-prācī. Thus it happens in the sparśa and mokṣa. The madhya-balana [is] in the dakṣiṇōttara direction from the tātkālīka-krānti-vṛt-prācī; therefore, from the centre it should be given in the balana-sūtra. In the śarāghra, the prāhla-vṛtta has its centre; therefore, from there, by the tātkālīka-prāhlābhi-pramāṇa, from the constructed circles, the sparśa-mokṣa-madhya [is obtained].

In the *Siddhāntaśekhara*, from the verse “mānacakrabāhṛtalaye tatpariyāminivi” etc., written in the Sanskrit proof, Śrīpati has spoken in accordance with the Ācāryas alone. In the *Siddhāntaśiromaṇi*, from “prāhlākhasūtreya vidhāya vṛttaṃ mānacakrabāhṛtaṃ ca sādhitāṃsam” etc., Bhāskarācārya has propounded in the form of deliberation (vicāra-rūpa) the very method of Śrīpati.

**[Verses 14-15]**

**[Introduction.]** In the Sūrya-siddhānta, the following verbal description of instruments (yantra-vivaraṇa) is given —

*tuṅgârkasamāyuktaṃ gola-yantraṃ prasādhayet  
gopayeetat-prakāśoktaṃ sarvagamyam bhaved dhi |*

One should prepare the spherical instrument (gola-yantra) endowed with the exalted Sun (tuṅgârka-samāyuktaṃ); having concealed it, [as] spoken of in the Prakāśa, it becomes indeed universally accessible (sarvagamyam).

*kāla-saṃsāadhanārthaṃ tathā yantrārīṇā sādhayet  
ekākī yojayed bījaṃ yantraṃ vismaya-kāriṇi |*

For the purpose of time-determination (kāla-saṃsādhana), one should thus accomplish [it] by the yantrārīṇa; the solitary one should apply the seed (bījaṃ) [to] the instrument, causing wonder.

*śaṅku-kuyaṣṭi dhanuś-cakreṣa-chāyā-yantrair naikathā  
gupta-deśāṅka-jñeyam kāla-jñānam atandritaḥ |*

By gnomon-staff (śaṅku-kuyaṣṭi), bow-wheel (dhanu-cakra), and shadow-instruments (chāyā-yantraiḥ) in manifold ways; the knowledge of time (kāla-jñāna) is to be known from the hidden place-marker (gupta-deśāṅka) [by one who is] vigilant.

*toya-yantra-kapālāṅka-gharmmy-udanatravānaraiḥ  
sa-sūtra-rekhu-gamais ca samaya-kālam prasādhayet |*

By water-instrument (toya-yantra), bowl (kapāla), aṅka of heat (gharma), water-stream (udanatra), and monkey [devices?] (vānaraiḥ), and with cord-line-rekhuga (sa-sūtra-rekhu-gamaiḥ), one should determine the proper time (samaya-kālam).

*paradārā-manuṣyāṇāṃ gulvataila-jalāni ca  
bījāni pāṃsavas teṣu prayogās te 'pi durlabhāḥ |*

The wives of others (paradāra) of men, oil (taila), and water — [these are] seeds (bījāni) [in] the sand (pāṃsavaḥ) of these; their applications are also difficult to obtain (durlabhāḥ).

*tā prapañcamadhiśchadd nyastaṃ kuṇḍe smalābhamis  
dviṭṭha-meja-laya-horātraṃ sphuṭaṃ yantraṃ kapālakaṃ |*

That covering of the world (prapañcam-adhiśchadd), placed in the bowl (kuṇḍe), smalābhamis [?]; the double-positioned (dviṭṭha) meja-laya day-night (horātra) — the clear instrument [is] the kapālaka.

*na yantraṃ tathā sādhu divā ca vimale ravau  
chāyā saṃsāadhanaiḥ proktaṃ kāla-sādhanaṃ uttamam |*

Not the instrument thus [alone is] good; by day, when the Sun is pure (vimale ravau), the shadow [is] declared by the saṃsādhana-methods [to be] the best time-determination (kāla-sādhana).

### | Mānādhyāya (Chapter on Measures)

*mānāni saura-cāndrārṅkṣa-sāvanāni grahāṇayanam aibhiḥ  
mānaiḥ prthak caturbhiḥ saṃkhyabahāroḍjña lokasya |*

The measures (mānāni): solar (saura), lunar (cāndra), stellar (nakṣa), and civil (sāvana) —

by these, the computation of planets (grahāṇayanam). By these four measures separately, the multitudinous arrangements (saṃkhyabahāroḍjña) of the world [are made].

**[Dvivedī's Hindi Commentary — Summary.]**

From this, the solar measure (saura-māna), lunar measure (cāndra-māna), stellar measure (nākṣatra-māna), and civil measure (sāvana-māna) — these four types of measures are spoken. By these very four measures, all the affairs of people are conducted. Which objects are grasped by which measure — all this is propounded (pratipādita).

Brāhma, divya, paitṛ, prājāpatya, bāhṛspatya, saura, sāvana, cāndra, nākṣa — these are nine measures. Of these measures, in the world of humans, only the saura, cāndra, sāvana, and nākṣa have primacy, because by these measures alone all human affairs are accomplished. In the Sūrya-siddhānta, Siddhāntaśekhara, Siddhāntaśiromaṇi, and all other treatises, the subject of measures is spoken of in the same manner. In this chapter of the Brāhmasphuṭasiddhānta, the methods of viewing the earth (bhūmādarśana) are also given.

**[Introduction.]** In the Smṛtis and in the Purāṇas, the doctrine of Rāhu-grasta eclipse (rāhugrasta-graḥaṇa) is present (pratipādana). Therefore, reconciling the two doctrines, the Ācārya has said —

*rāhus tac chadayati praviśati yac chuklapacchadayante  
chāyā tamasi-dvairaghra-dānām kamala-yoniḥ |  
candro 'smuyojah syo yadinaimaya-bhāskarasya māsāt  
chādayati śarmi-tatāpo rāhuś chadayati tat savituḥ |*

Rāhu covers that [Moon], entering [it] which the bright and dark [portions] cover. The shadow (chāyā) [is] the tamasi [force] of the enemy of the lotus-born [i.e., Rāhu, enemy of Viṣṇu?].

The Moon (candra), the asmu-yojah, the syo [?], from the māsa of the sun-of-the-yadinaimaya [?] — [this] Rāhu covers the scorching-heat (śarmi-tatāpah) [of] that Sun (savituḥ).

In the Golādhyāya of the Siddhāntaśiromaṇi also, the unspoken (apokṛta) statement of Bhāskara [is as follows] —

*dvi-deśa-kāla-varayāṇi medānām chādako rāhur iti sa varṇita  
yan māninaḥ kevala-golavidhās tat-sahitā vedy araṇābāhyām |*

Rāhu, the eclipser (chādakaḥ) of the vara of the two-place-time (dvi-deśa-kāla-varayāṇi medānām), thus he is described. Those who know only the spherical method (kevala-golavidhāḥ), together with that [knowledge], should know the araṇā-bāhyām [?].

*rāhuḥ kṣubhā-maṇḍalagaḥ śaśāṅkaḥ śaśāṅkaś chādayatīva bimbam  
tamo-mayaḥ śamyuva-pradānāt sarvagamānām aviruddham etat |*

Rāhu, situated in the agitated orb (kṣubhā-maṇḍalagaḥ), [eclipses] the Moon (śaśāṅkaḥ); the Moon appears to cover the disc (bimbam). [Rāhu is] composed of darkness (tamo-mayaḥ); from the bestowal of śamyuva [?], this is not contradictory (aviruddham) for all who go [i.e., all observers].

**[Dvivedī's Commentary.]** This reconciliation (samanvaya) has been made. In the Siddhāntaśekhara, for the refutation (khaṇḍanārtha) of the Rāhu-grasta eclipse, a chapter named “Rāhu-nirākaraṇādhyāya” has been placed. In this, Śrīpati also has made the reconciliation with the following śloka —

*viṣṇu-lūna-viśirasaḥ kila pañcad ettavān varam imaṃ parameṣṭhī  
hema-dāna-vidhinā tavatupatis tam-arati-mahas tad uparāge |*

The Supreme Lord (parameṣṭhī) indeed gave this excellent boon (varam imaṃ) to the headless Viṣṇu-lūna [i.e., the decapitated Rāhu], [that] by the method of donating gold (hema-dāna-vidhinā), at that eclipse (tad uparāge), [there is] the great darkness (tam-arati-mahaḥ) [for him?].

*bhūmecchāyā praviṣṭaḥ sthagayati śaśinaṃ śuklapakṣāvasāne  
rāhur bhu-prasādāt samavigatavastu-tāmo vyasutulyaḥ |*

Entering the earth's shadow (bhūmecchāyā praviṣṭaḥ), [Rāhu] obscures the Moon (sthagayati śaśinaṃ) at the end of the bright fortnight (śuklapakṣāvasāne). Rāhu, by the grace of the earth (bhu-prasādāt), [being] equal to the vyasu [?], the darkness [is] the real state of



things (samavigatavastu-tāmaḥ).

*urvyāṃ sthaṃ bhānu-bimbaṃ salila-maya-tanor yathaivôktaṃ bimbam  
saṃsūyaivaṃ ca māsavy uparati-samaye svasya sâhitya-hetoḥ |*

Standing on the earth (urvyāṃ sthaṃ), the solar disc (bhānu-bimbam), [this] disc as described [earlier], of the water-bodied one [i.e., the Moon] — thus indeed at the time of lunar cessation (māsavy-uparati-samaye), for the sake of its conjunction (svasya sâhitya-hetoḥ).

### [Dvivedī's Commentary on Spherical Construction.]

In the Golabandhâdhikâra, having constructed the great circles (mahad-vṛttâḥ: pūrvâpara-vṛtta, yâmyôttara-vṛtta, sthitija-vṛtta, etc.) and the small circles (laghu-vṛttâḥ: the outer royal circles of the twelve signs from Meṣa, etc.), the Ācārya has demonstrated the nature (svarūpa) of the maximum equation (paramalambana-nati) and computed the knowledge of the circles (vṛttôpāyanam).

As the Ācārya has spoken of the golabandha in the Brāhmasphuṭasiddhānta, so also has Śrīpati in the Siddhāntaśekhara, and Bhāskarâcārya in the Golâdhyâya of the Siddhāntaśiromaṇi. In the graha-gola and nakṣatra-gola, five fixed circles (sthira-vṛtta) are spoken: pūrvâpara-vṛtta, sthitija-vṛtta, yâmyôttara-vṛtta, unmaṇḍala, and viṣuvad-vṛtta. All these are equal to the kalâ-maṇḍala.

And the moving circles (calavṛtta) of the planets: mandanîcôcavṛtta = 7, the śîghranîcôcavṛtta of the manda-etc. planets = 5. The manda-prativṛtta = 7, the śîghra-prativṛtta = 7. The hamaṇḍala and hṛd-kṣepa of the seven planets [are also given].

## Chapter 17: Advanced Planetary Computations

(Original pages 1136–1146; covering mean and true longitudes, *parilekha* drawings)

### Problem Statement

**15.** The half-chord segments (*ardhajyā-khaṇḍa*) of the arcs for the purpose of [computing] the Sun, the Moon, Venus, Mars, Jupiter, Saturn, and the nodes of the Moon — the sum of these chord-segments is the *bhogya-phala* [incremental motion].

**16.** The difference of the *gata-bhogya-khaṇḍa* [past and future segments], divided by the minutes of arc, multiplied by the elapsed [time in] minutes — that, combined with half the difference, [added to] the *bhogya* [motion], is the mean *bhogya*.

**Mathematical note:** The *bhogya-khaṇḍa* method computes the incremental motion of a planet by dividing the orbital arc into segments. If  $k_1, k_2, \dots, k_n$  are the chord-segments for each planet's arc, the *bhogya-phala* is  $umk_i$ . The mean motion is then interpolated using the proportion of elapsed time. Brahmagupta uses a *triṣṭā* (radius) of 150, whereas Bhāskara II later adopted 120 in the *Siddhānta-śiromaṇi*.

### Parilekha (Planetary Diagrams)

**[Pañcasiddhāntikā 3.]** The Sun's [daily] motion is five [ghaṭikās]; the pairs [of planets] and the minutes [are as] declared by Pitāmaha [Brahmā]. Half a month, half [the motion], the nodes, the apogees, and the quantities — *these* are established by calculation.

This verse from the *Pañcasiddhāntikā* of Varāhamihira (c. 505–587 CE) cites the *Paitāmaha-siddhānta* — the oldest of the five astronomical systems summarized in that work. Brahmagupta critically examines these earlier parameters, noting discrepancies with observational data. The phrase *pita-mahoddhiṣṭāni* (declared by Pitāmaha) refers to the legendary divine origin ascribed to this *siddhānta*.

### Special Rules

Between the two times of sunrise and sunset, the time when the [shadow] increases and takes support [of the gnomon], blazing with the form of fire — [it is] condemned in all rituals [as inauspicious].

As much as the difference between the circles of the [ecliptic and equatorial] paths, so much is the *sambādha* [obstruction] — this destroys all rituals.

The *pātāntara-kāla* (intermediate time of the node passage) is declared inauspicious for all rituals including bathing, charity, recitation, *śrāddha*, and *homa*. When the Sun and Moon have equal brightness near the *graha-sandhi* (planetary conjunction), the *pāta* (node passage) occurs twice in a short time; but when the Sun is near apogee, the equality of brightness persists for a longer duration.

## Chapter 18: The Pulverizer (Kuṭṭaka)

*The most celebrated chapter: linear Diophantine equations, operations with zero, and the “square-nature” (vargaprakṛti) problem*

## Purpose of the Kuṭṭaka

**[Introduction.]** The term *kuṭṭaka* (“pulverizer”) refers here to the method of solving linear Diophantine equations of the form  $ax + by = c$ . Here  $a$  is the *bhājya* (dividend),  $b$  is the *bhājaka* (divisor),  $c$  is the *śeṣa* (remainder),  $x$  is the *guṇaka* (multiplier), and  $y$  is the *koṭi* (quotient).

Brahmagupta was the first to give systematic rules for arithmetic operations involving zero and negative numbers, as well as the general solution to the pulverizer equation.

**14.** The two [quantities] — the *śaka-indra-kāla* [elapsed years of the Śaka era] and the *śeṣa-varṣa-yuga* [residue of years in the yuga] increased by five — make them half the *ardha-sita* [half the deficit]; the *bhugrā* [earth-number] for that pair [is found] by the pulverizer.

**15.** The *saṃkṣipta* [reduced quantity], two, three, the motions, the *tithi-bhū* [lords of the tithis], multiplied by nine, the *eṣya* [desired quantity] — the *digra-sabhāga* [directional parts] divided by seven, the Moon’s [motion], [these are] the wealth of the planet’s self.

**On the astronomical setting:** In the *kuṭṭaka* method as applied to astronomy, the dividend (*bhājya*) typically represents the number of days in a yuga (e.g., 1577917500 civil days), the divisor (*bhājaka*) represents the number of planetary revolutions in that yuga, and the remainder (*śeṣa*) represents the elapsed portion. The method finds the multiplier (*guṇaka*) such that the congruence is satisfied, yielding the number of elapsed days corresponding to a given planetary longitude.

## The Pulverizer Method

**40.** The *bhājya* [dividend,  $a$ ] is [divided] by the *bhājaka* [divisor,  $b$ ] together with the *śeṣa* [remainder,  $c$ ] until unity becomes the operation. The first [quotient] is united with one [rupas]; then the second operation becomes [clear].

**[English translation of commentary:]** The commentary explains the pulverizer method as follows: “The pulverizer with unity (*rūpa-kuṭṭaka*) is the method by which, starting from the desired [quantity], the root-quantity is found. The first quotient, combined with unity (*rūpa*), gives the basis; then for the purpose of the second [step], unity becomes the operative quantity. In this manner, roots are extracted from the non-aggregate [quantities].

**Example:** ‘The fourteen-second rows [of a formation], O conqueror of the world, tell that quantity quickly if you know the pulverizer.’ Here, for divisor 34 the remainder is 2; for divisor 13 the remainder is 10. Hence the greater remainder-divisor is 13, the lesser is 34. Dividing the greater by the lesser gives remainder 13; then by mutual division the chain (*vallī*) is produced. The penultimate [quotient], multiplied by its own higher [term] and united [with the lower] — this is the final [step] of the pulverizer method.”

**Algorithmic summary:** Given  $ax + by = c$  where  $\gcd(a, b) \mid c$ :

1. Apply the Euclidean algorithm to  $(a, b)$  to obtain quotients  $q_1, q_2, \dots, q_n$ .
2. Construct the *vallī* (chain) of partial quotients.
3. Starting from the bottom with 1, work upward:  $x_0 = 1, x_1 = q_n, x_{k+1} = q_{n-k} \cdot x_k + x_{k-1}$ .
4. The penultimate value gives the multiplier  $x$ ; then  $y = (c - ax)/b$ .

This is algebraically equivalent to the extended Euclidean algorithm.

## Operations with Zero

**[On zero:]** The aggregation and diminution of positive, negative, and zero [quantities]; the rule for multiplication; and for division — [are declared].

**[Addition and Subtraction:]** A negative [quantity] joined with a negative becomes more negative; a positive with a positive [becomes] more positive; zero joined with zero [remains] zero. From a positive, a negative is subtracted; from a negative, a positive is subtracted. A quantity united with zero remains in its own form.

**[Multiplication:]** A quantity multiplied by zero becomes zero. [But] a quantity divided by zero becomes zero [*śūnya* — Brahmagupta’s controversial rule: he treats  $a/0 = 0$ , later corrected by Bhāskara II].

The rule for the aggregation of unknowns (*avyakta*), as much as desired, and the rule for the multiplication of unknowns, and the determination of the measure of unknowns — [are given].

**Historical note on zero:** Brahmagupta (628 CE) was the first mathematician in history to establish rules for arithmetic operations with zero as a number. His rules:

- $a + 0 = a$ ;  $a - 0 = a$ ;  $0 + 0 = 0$ ;  $0 - 0 = 0$
- $a \times 0 = 0$ ;  $0 \times 0 = 0$
- $a0 = 0$  (*this rule is incorrect; division by zero is undefined*)
- $0a = 0$  for  $a \neq 0$

He also gave rules for operations with negative numbers (*ṛṇa*):  $(-a) \times (-b) = +ab$ ,  $(-a) \times (+b) = -ab$ , etc. These rules were revolutionary for the 7th century.

## The Square-Nature (Vargaprakṛti) Problem

**[Square-nature:]** In an isosceles triangle, the sum of the squares of the sides — [and] the product of the sum and difference of the *karaṇīs* [surds].

The statement of the square-equation [Pell’s equation]; the elevation of the root of the square-equation [its solution].

**[English:]** The “square-nature” (*vargaprakṛti*) refers to the equation  $Nx^2 + 1 = y^2$ , known in modern mathematics as Pell’s equation. Here  $N$  is the *prakṛti* (multiplier),  $x$  is the *hrasva-mūla* (lesser root), and  $y$  is the *dīrgha-mūla* (greater root).

Brahmagupta gives the *bhāvanā* (composition) method by which, from two pairs of lesser and greater roots, a new pair can be generated. If  $(x_1, y_1)$  and  $(x_2, y_2)$  are solutions for the same  $N$ , then:

$$x' = x_1y_2 + x_2y_1, \quad y' = Nx_1x_2 + y_1y_2$$

is also a solution. This is the *samāsa-bhāvanā* (additive composition).

**The bhāvanā composition:** Brahmagupta’s *prakṛti-samāsa* (composition of roots) is one of the most remarkable achievements in early algebra. Given two solutions  $(x_1, y_1)$  and  $(x_2, y_2)$  of  $Nx^2 + k_1 = y^2$  and  $Nx^2 + k_2 = y^2$  respectively, then:

$$x_3 = x_1y_2 + x_2y_1, \quad y_3 = Nx_1x_2 + y_1y_2$$

satisfies  $Nx_3^2 + k_1k_2 = y_3^2$ . When  $k_1 = k_2 = 1$ , this generates an infinite family of solutions from a single minimal solution. For example, with  $N = 92$ , the minimal solution is  $x = 120$ ,  $y = 1151$ , since  $92 \cdot 120^2 + 1 = 1151^2$ .

## Reverse Motion (Viloma-gati)

**14.** Half the *rāśi* [sign], increased by its *śeṣa* [residue], and diminished by the *ardha-māsa* [half-month] etc.; [and] united with its *śeṣa* — he who states the *yuga-gata* [planetary position for the yuga] is the knower of the pulverizer.

**On reverse computation:** The *viloma-gati* method computes backwards from a given planetary longitude to find the corresponding elapsed time (*ahargaṇa*). Given a planet's mean longitude, one first determines the residual motion within the current sign (*rāśi*), then applies the pulverizer to find how many days have elapsed since the epoch. This is the inverse problem of the forward computation.

## Chapter 19: Gnomon and Shadow

*Shadow computations using the gnomon , time determination,  
and related astronomical problems*

### First Problem Statement

Divided by the divisor [*hara*], what is that quantity [*rāśi*] which becomes pure? First, of the multiplier and divisor, the greater is brought [into consideration] by means of the *apavarta* [reduction].

**[English translation:]** When two [numbers] divide each other mutually, that which [leaves] a remainder — by that they become two multiplier-divisor [pairs]. Then, having placed the quotients obtained from the mutual division of the multiplier and divisor in the heart [i.e., in a column], according to the pulverizer rule demonstrated previously. This operation is to be performed until finally unity remains as the remainder. That remainder, multiplied by some multiplier and diminished by unity, when divided by the divisor related to that remainder, becomes pure. Having done so, the multiplier is to be placed below the previously stated quotients; below that, the result is placed. In this manner the chain (*vallī*) is produced.

### Determination of Gnomon and Shadow

The determination of the shadow of houses etc. from the [Sun's] shadow. The solution of the problem: "If [the shadow] is equal to the height of the desired cave" — [its computation is given].

**On shadow problems:** The gnomon (*śaṅku*) is a vertical staff of standard height (typically 12 *aṅgulas*). Its shadow (*chāyā*) cast by the Sun is used to determine:

1. The Sun's altitude:  $\sin(\text{altitude}) = \frac{\text{śaṅku}}{\sqrt{\text{śaṅku}^2 + \text{chāyā}^2}}$
2. The time of day from the shadow's length
3. The dimensions of objects from their shadows
4. The *palabha* (equinoctial shadow) for latitude determination

If the shadow equals the height of a cave (i.e., the Sun's altitude is 45°), the horizontal distance from the gnomon to the cave equals the cave's height.

### Height Determination

The determination of the height of a nearby elevated object and of a distant one; and the statement of the motion of light [rays], and the measure of the interval between the two.

**[Height from shadow — examples:]** (a) The determination of the answer to the problem of a house, a man, the interval, and water — concerning a *haṭa-caitya* [pillar shrine] etc.

(b) The problem concerning a seed in a cave, water, and spreading — its solution is stated.

(c) By the method of double-difference, the answers to three problems from the shadow of one house are given.

(d) From the shadow, the determination of shadows of houses etc.

**The double-difference method:** For determining heights from shadows: If two observations of a gnomon and its shadow are made at different times or positions, the height  $h$  of an object and its horizontal distance  $d$  can be found from:

$$h = \frac{s_2 \cdot g}{s_2 - s_1} \cdot \frac{d_2 - d_1}{g}, \quad d = \frac{h \cdot s_1}{g}$$

where  $g$  is the gnomon height,  $s_1, s_2$  are shadow lengths at two positions, and  $d_1, d_2$  are the distances of the observation points from the object's base. This is equivalent to similar-triangle methods in modern trigonometry.

## Chapter 20: Advanced Shadow Problems

*Extended shadow computations: obscuration, refraction,  
and celestial sphere applications*

Of the *chad-dirāśi* [obscured signs], the day-and-night strengths — the rise of the signs, the statement of their unequal pairs.

The position of the *carāgra* [ascensional difference] is stated. The position of the gnomon and shadow is stated.

Of the *hag-gola* [terrestrial sphere], the cause of the *lambana* [parallax] and *avanati* [depression] by means of the *haryāhara* [division-operation] is shown. The statement of the *para-lambana* [secondary parallax] and *avanati*.

**[English:]** By another method, the position of those two [gnomon and shadow] and of the *śaṅkula* [combined gnomon-shadow] is stated.

The statement of the *hak-karma* [obscuration computation]. The fixed pulverizer of the planets and the celestial sphere. The moving pulverizer of the planets. The conclusion of the chapter.

Here the computation of shadows is shown for the knowledge of the height of a lamp-flame, the boundary of light, and the Earth. Something is also said concerning the reflection of light.

**On the *chad-dichāyā*:** This chapter extends the shadow calculations of Chapter 19 to more complex astronomical situations. The *chad-dirāśi* refers to zodiacal signs that are “obscured” (below the horizon), and the computation of their day/night strengths is relevant to astrological applications. The *carāgra* (ascensional difference) measures the unequal rising times of signs at different latitudes, computed from:

$$\text{carāgra} = \arcsin \left( \frac{\tan(\phi) \cdot \tan(\delta)}{\cos(\alpha)} \right)$$

where  $\phi$  is the observer's latitude,  $\delta$  is the celestial declination, and  $\alpha$  is the right ascension.

## Chapter 21: Spherical Astronomy (Golādhyāya)

*The opening of the Spherical Astronomy Book: armillary spheres, celestial and terrestrial globes, astronomical instruments*

### Purpose of the Chapter

The praise of the celestial sphere (*gola*) is stated. The cause of the proper position of one's own sphere [is given].

The praise of the sphere of motion [is stated]. The purpose of the commencement of instruments [is stated].

#### [Contents of the Golādhyāya:]

1. The establishment of instruments (*tantra*)
2. The use of water etc. [in instruments]
3. The statement concerning arcs (bows)
4. The bearing of instruments toward the Sun etc.
5. The determination of the form of the arc for a desired ghaṭikā
6. The determination of the testing ghaṭikā
7. The knowledge of time by means of the instrument's depression and elevation
8. Special rules concerning the bow-instrument

### Arrangement of the Terrestrial Sphere

The arrangement of the terrestrial sphere is stated. The description of the arrangement of the divine and demoniacal [cities].

**[The celestial cities:]** The establishment of the Aries-cycle and the [rate of] motion for the divine and demoniacal [cities]. The statement of the arrangement of the wheel's revolution. Of the days of the gods, for the purpose of the Sun's revolution. The zodiacal arrangement of the divine and demoniacal [cities]. The determination of the measure of the day of the father [Brahmā] for the divine and demoniacal.

**The celestial geography:** Brahmagupta describes the traditional Indian celestial geography. At the North Pole is *Meru*, the abode of the gods (*deva*); at the South Pole is *Vadavā-mukha*, the realm of the demons (*asura/daitya*). The central terrestrial sphere (*bhū-gola*) has the observer's horizon, the prime meridian (through Ujjain), and the equator.

The divine day (one day of Brahmā) equals 1000 *caturyugas* (4,320,000,000 solar years). The solar year is defined by the Sun's return to the same fixed star — the *nakṣatra-vatsara* (sidereal year). The *divya-dina* (divine day) of 360 solar years is used for calculating planetary mean motions.

## Prime Meridian and Longitude Difference

The arrangement of the terrestrial sphere and the *lagna-valaya* [horizon circles] is [stated]. The statement of the *nirakṣa-deśāntara* [equatorial longitude difference] in *yojanas*.

The description of the two *kṣudra-dyau* [small celestial circles].

**[On the *deśāntara*:]** The prime meridian (*nirakṣa-deśa*) passes through Ujjain, where the celestial equator (the east-west circle) is traversed equally by all planets. For any other location, the longitude difference (*deśāntara*) is measured in *yojanas* — for every four *yojanas*, there is one *laghu-ghaṭikā* (small ghaṭikā, about 4 minutes of time).

Accordingly, the rising times of planets vary by the *deśāntara* ghaṭikās. The *deśāntara-kāla* (longitude-difference time) is computed as:

$$\text{deśāntara} = \frac{\text{yojanas from Ujjain}}{4}$$

in *ghaṭikās* (1 ghaṭikā = 24 minutes).

**The *deśāntara* formula:** Brahmagupta gives the standard Indian rule for computing longitude difference from Ujjain (the prime meridian): every 4 *yojanas* east or west corresponds to a 1-ghaṭikā (24-minute) difference in local time. Since the Earth's circumference was taken as approximately 5000 *yojanas*, this gives 1250 ghaṭikās for a full circle — close to the actual  $60 \times 24 = 1440$  minutes, though the *yojana* was not a precisely standardized unit.

The *deśāntara* is essential for computing the local time of planetary risings, setting of the *lagna* (ascendant), and the timing of eclipses at different longitudes.

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